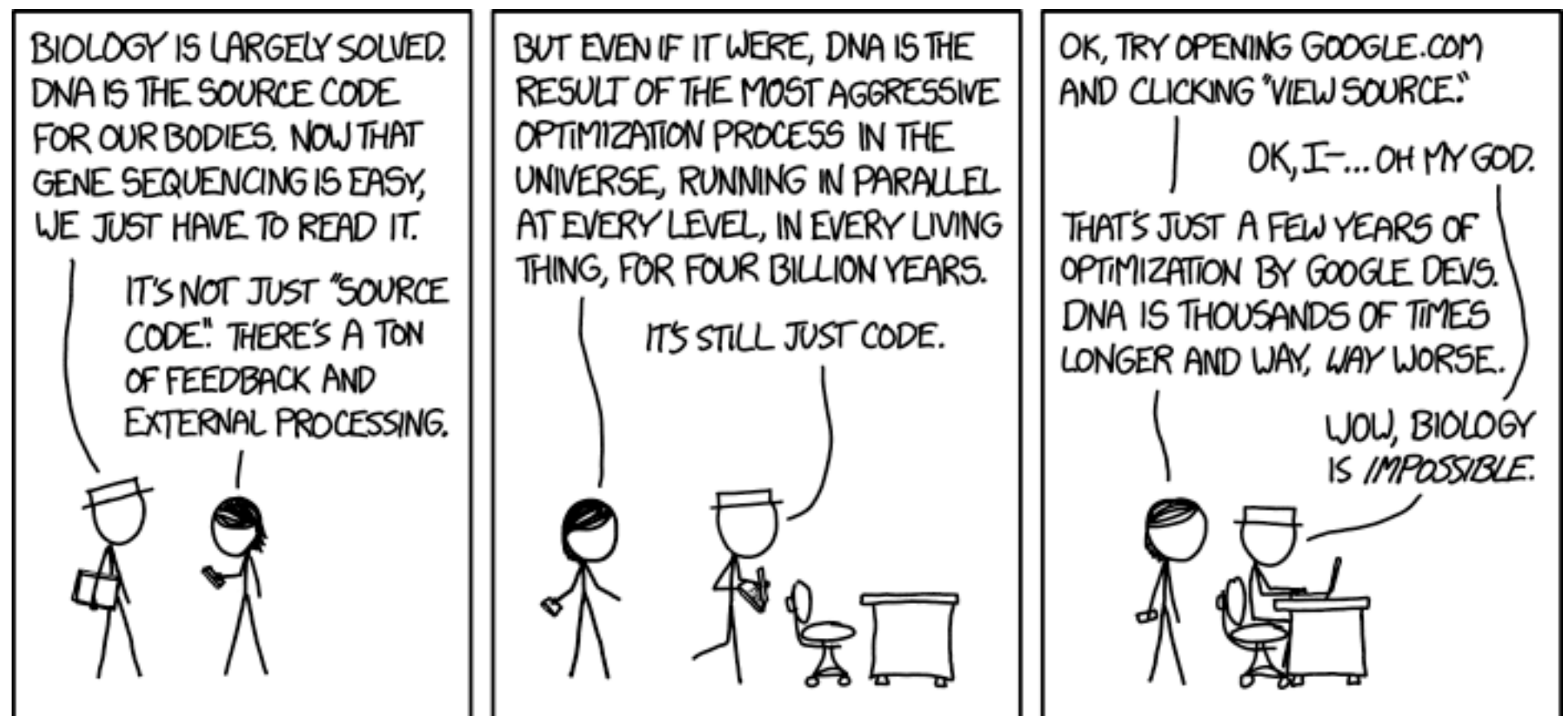


Fitting time-dependent models to the transcription of RpoS-dependent genes in E-coli

Jo Hardin
Pomona College
July 15, 2024



Sensitivity to RpoS

Image credit: <https://doi.org/10.1128/msystems.00663-23>

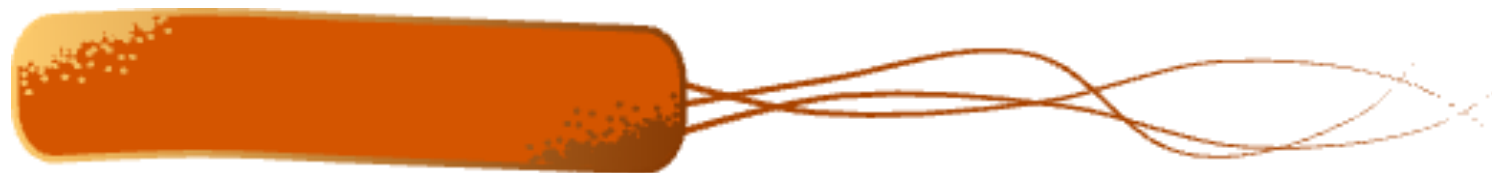


Image credit: <https://bioicons.com/>

- The alternative sigma factor RpoS is a central regulator of many stress responses in *Escherichia coli*.
- We studied transcriptional changes across differences in RpoS.
- Experiment: 3 different levels of RpoS

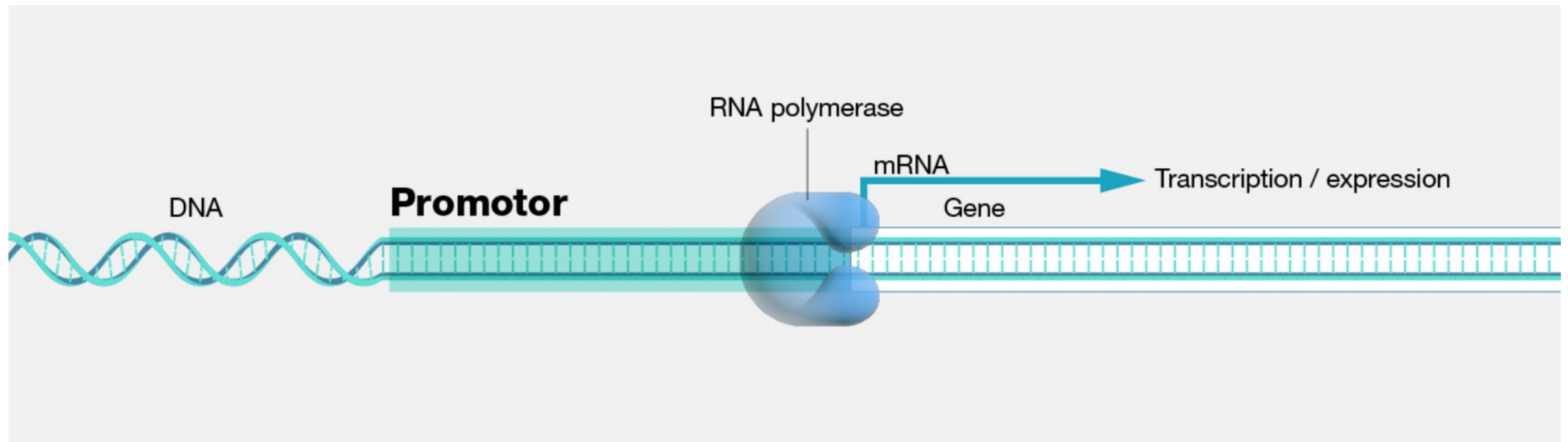


Image credit: <https://www.genome.gov/genetics-glossary/Promoter>

First set of
experiments

- RNA-Seq: differential expression of 1,044 genes (23% of genes) between the two conditions
- 605 are upregulated
- 439 are downregulated
- RpoS is either present or absent

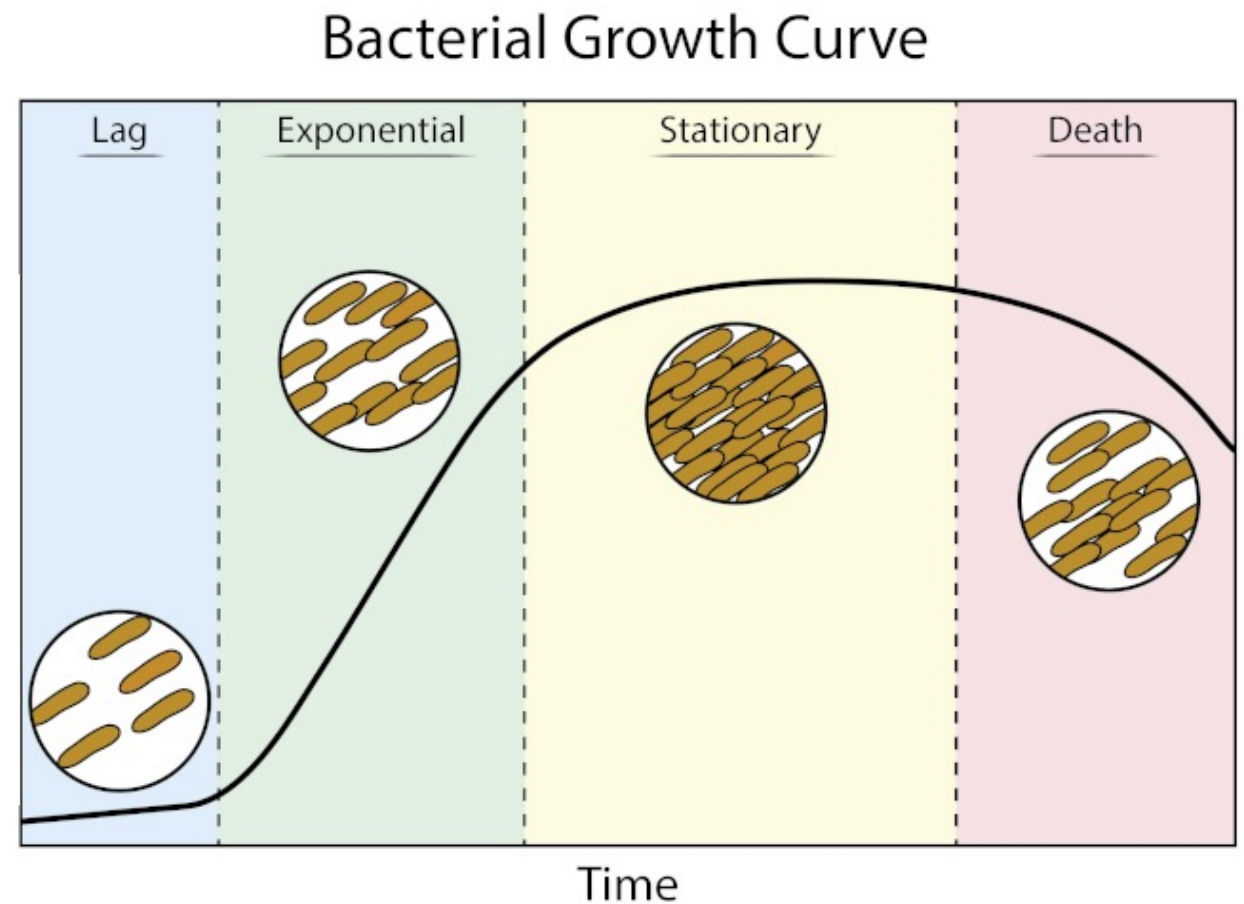
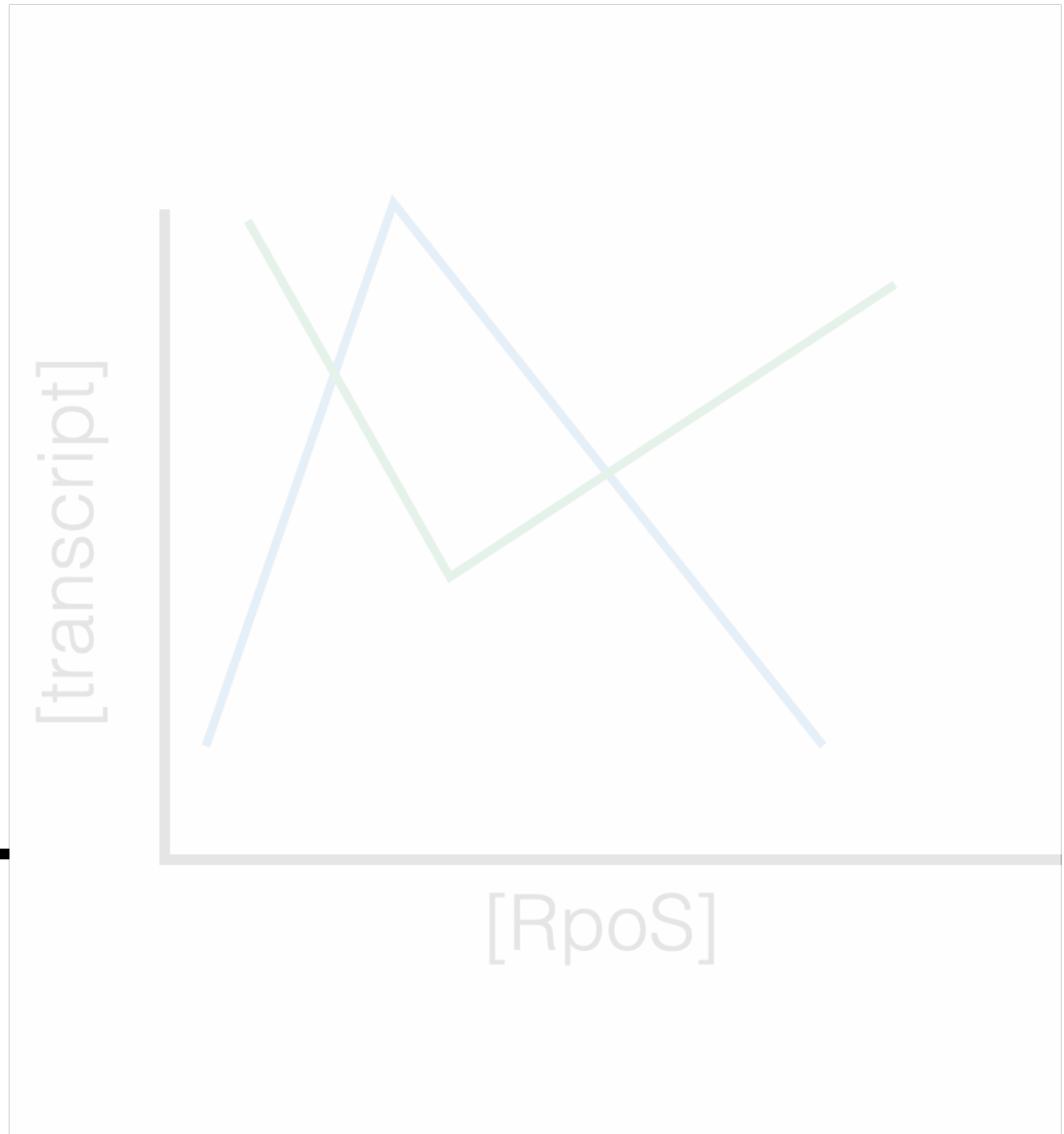
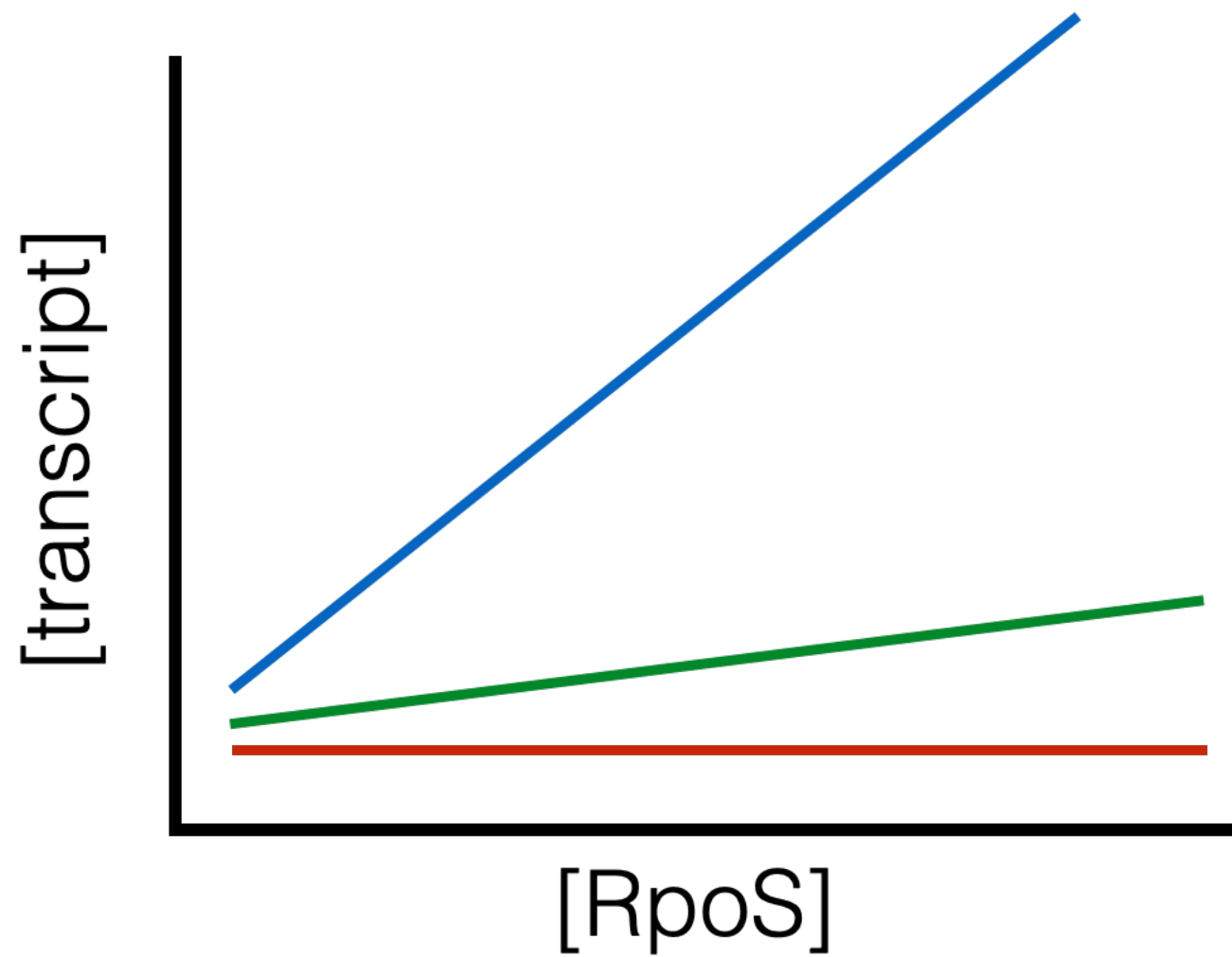
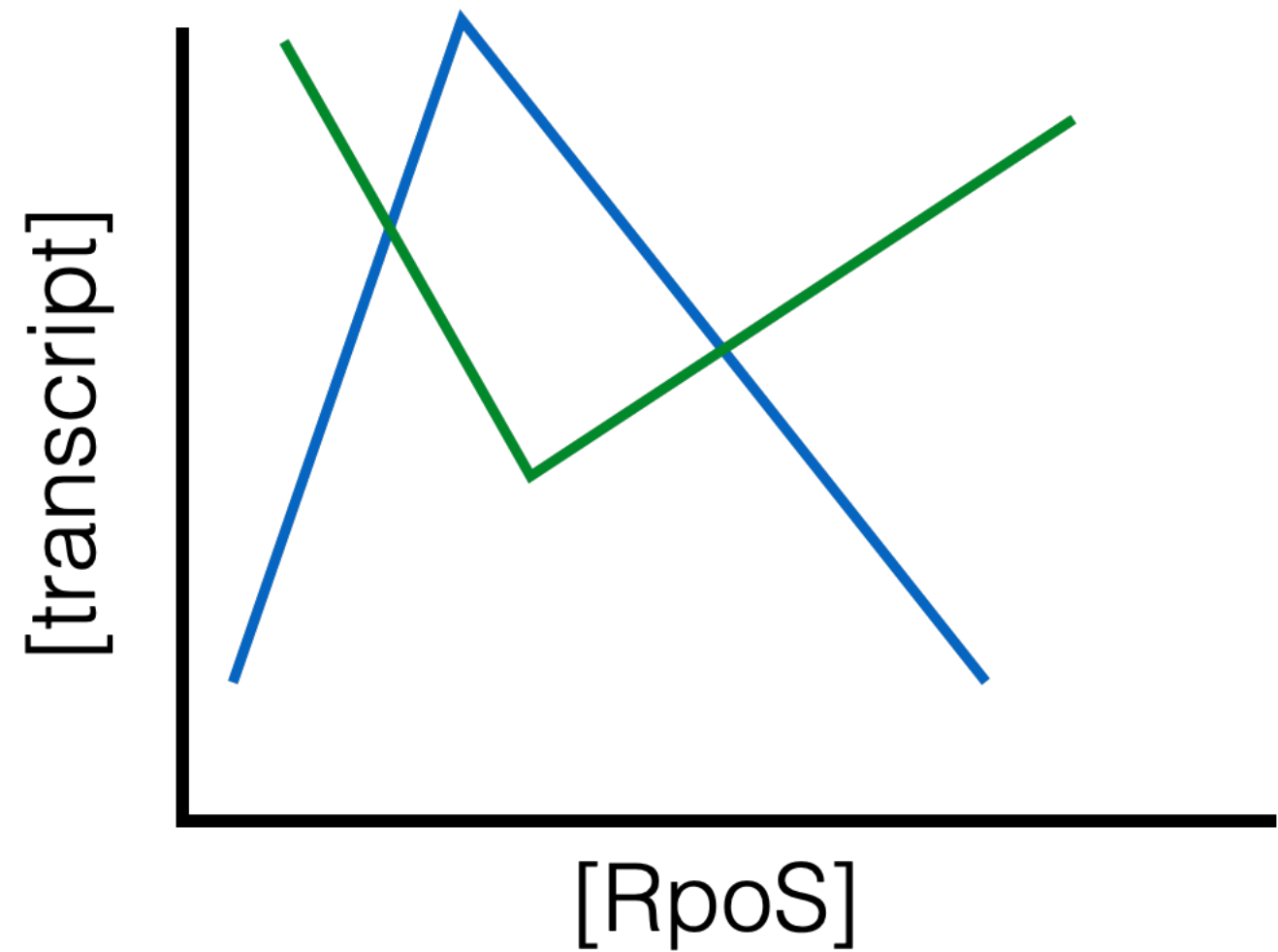
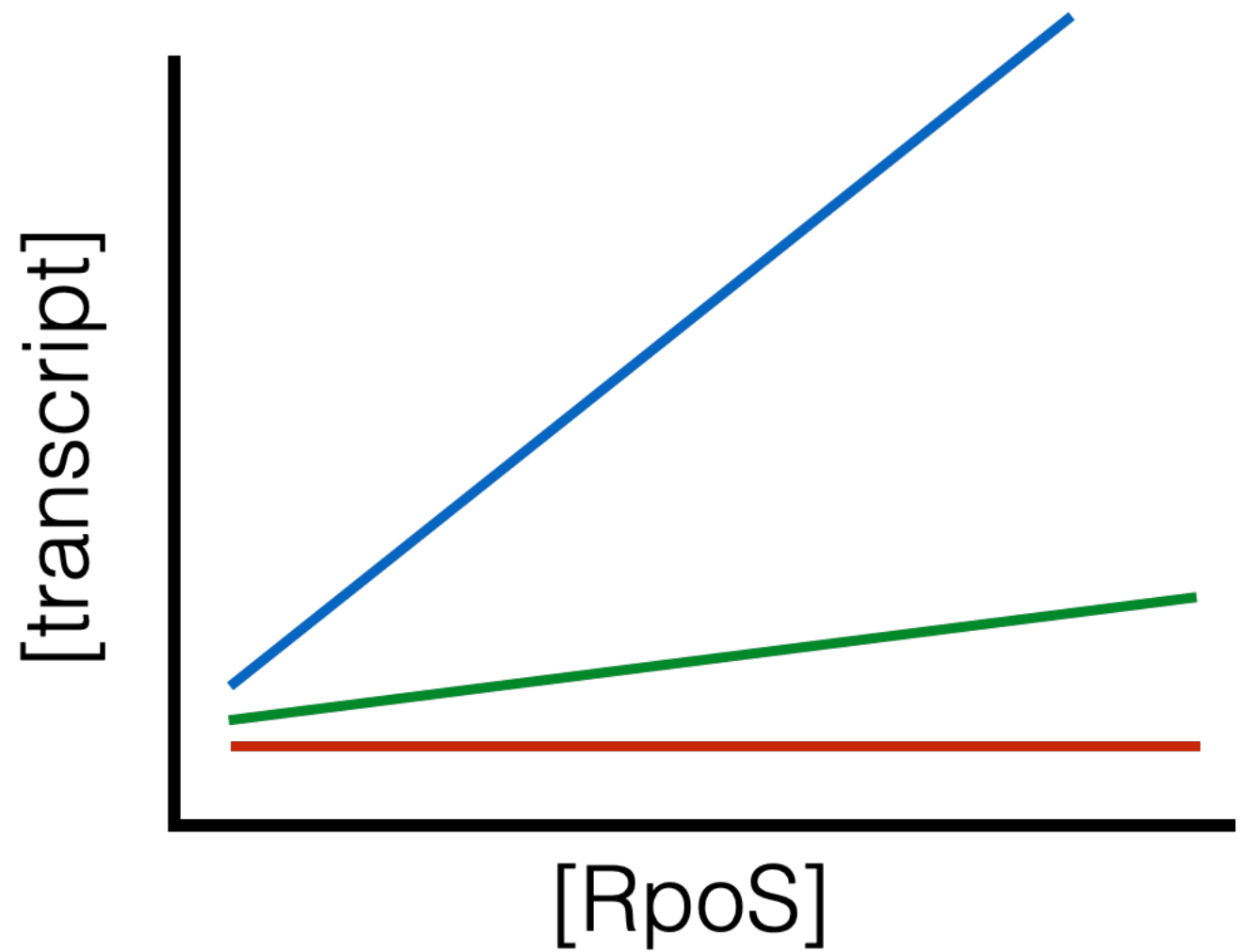


Image credit: <https://www.jove.com/v/10511/growth-curves-cfu-and-optical-density-measurements>

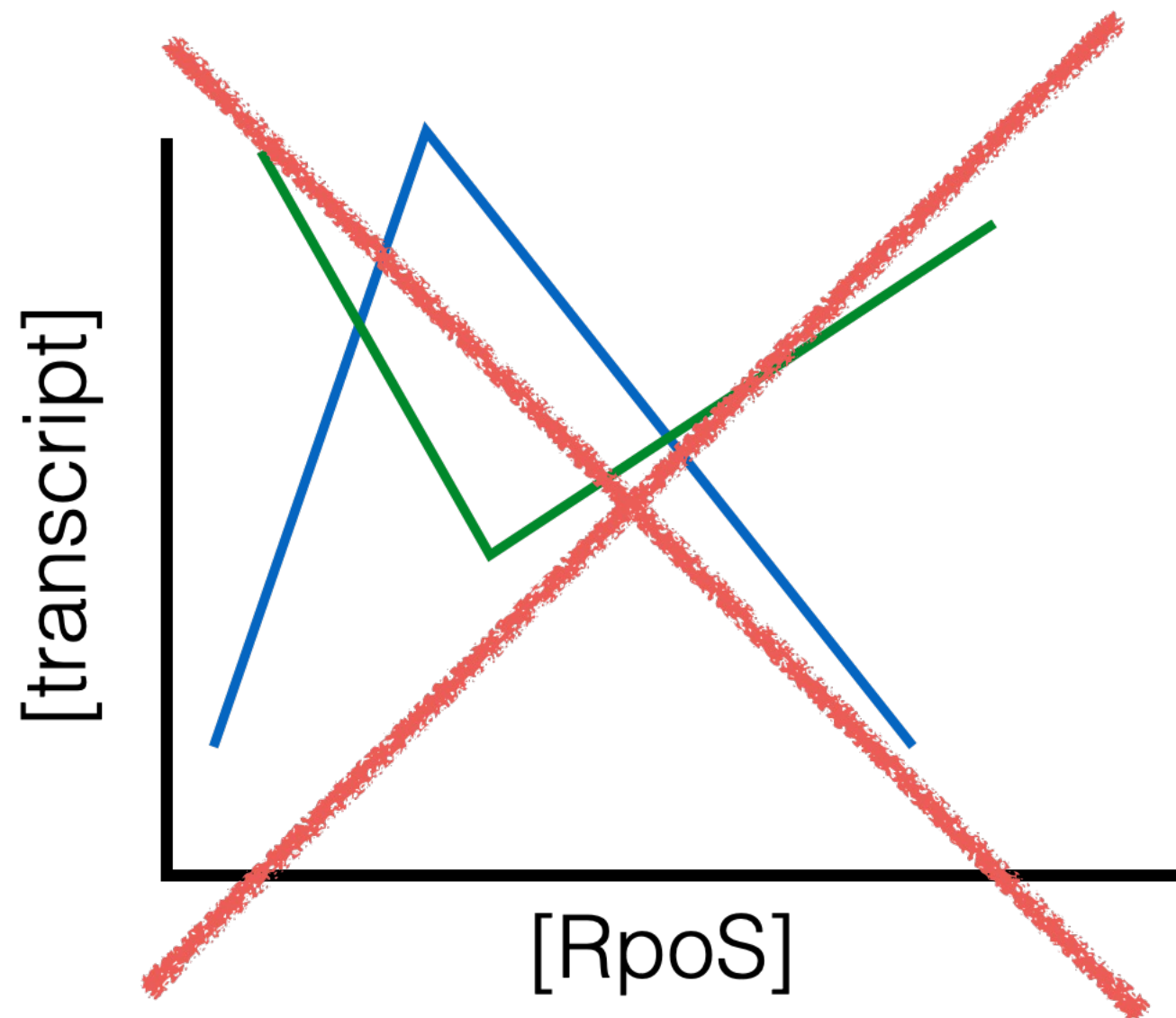
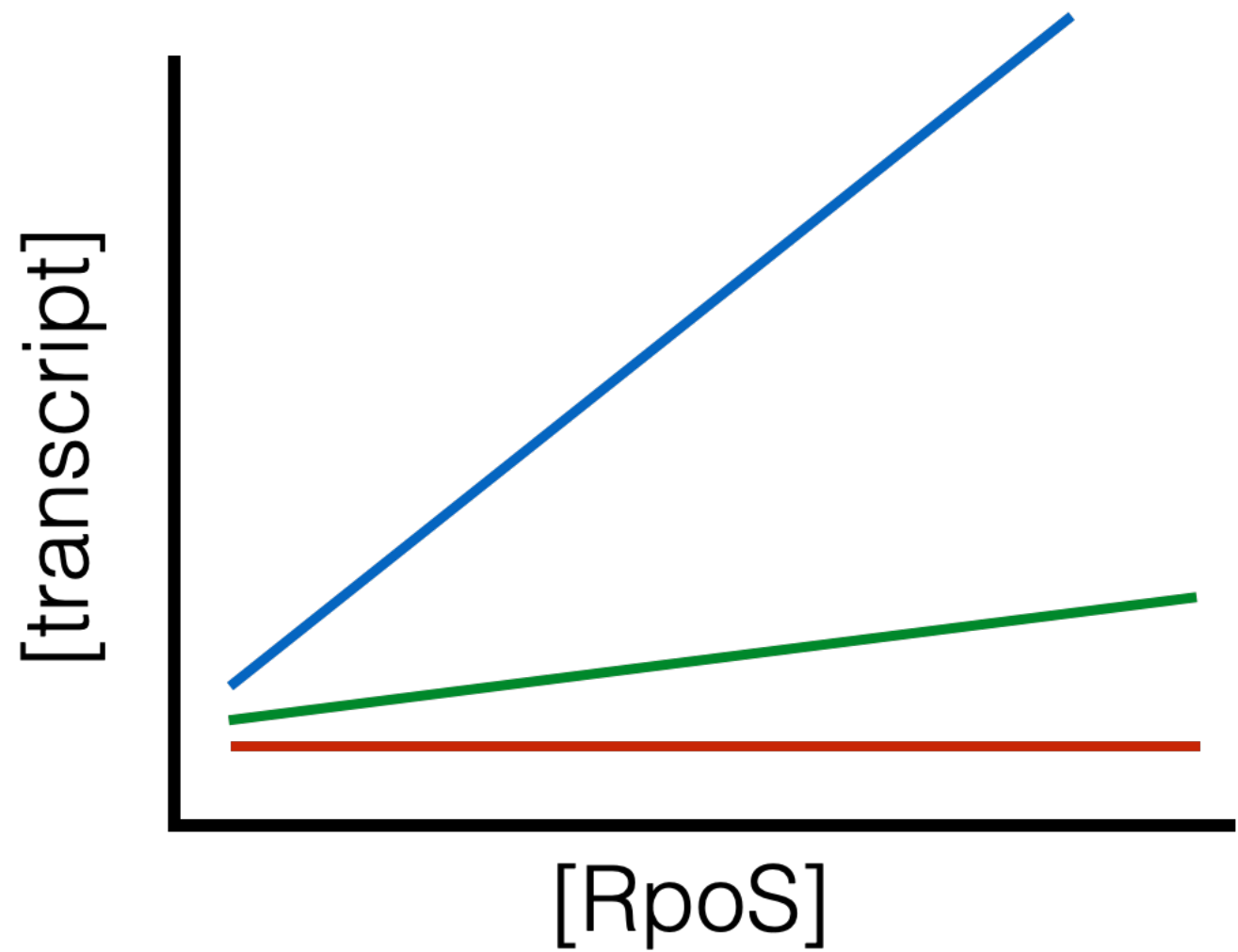
wt vs $\Delta rpoS$ in Stationary Phase



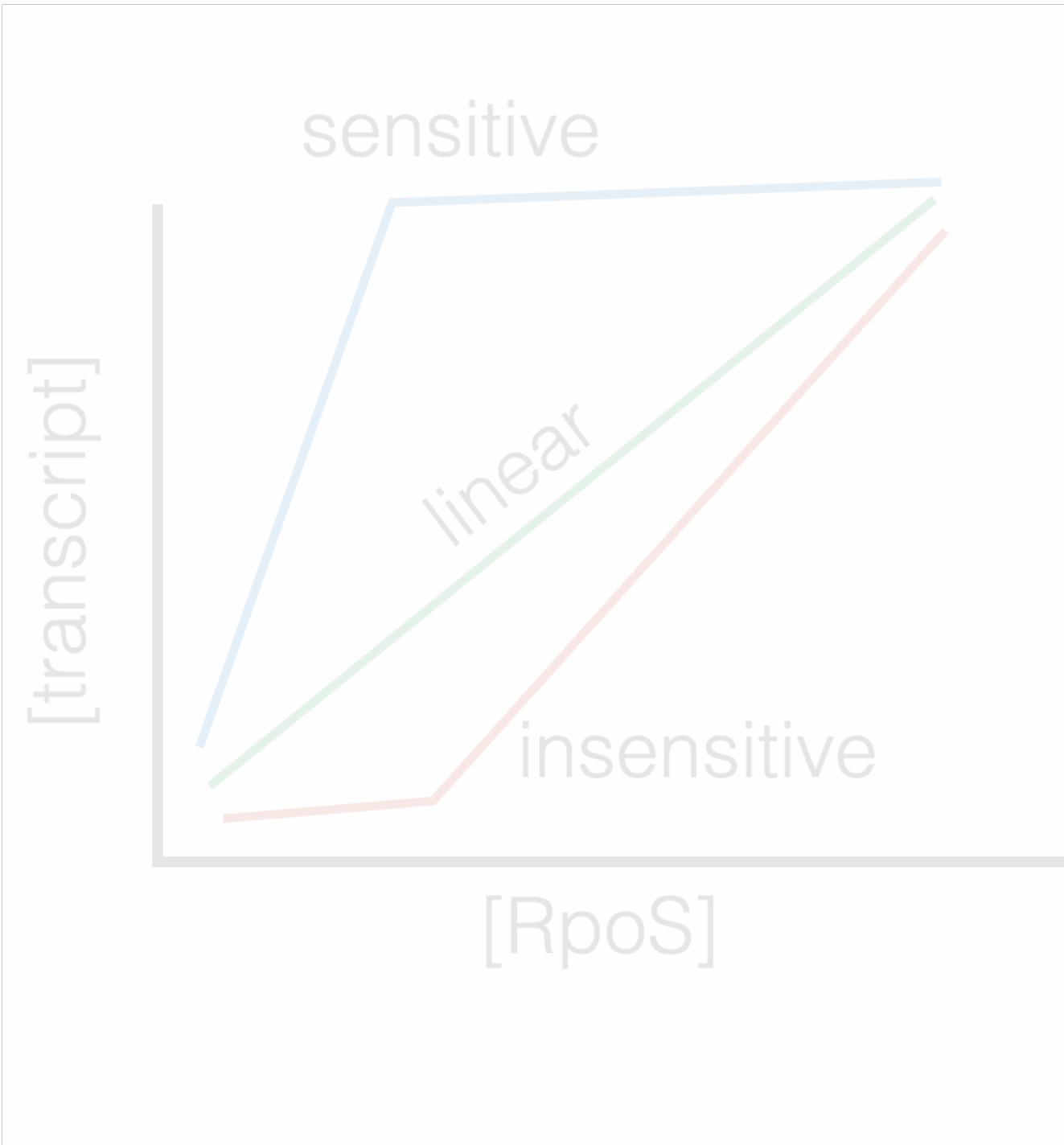
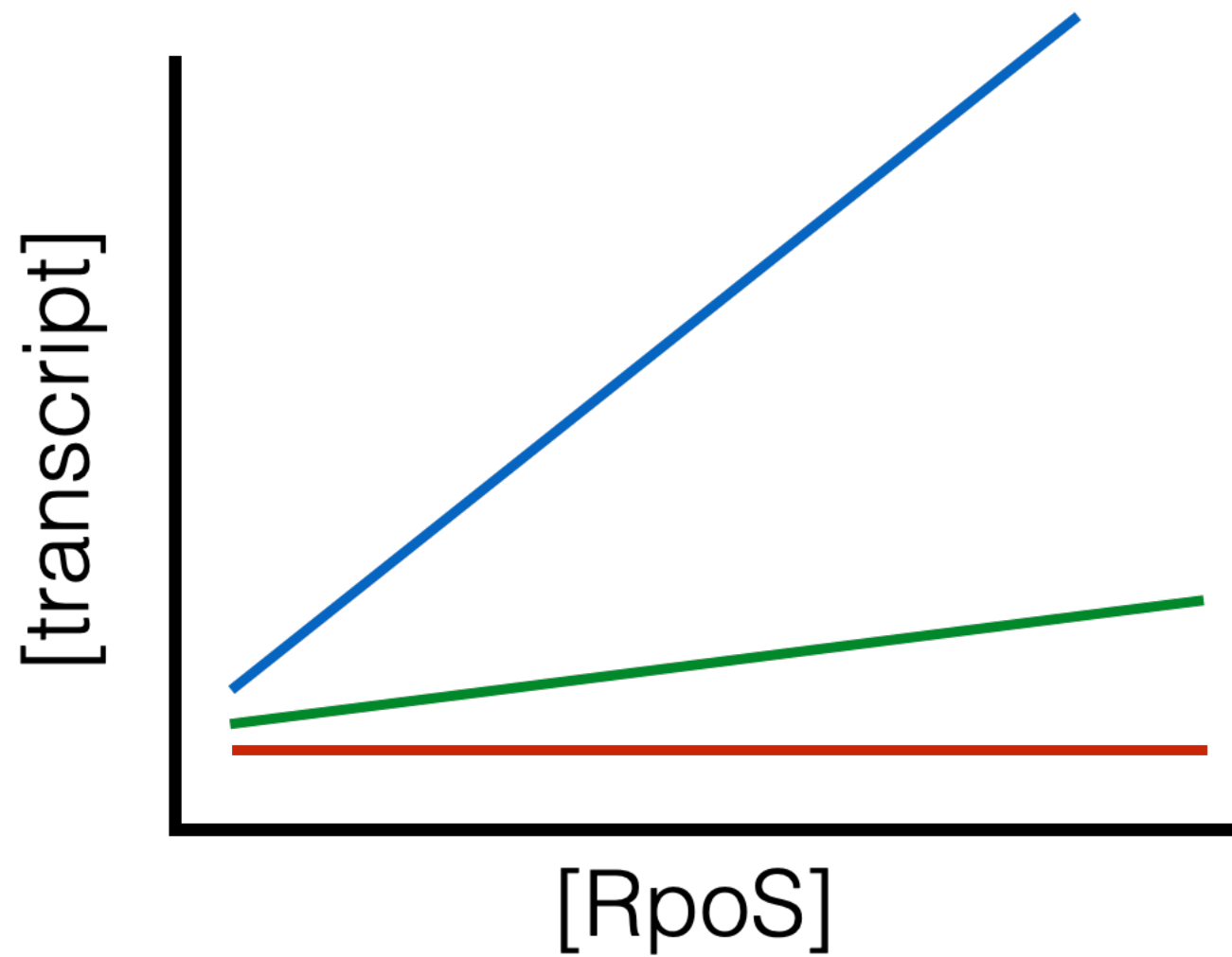
Possible models of transcription
as a function of: **RpoS**



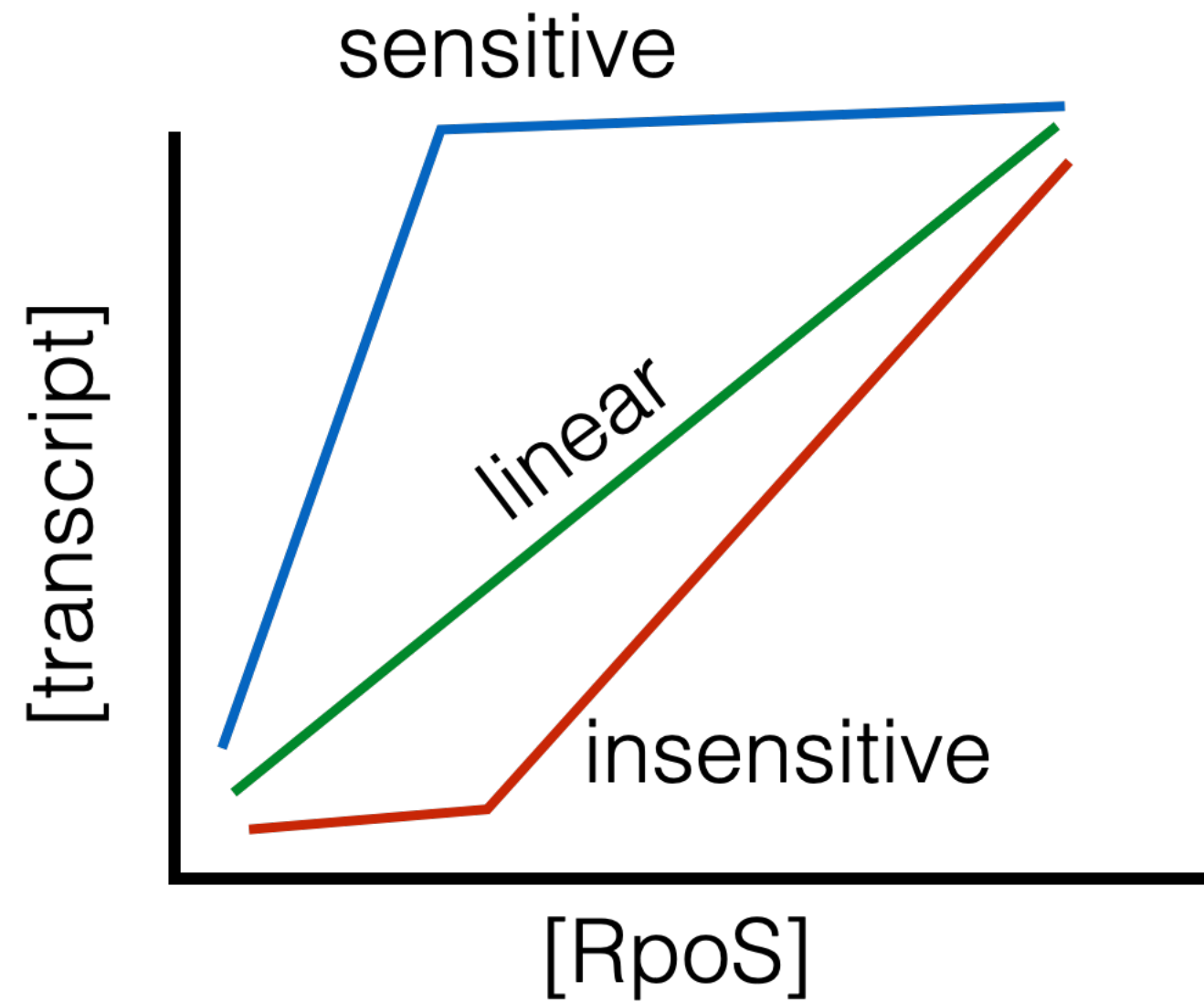
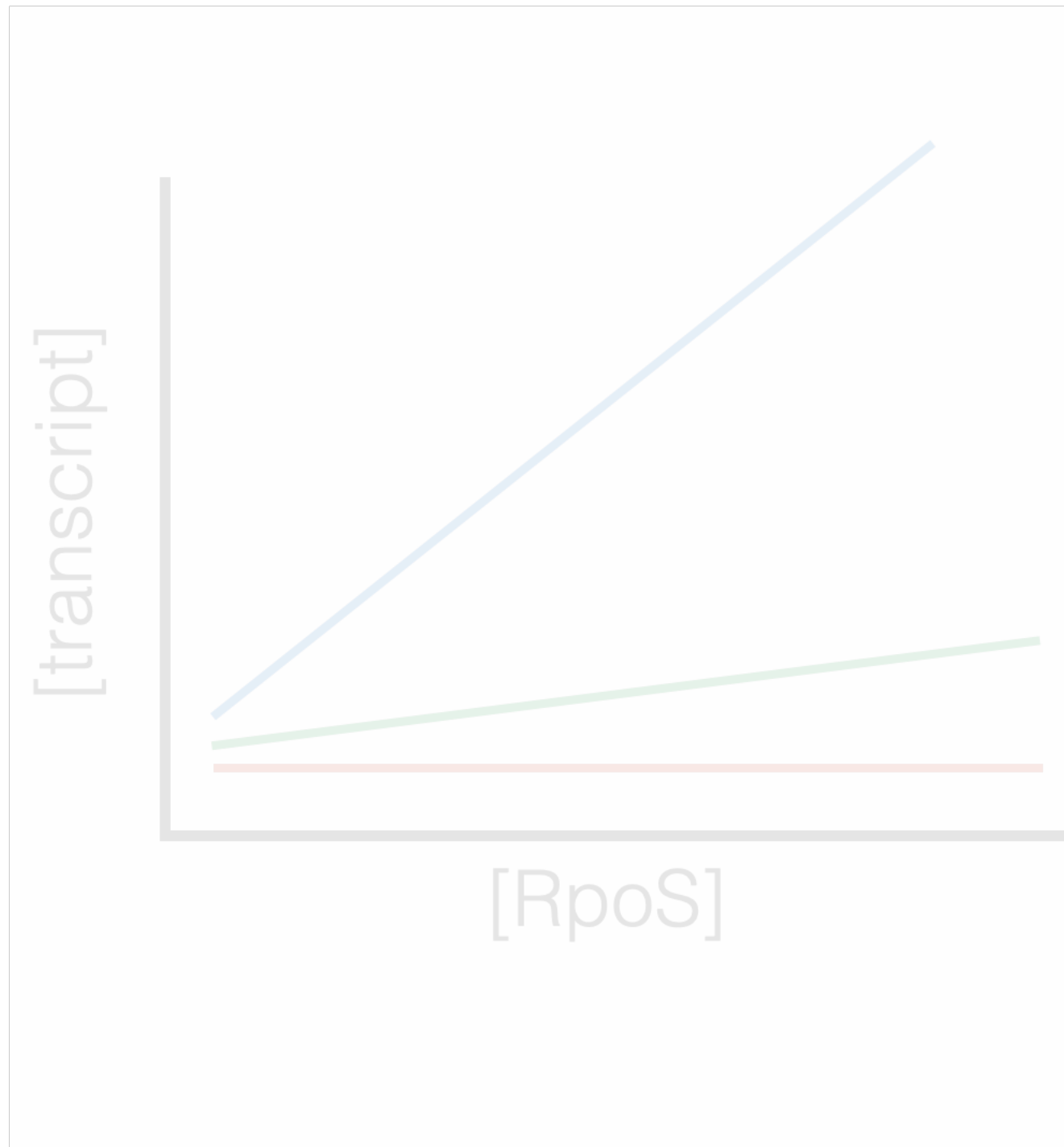
Possible models of transcription
as a function of: **RpoS**



Possible models of transcription
as a function of: **RpoS**

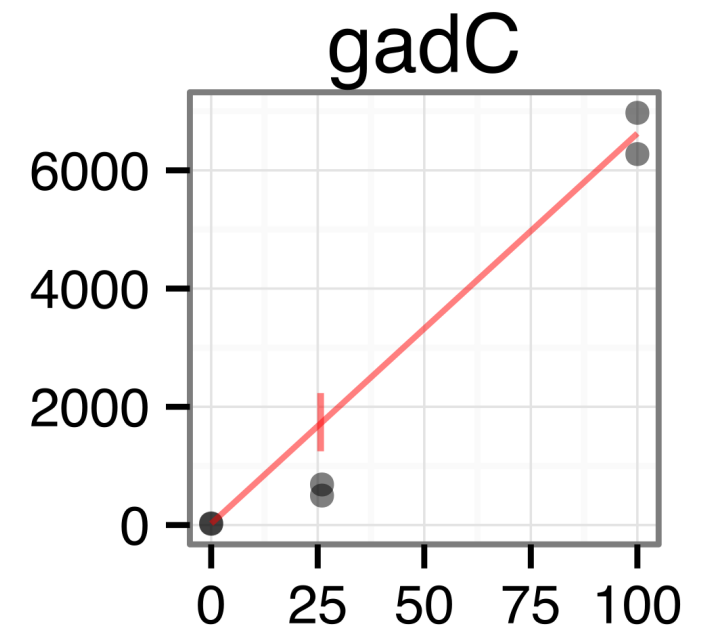
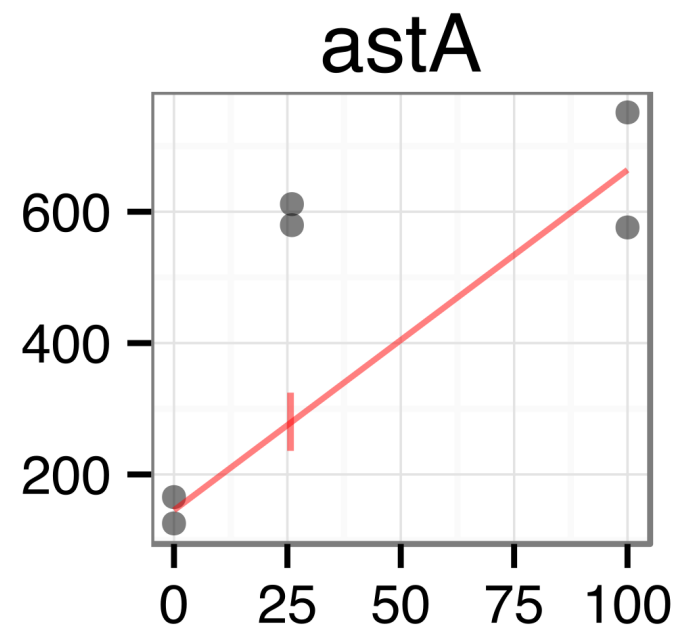
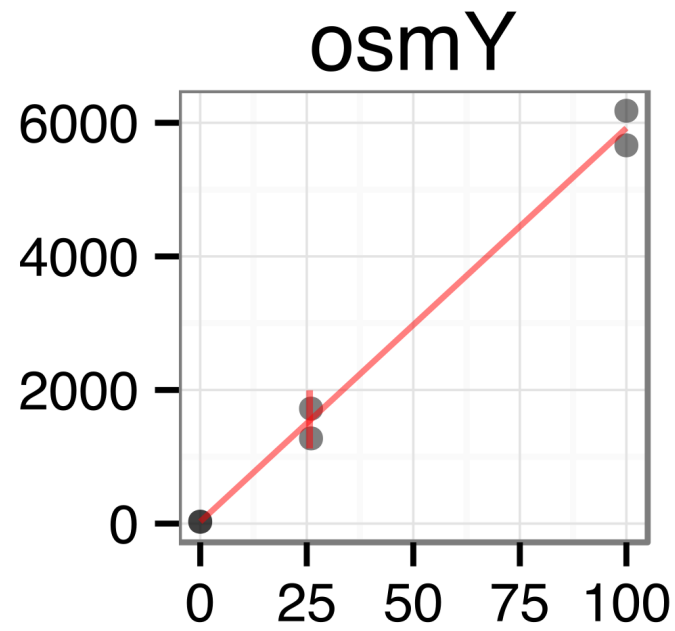


Possible models of transcription
as a function of: **RpoS**



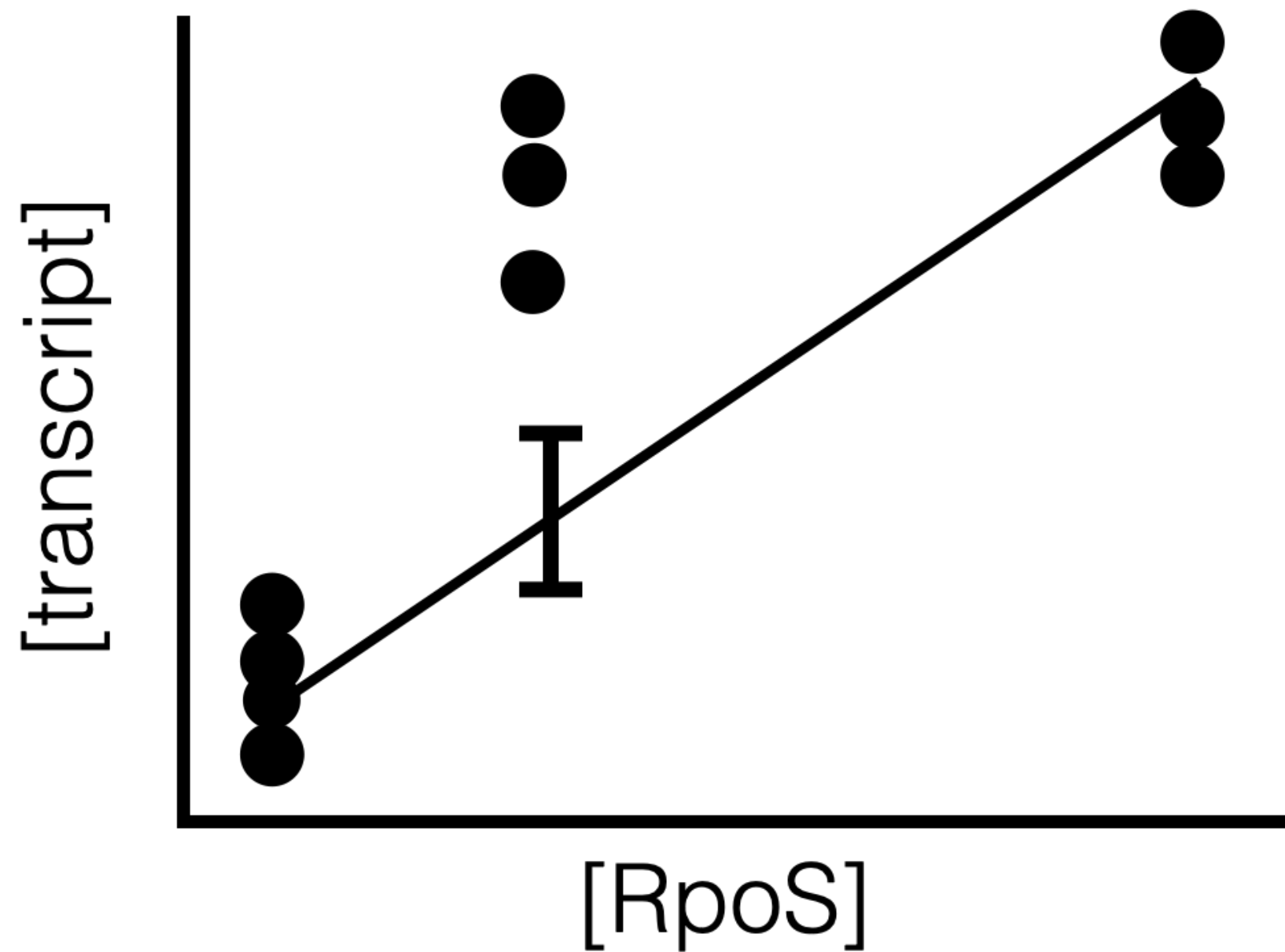
Possible models of transcription
as a function of: **RpoS**

Normalized
expression

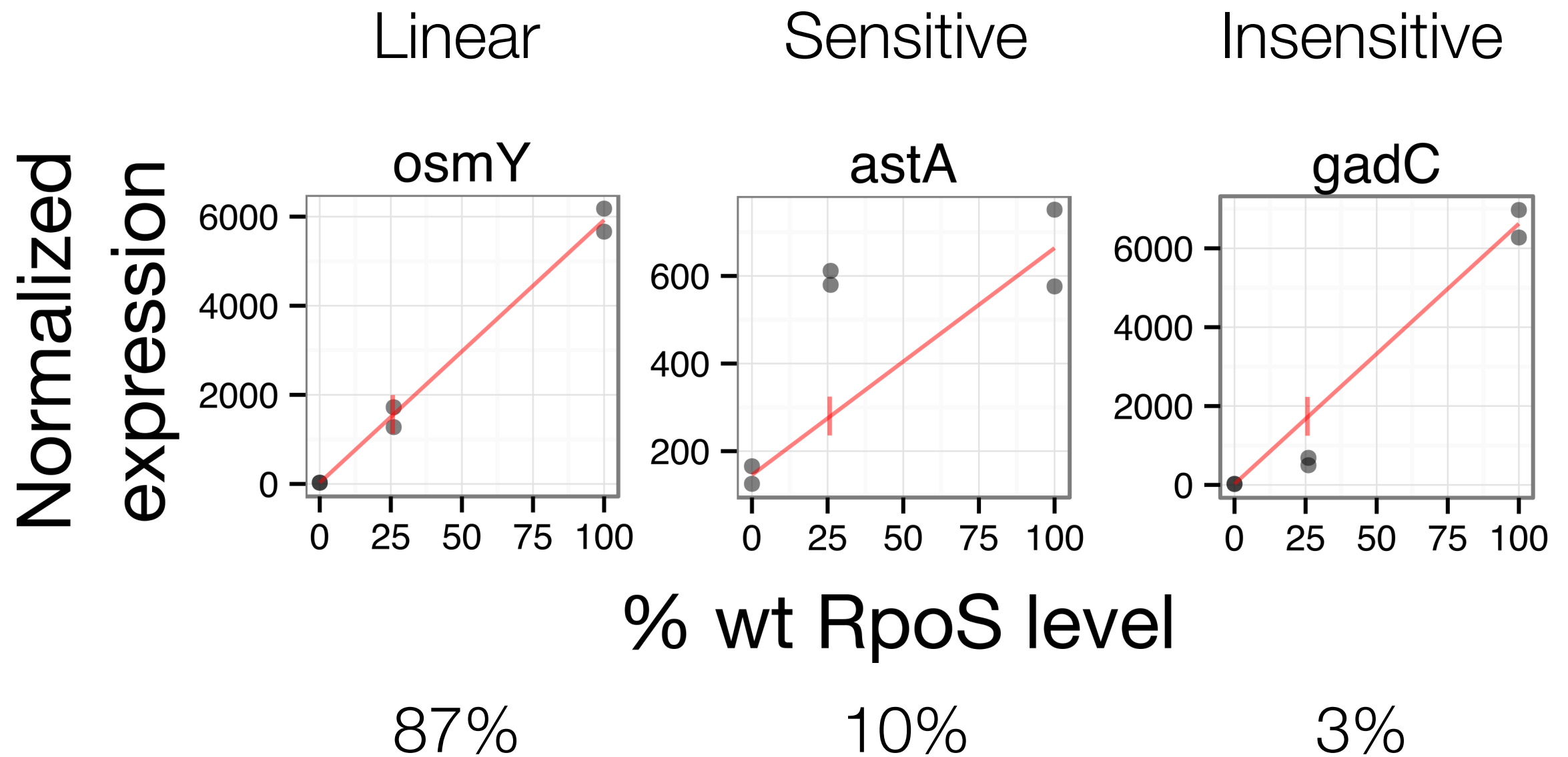


% wt RpoS level

Actual patterns of
response to RpoS



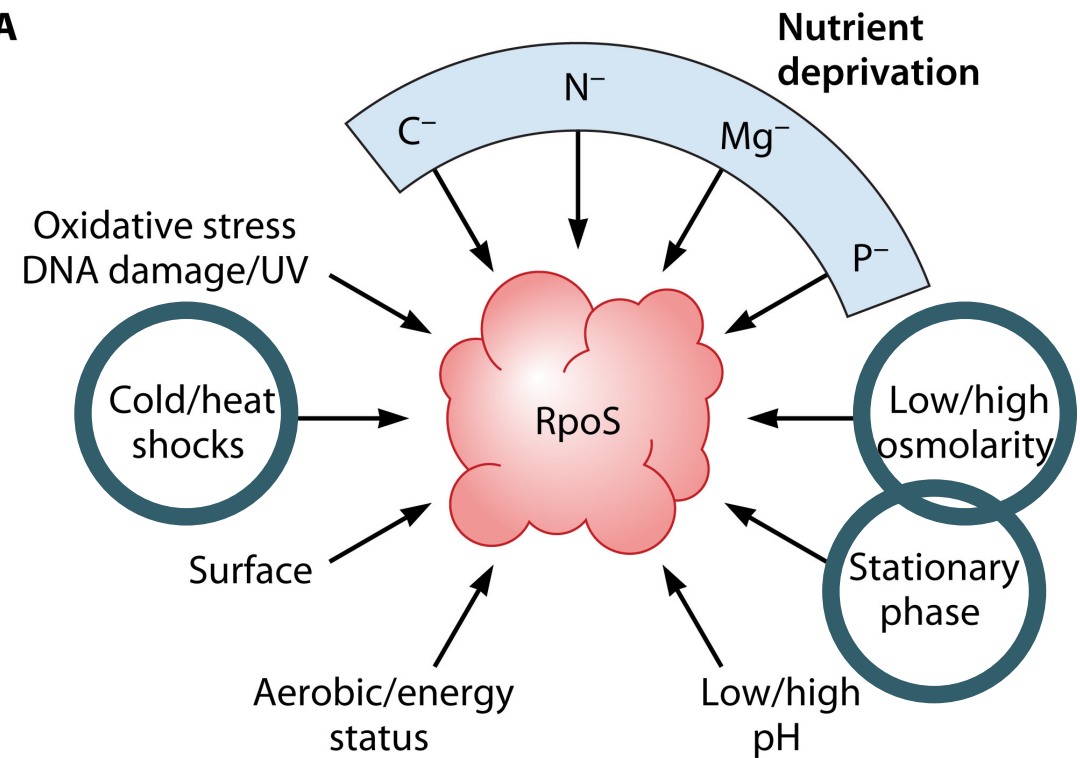
Testing for the sensitivity of
genes in the RpoS regulon



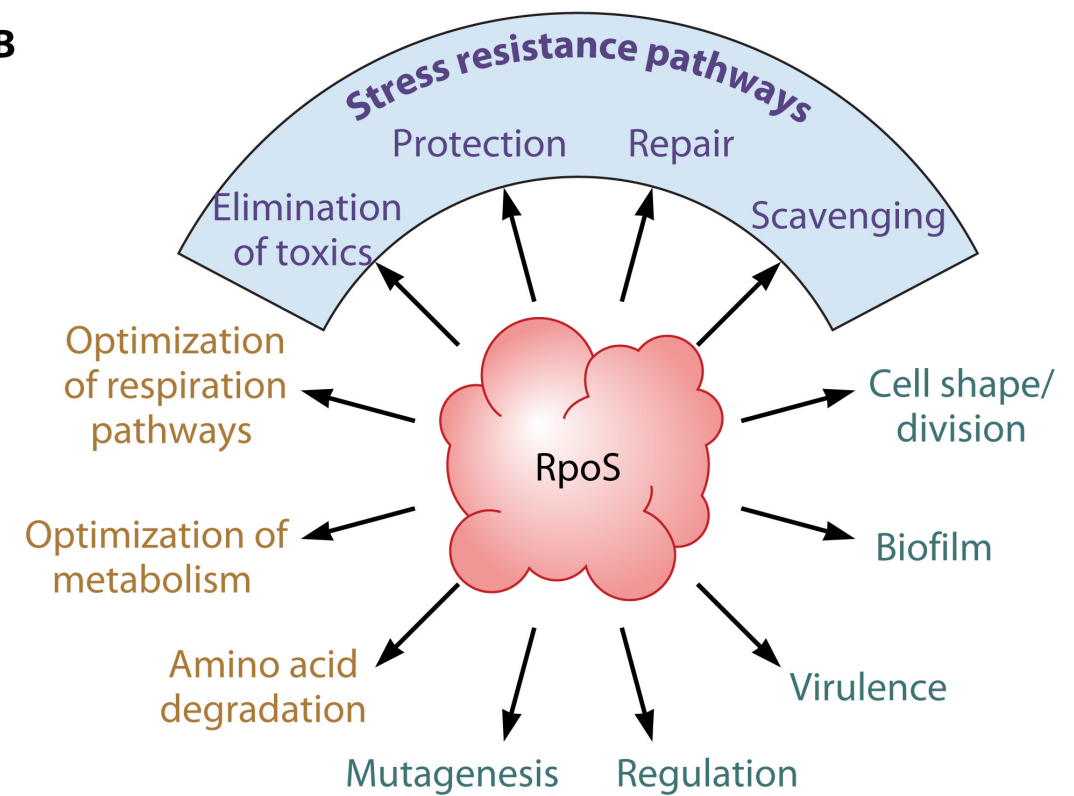
Actual patterns of
response to RpoS

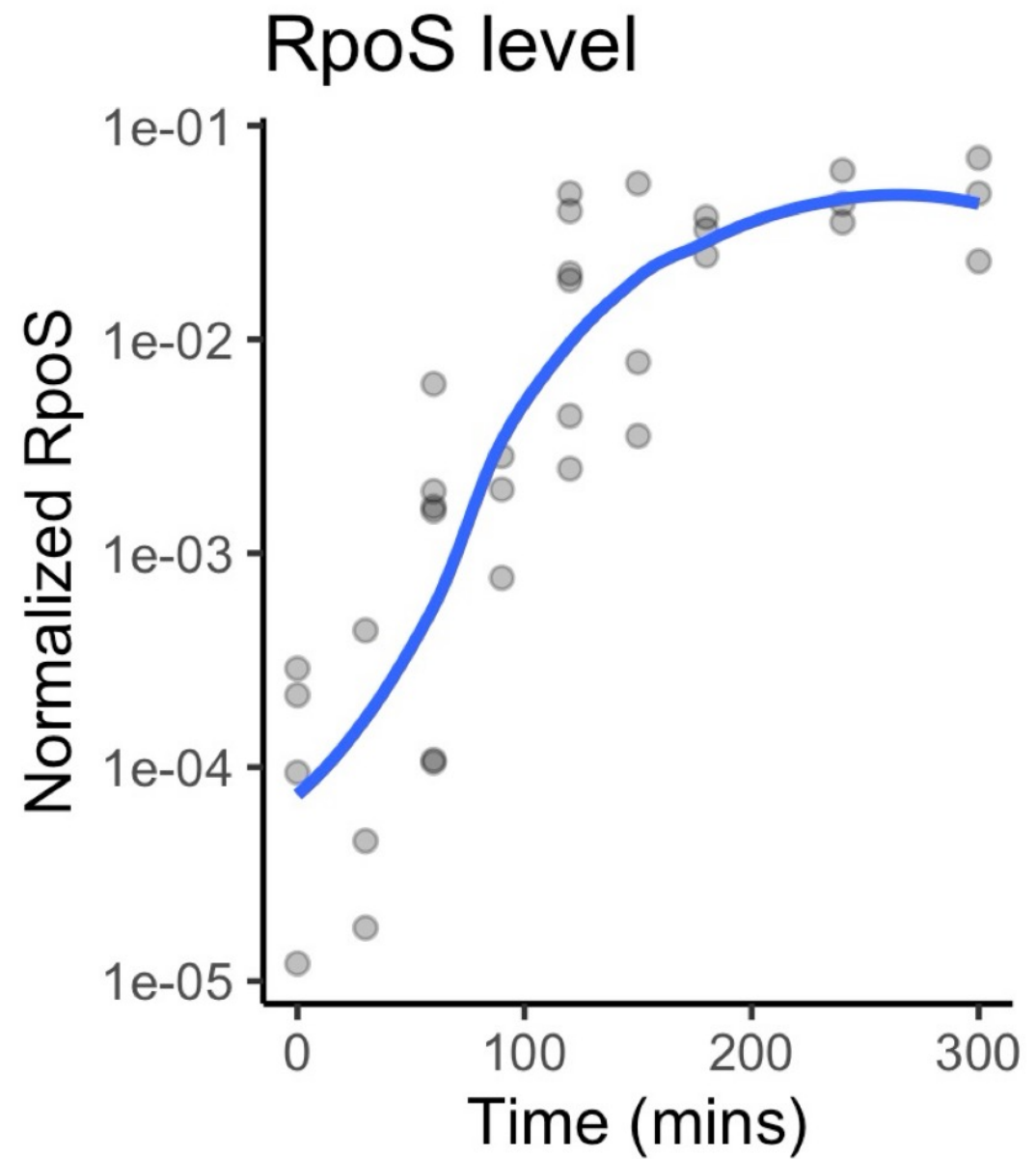
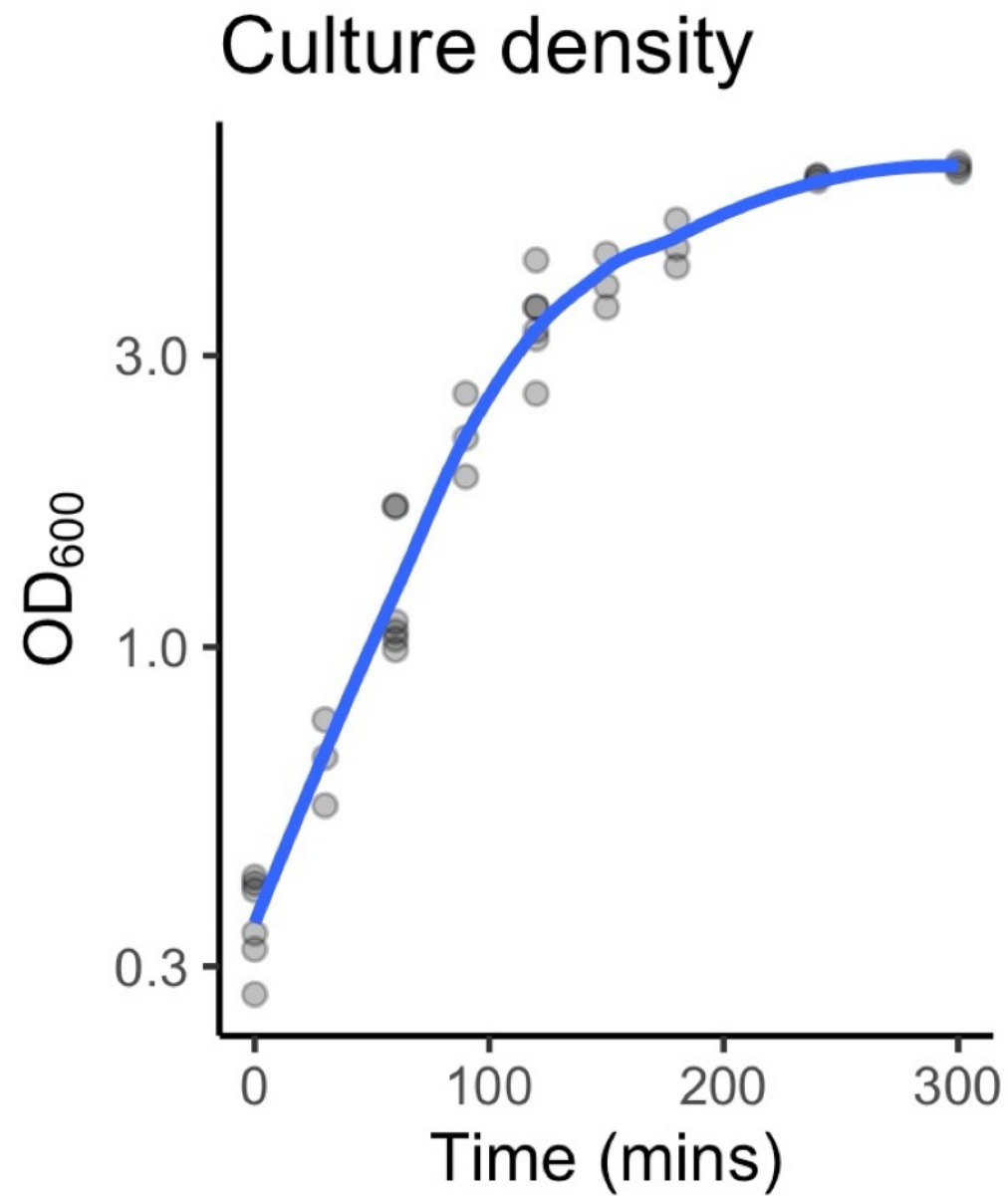
Second set of experiments

A



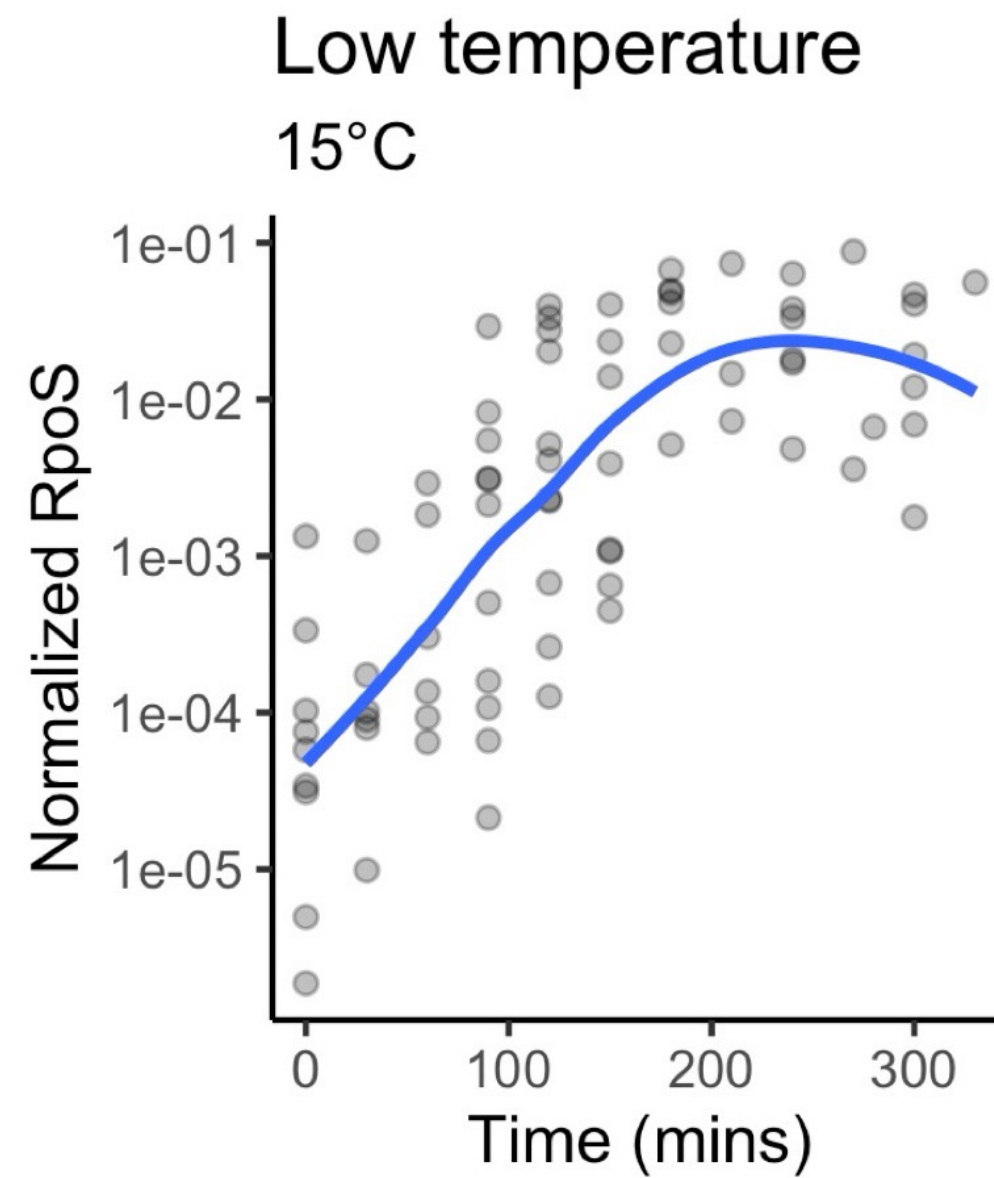
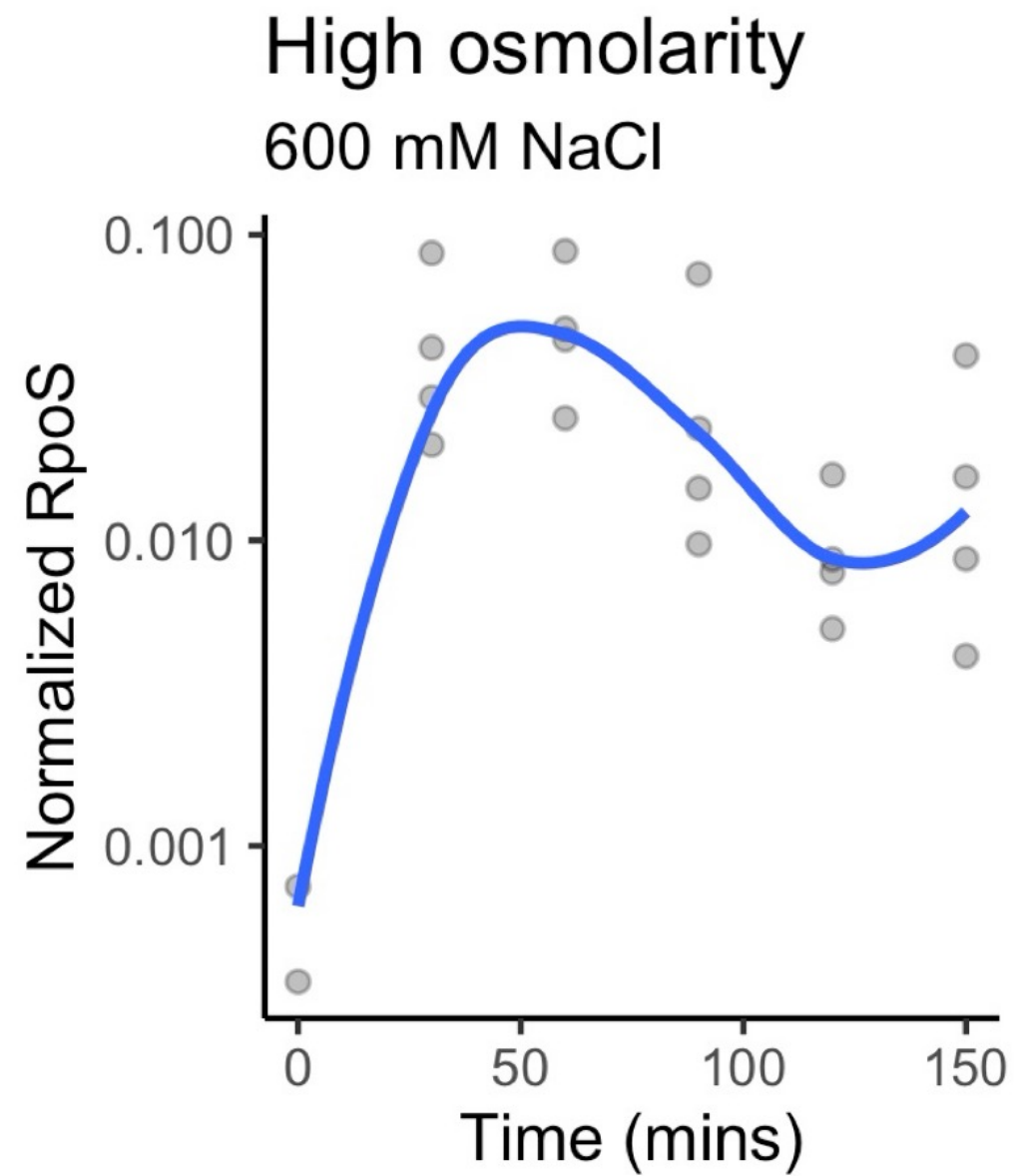
B





The stress of Stationary
Phase

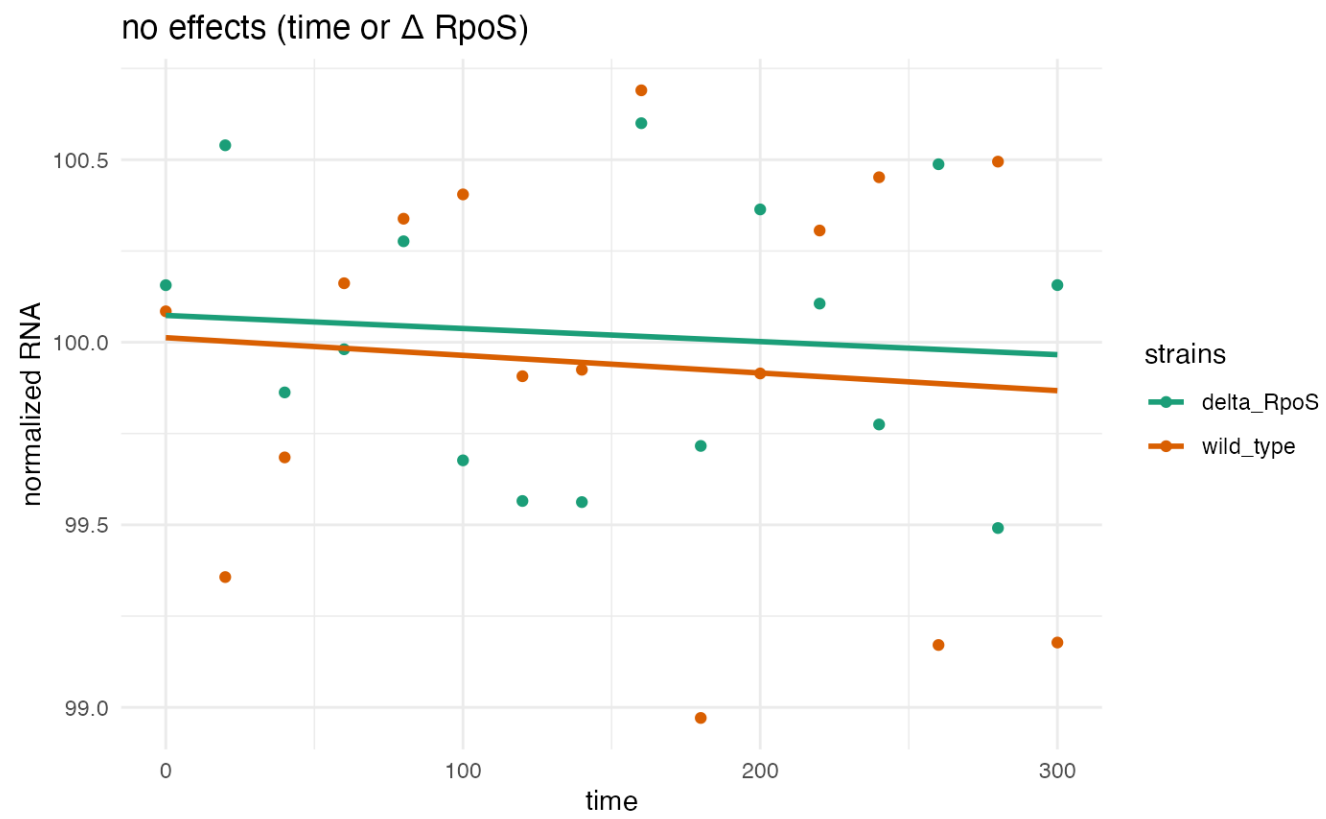
37°C LB media



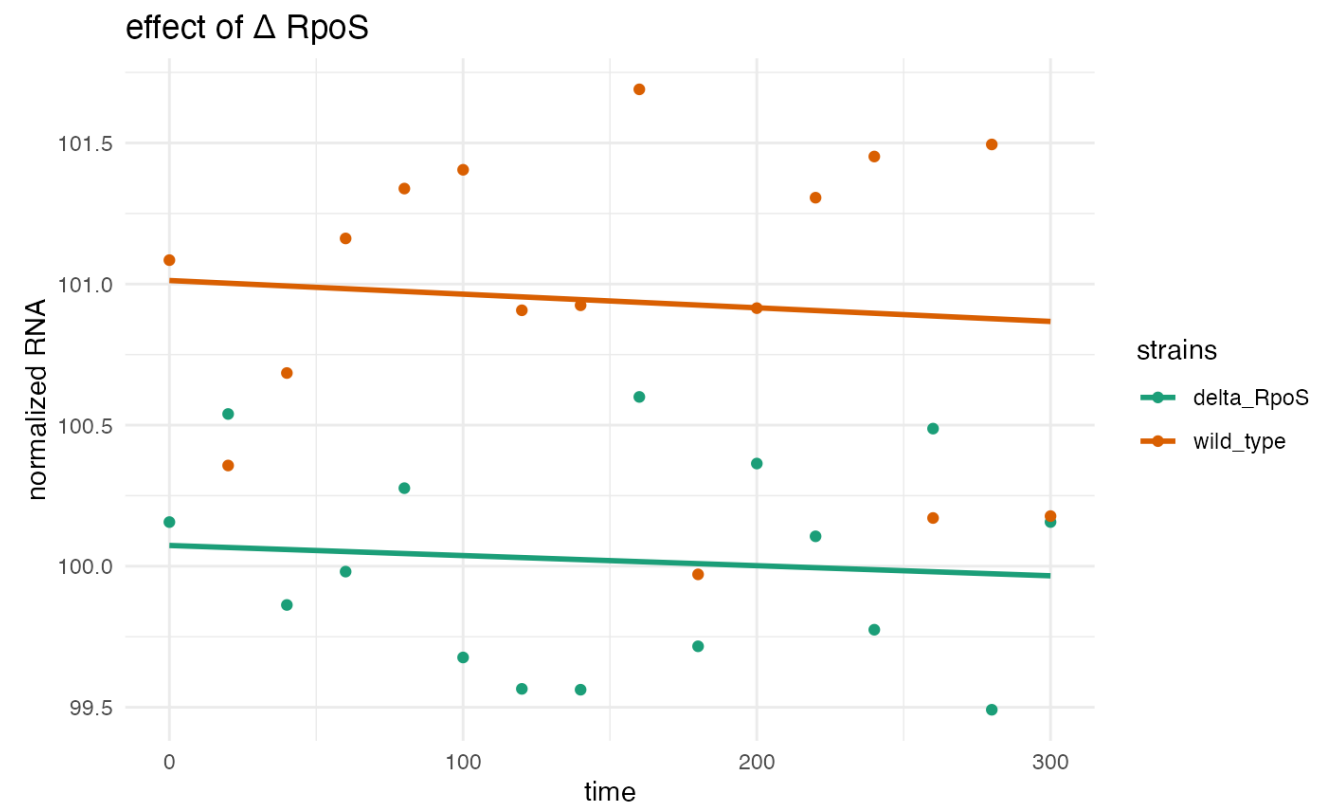
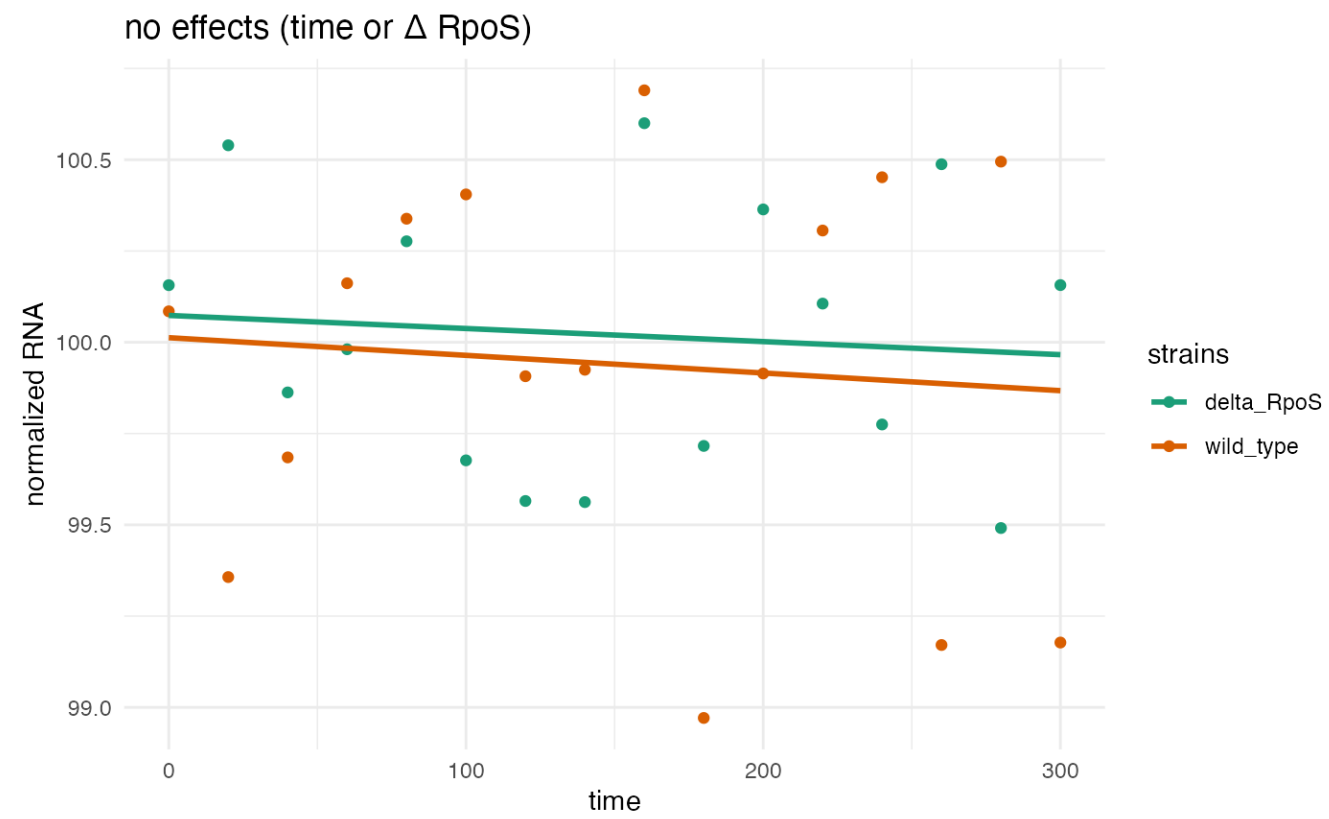
What about other
stresses?

- Grew wild-type and $\Delta rpoS$ *E. coli* in LB at 37°C, started taking samples at OD₆₀₀ = 0.3, took samples for 5 hours (300 min).
- Sampled total RNA at different time points.
- Performed RNA-seq on each sample.

Experimental design

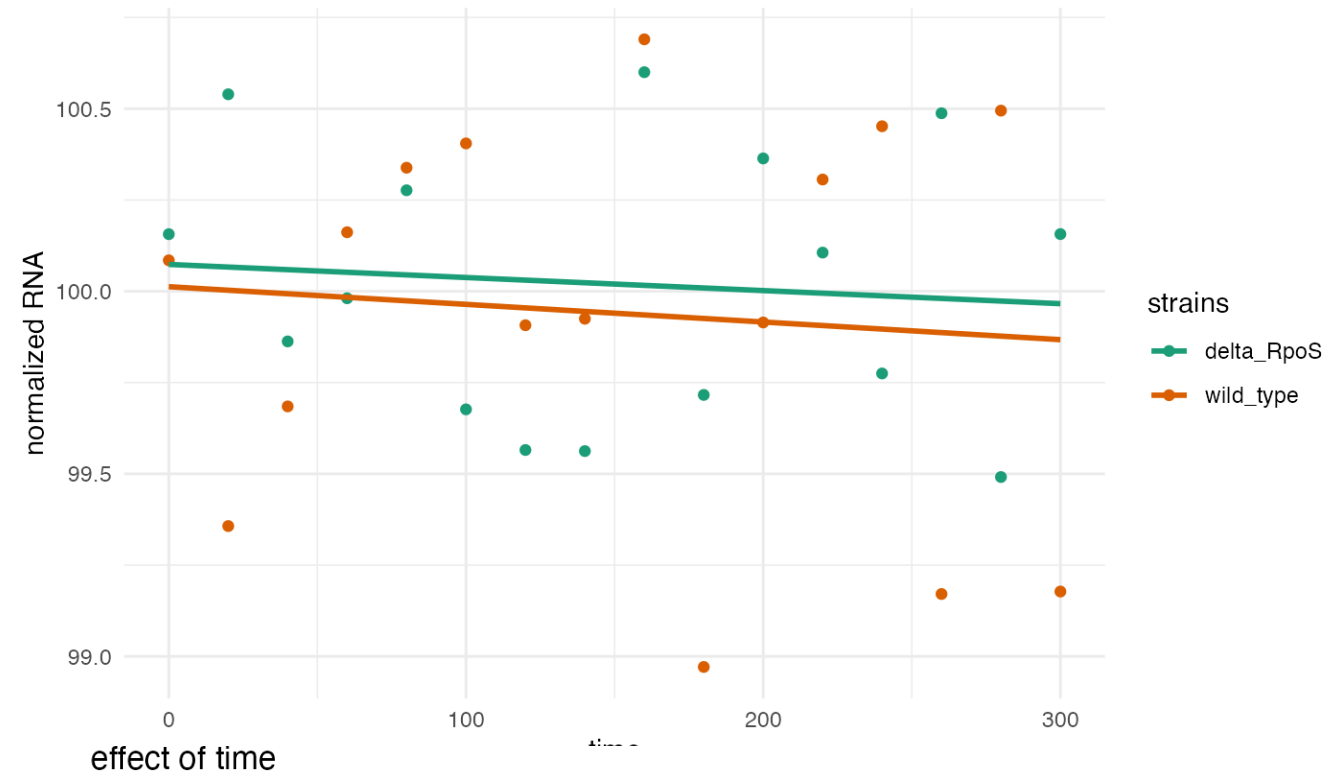


Differential Expression
due to Interaction

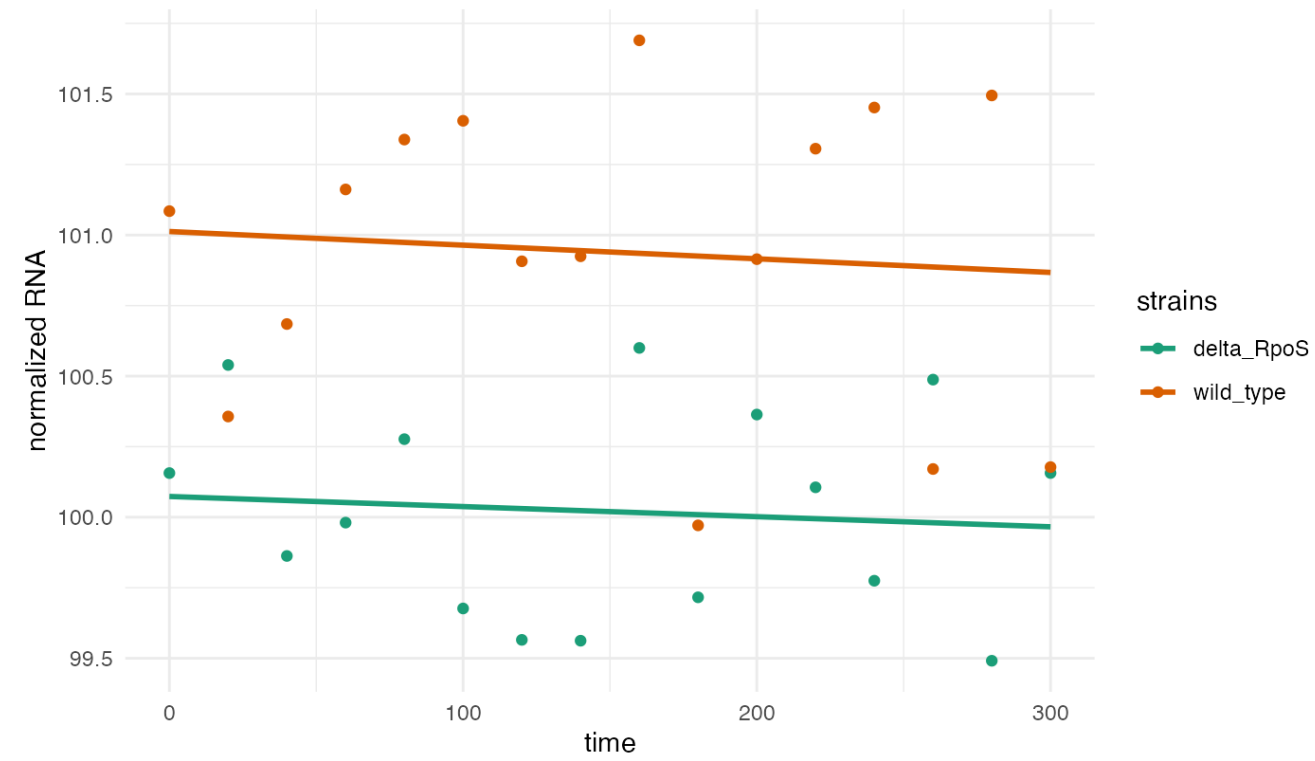


Differential Expression
due to Interaction

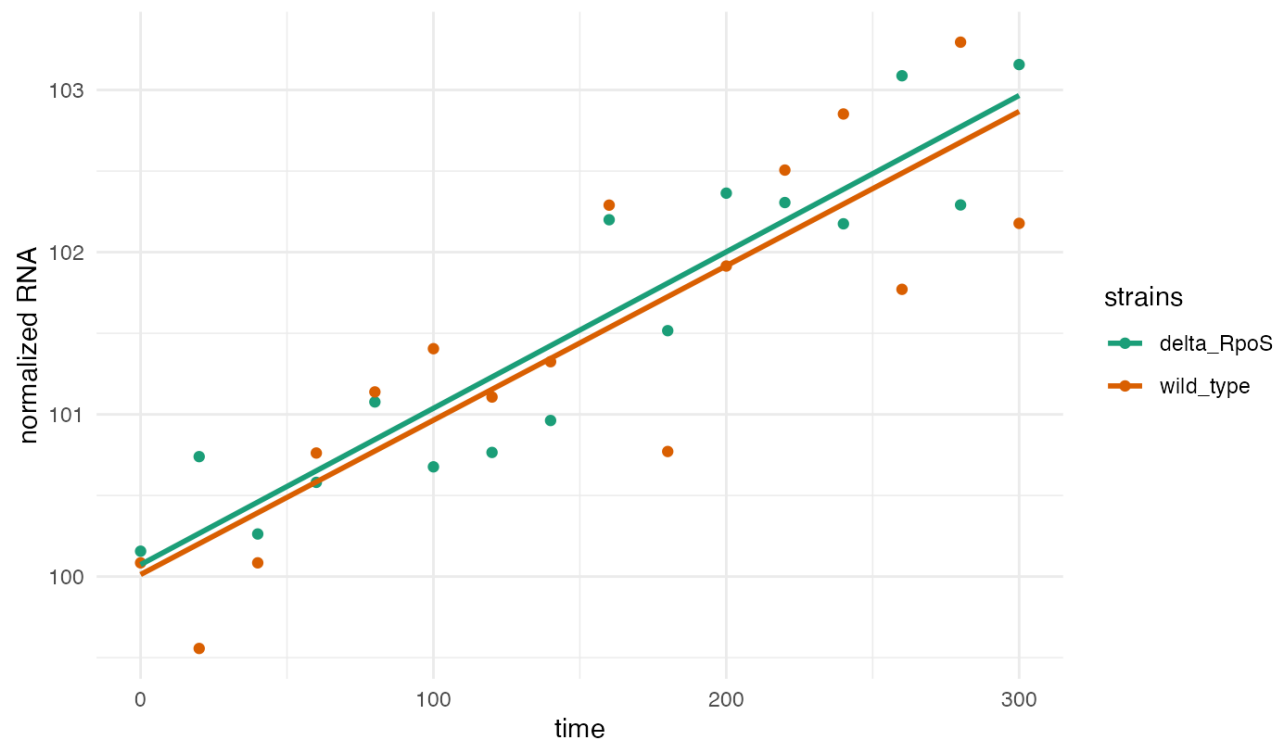
no effects (time or Δ RpoS)



effect of Δ RpoS

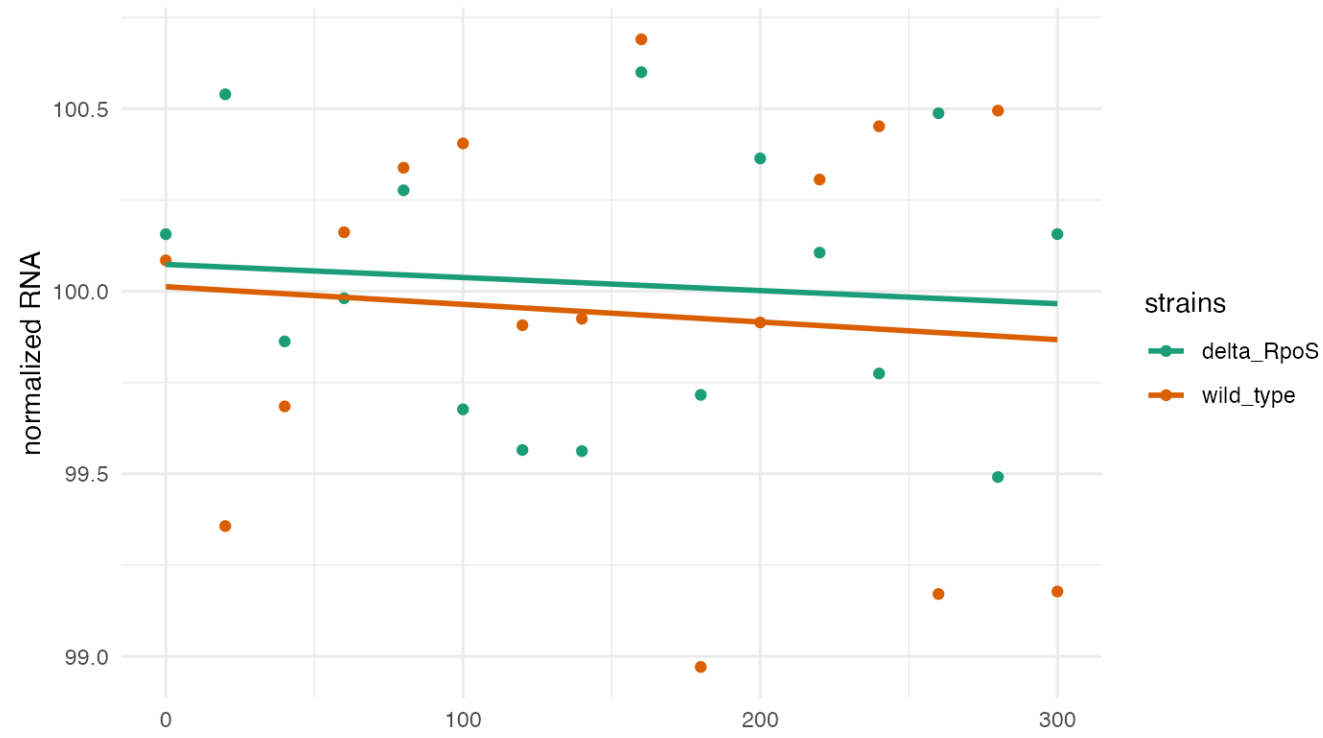


effect of time

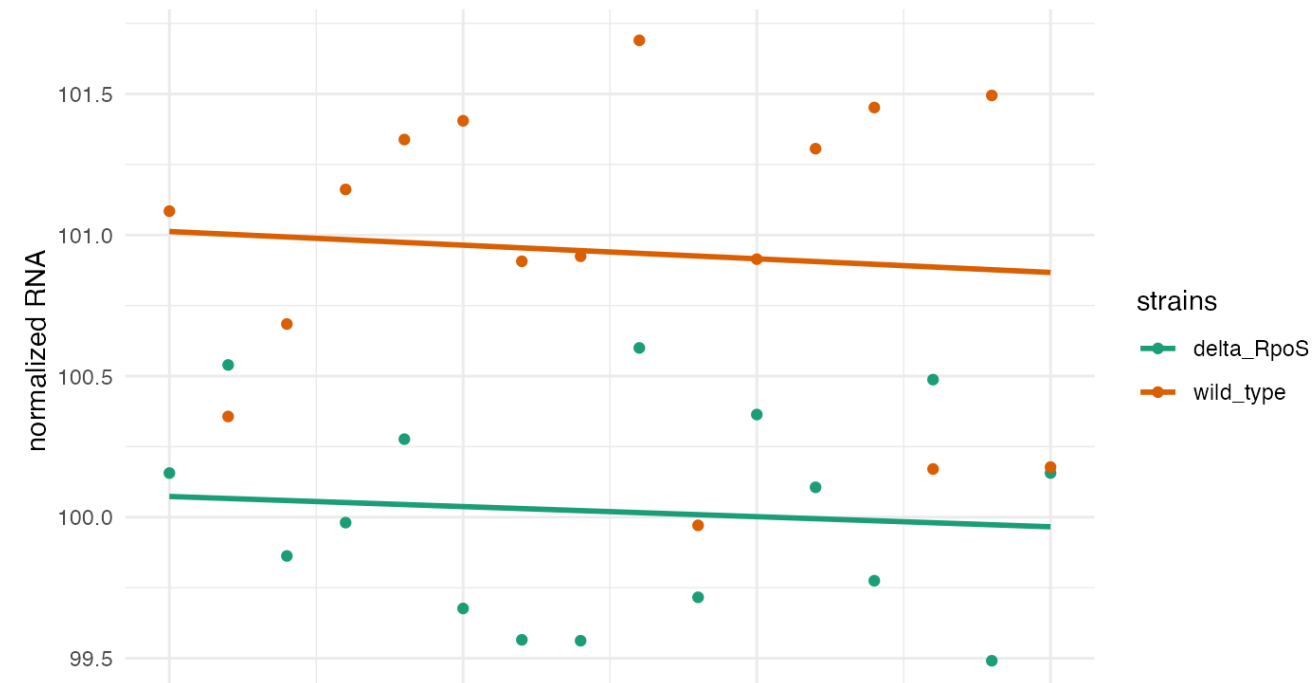


Differential Expression
due to Interaction

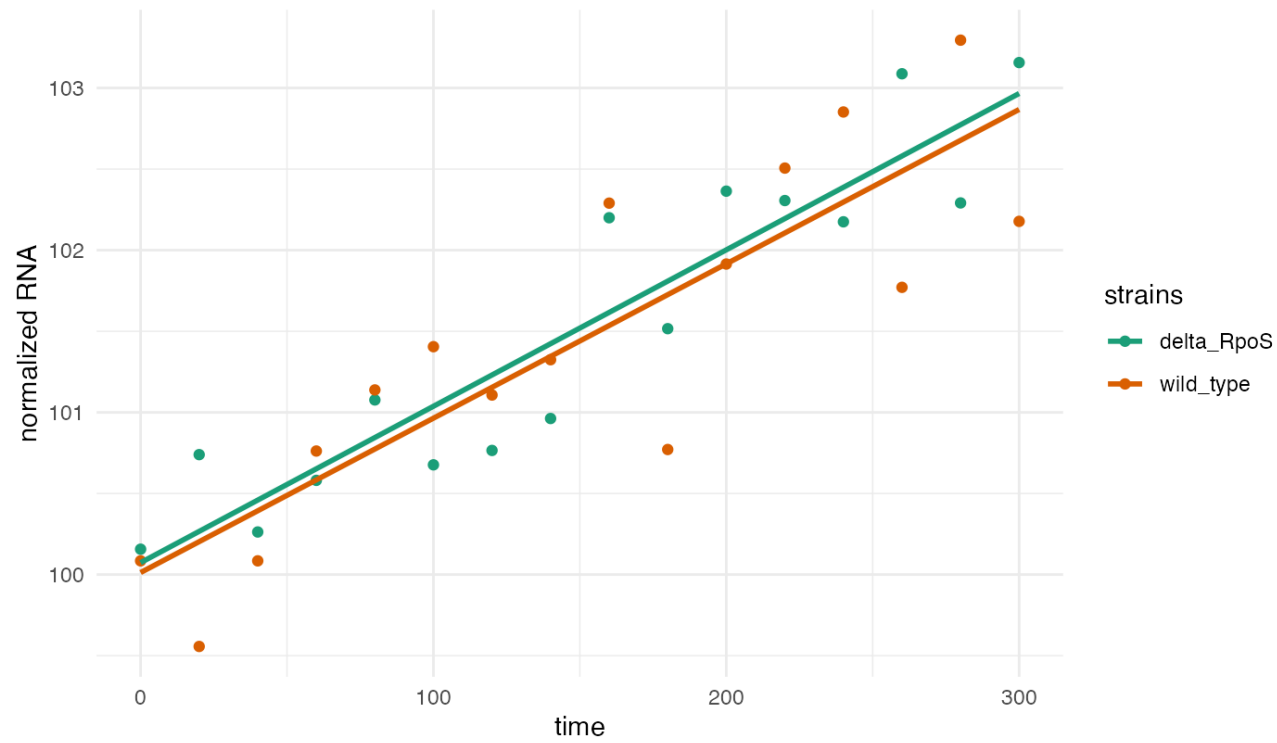
no effects (time or Δ RpoS)



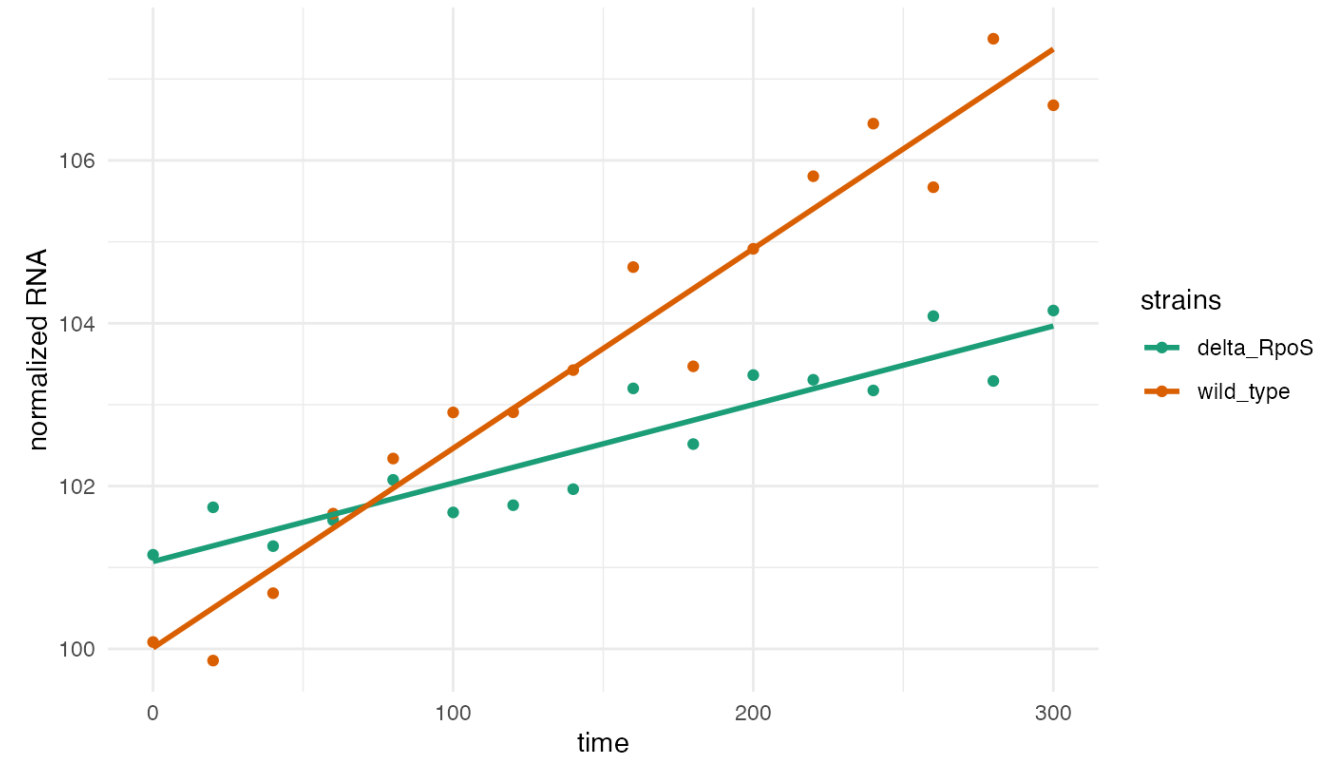
effect of Δ RpoS



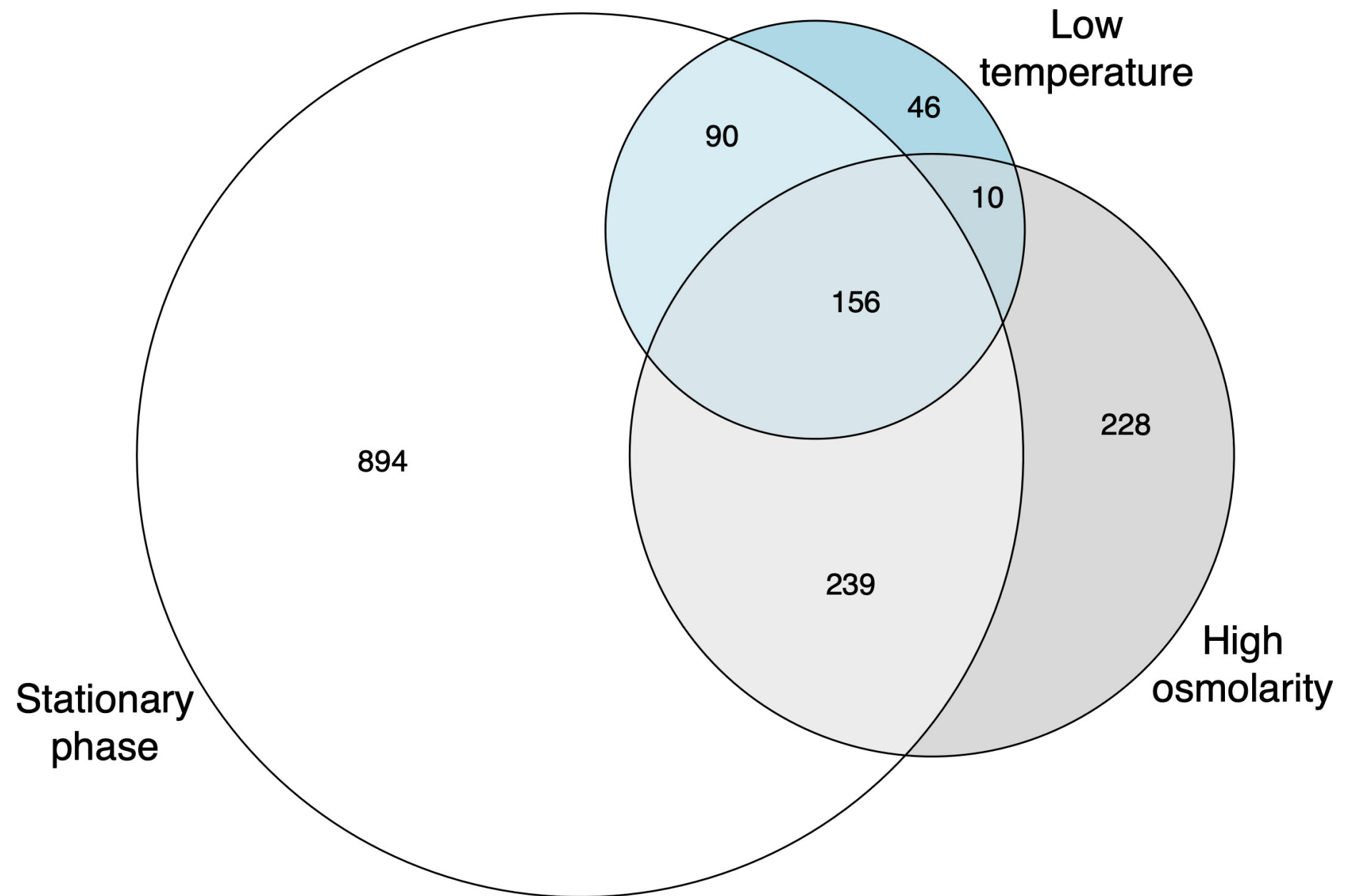
effect of time



Interaction: effect of time and Δ RpoS

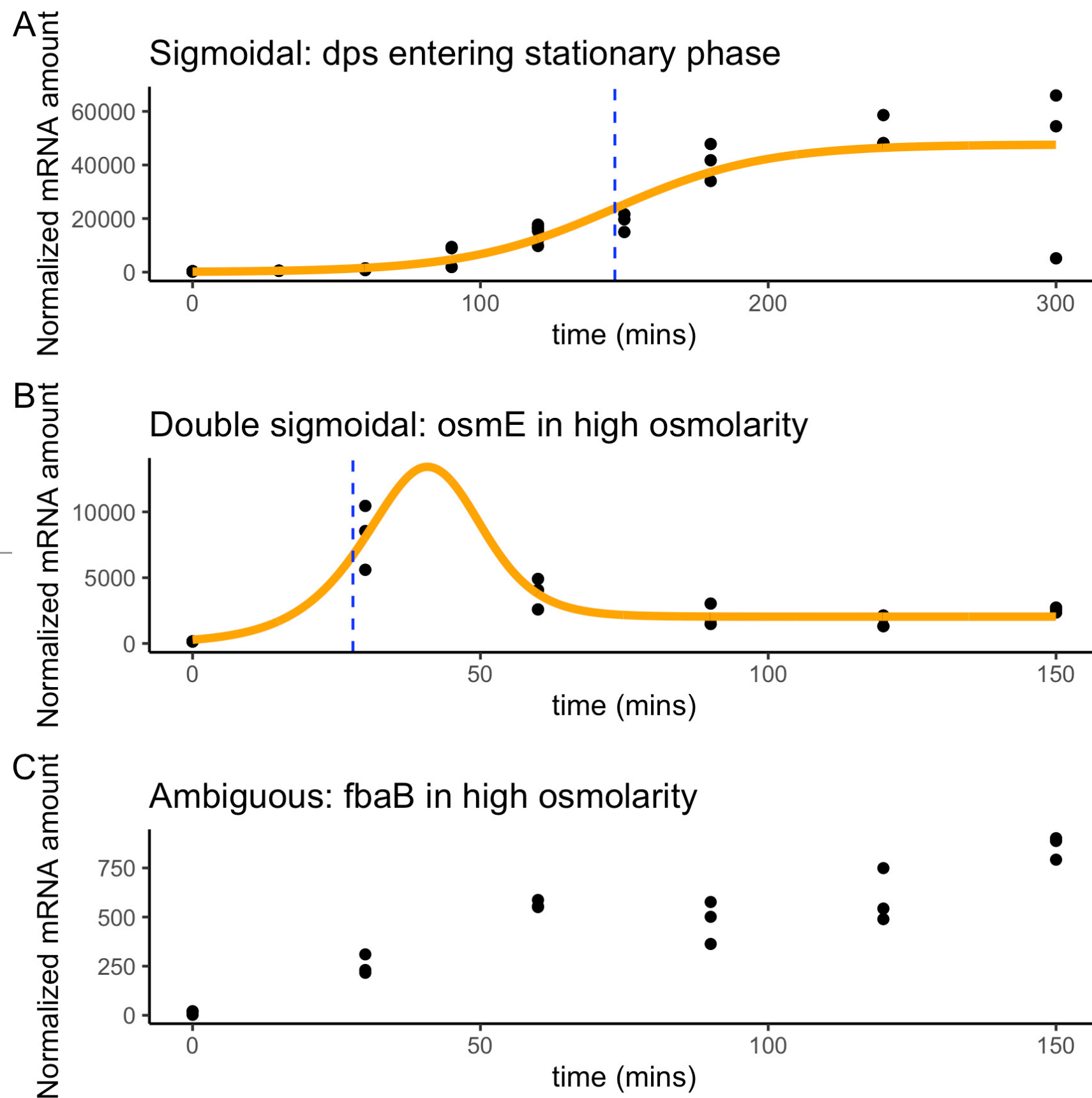


Differential Expression
due to Interaction

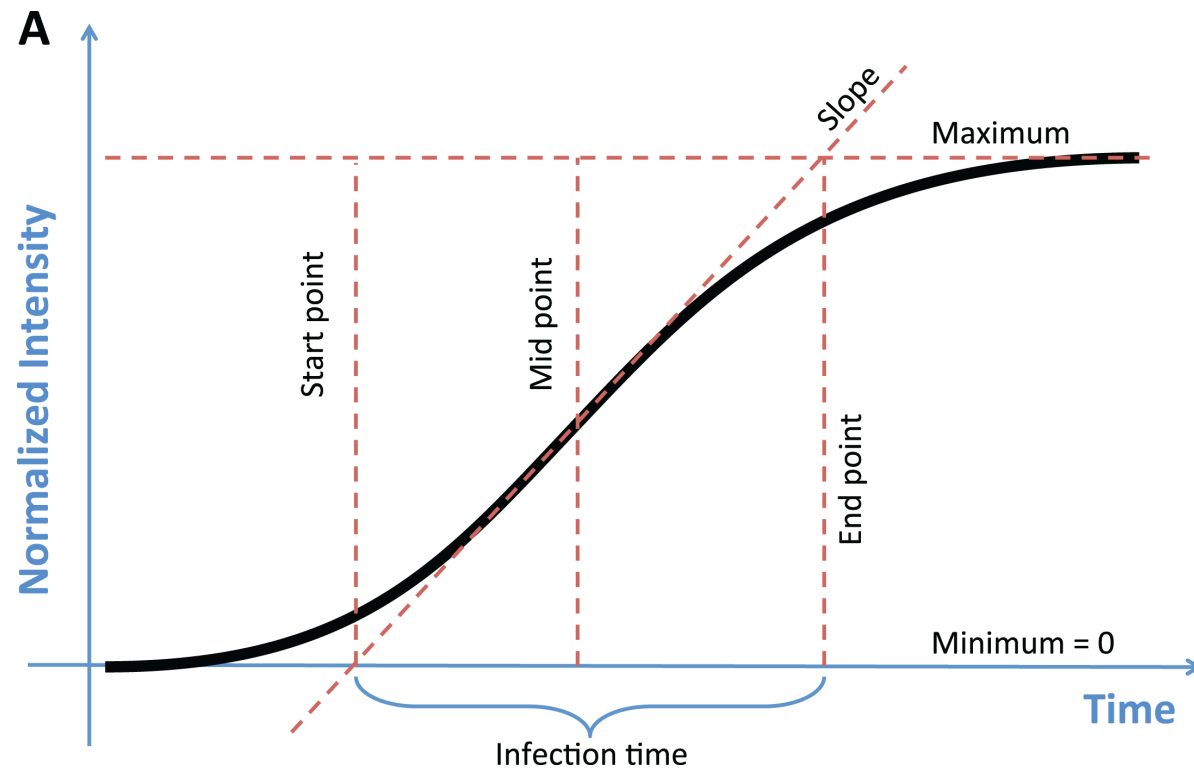


Differentially expressed
genes

sicegar



Sigmoid

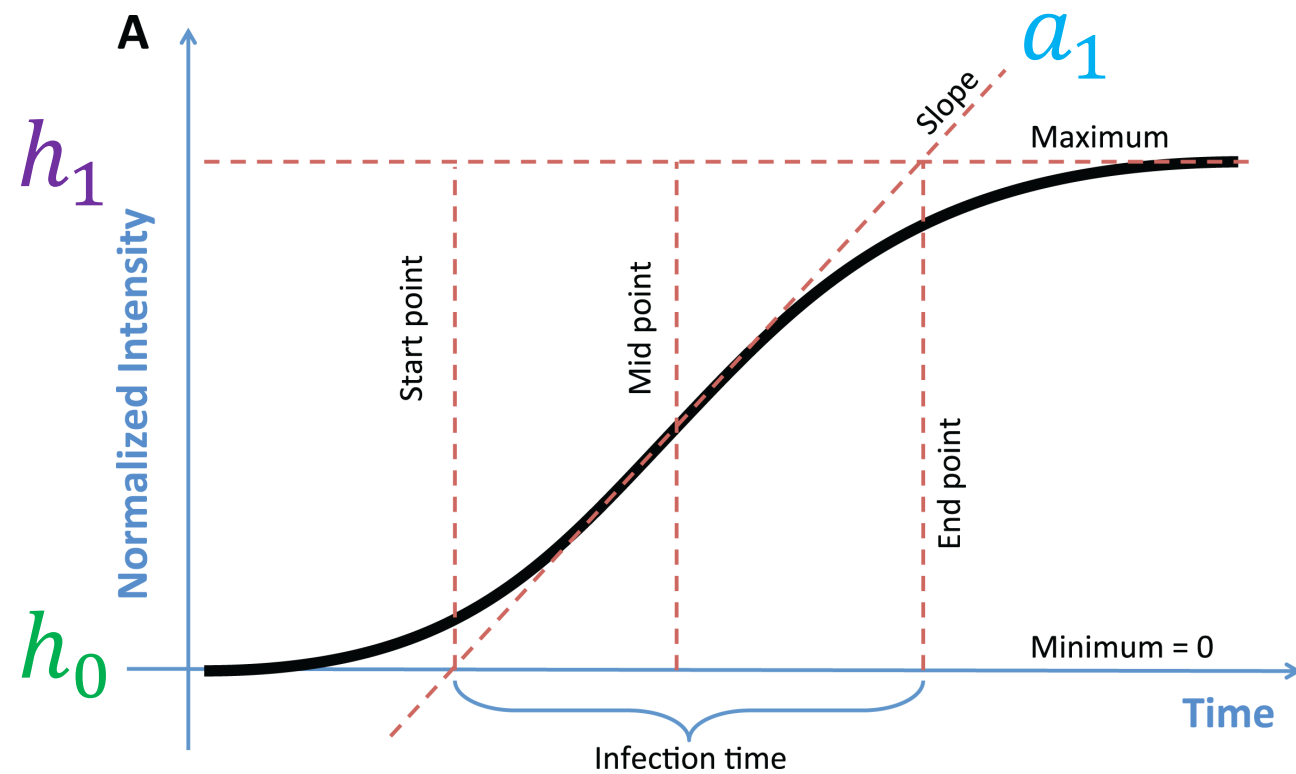


$$f(x) = h_0 + \frac{(h_1 - h_0)}{1 + e^{-a_1(x - t_1)}}$$

Sigmoid and Double Sigmoid Models

Image credit: doi: [10.7717/peerj.4251](https://doi.org/10.7717/peerj.4251)

Sigmoid

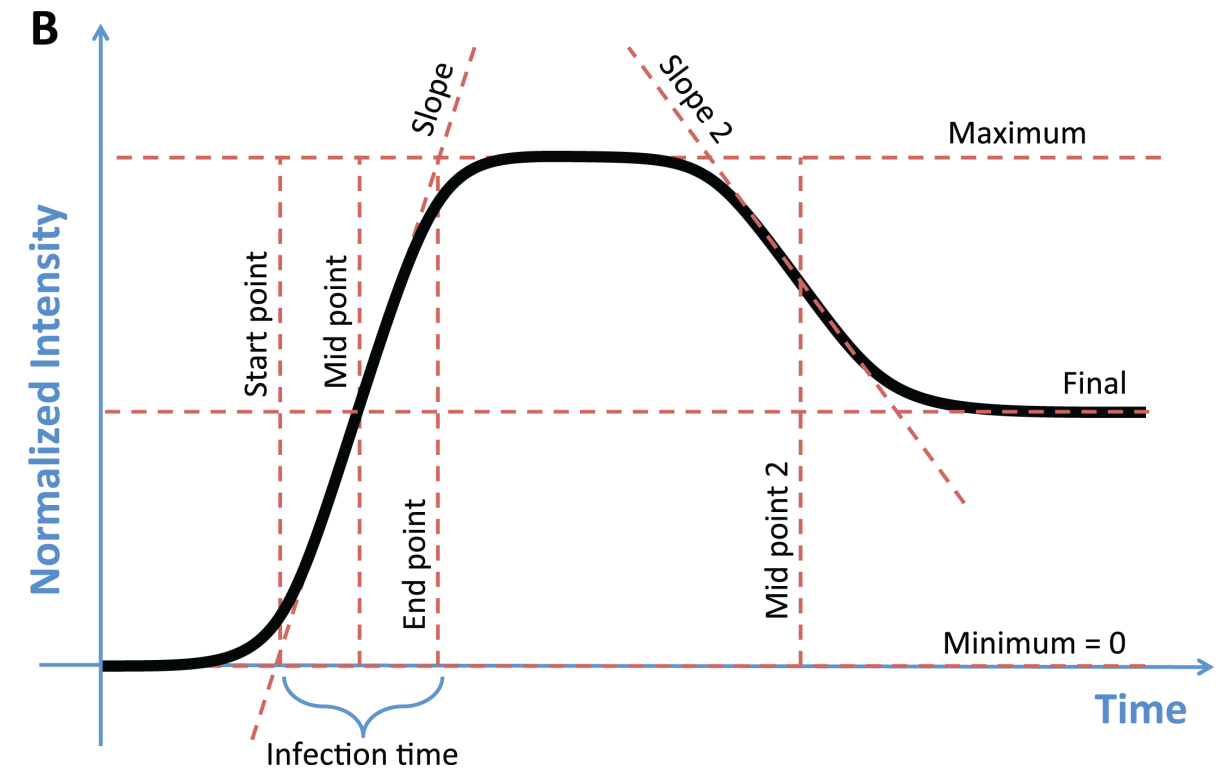


$$f(x) = h_0 + \frac{(h_1 - h_0)}{1 + e^{-a_1(x - t_1)}}$$

Sigmoid and Double Sigmoid Models

Image credit: doi: [10.7717/peerj.4251](https://doi.org/10.7717/peerj.4251)

Double Sigmoid



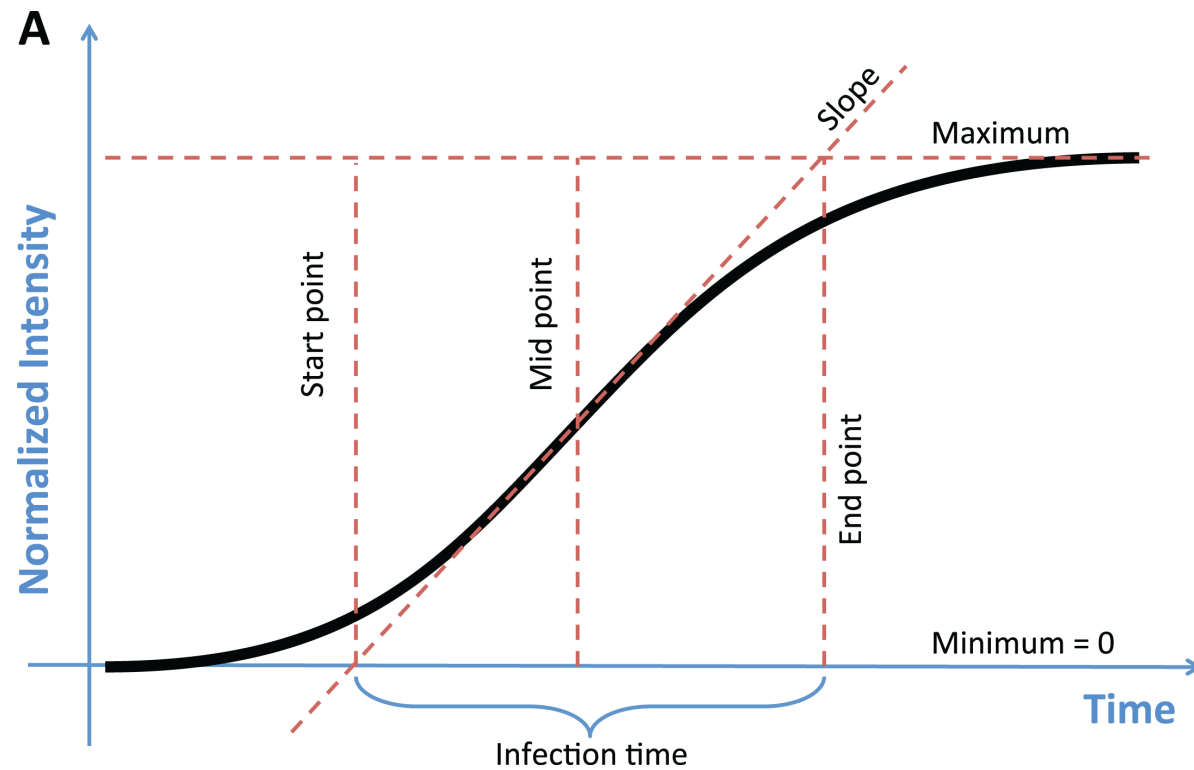
t_1 ???

$$f(x) = \frac{1}{h_1} \left(h_0 + \frac{(h_1 - h_0)}{1 + e^{-a_1(x - t_1)}} \right) * \left(h_2 + \frac{(h_1 - h_2)}{1 + e^{a_2(x - t_2)}} \right)$$

Sigmoid and Double Sigmoid Models

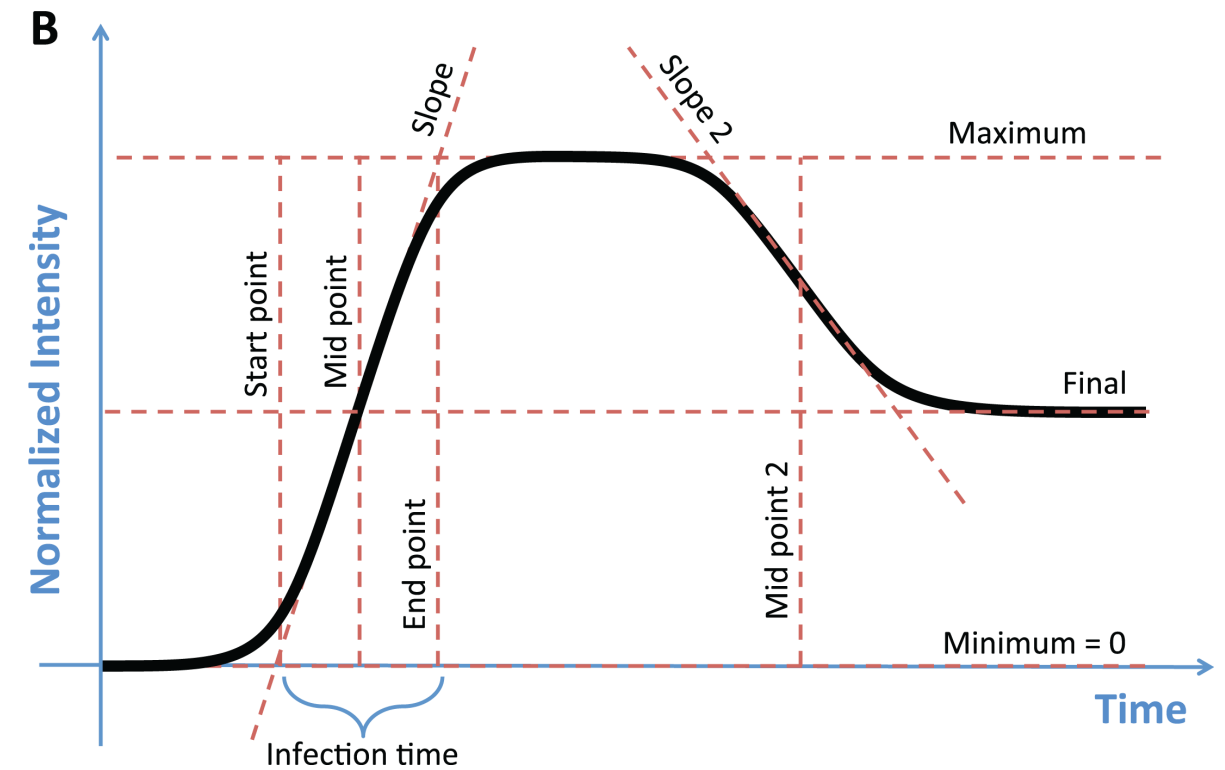
Image credit: doi: [10.7717/peerj.4251](https://doi.org/10.7717/peerj.4251)

Sigmoid



$$f(x) = h_0 + \frac{(h_1 - h_0)}{1 + e^{-a_1(x - t_1)}}$$

Double Sigmoid

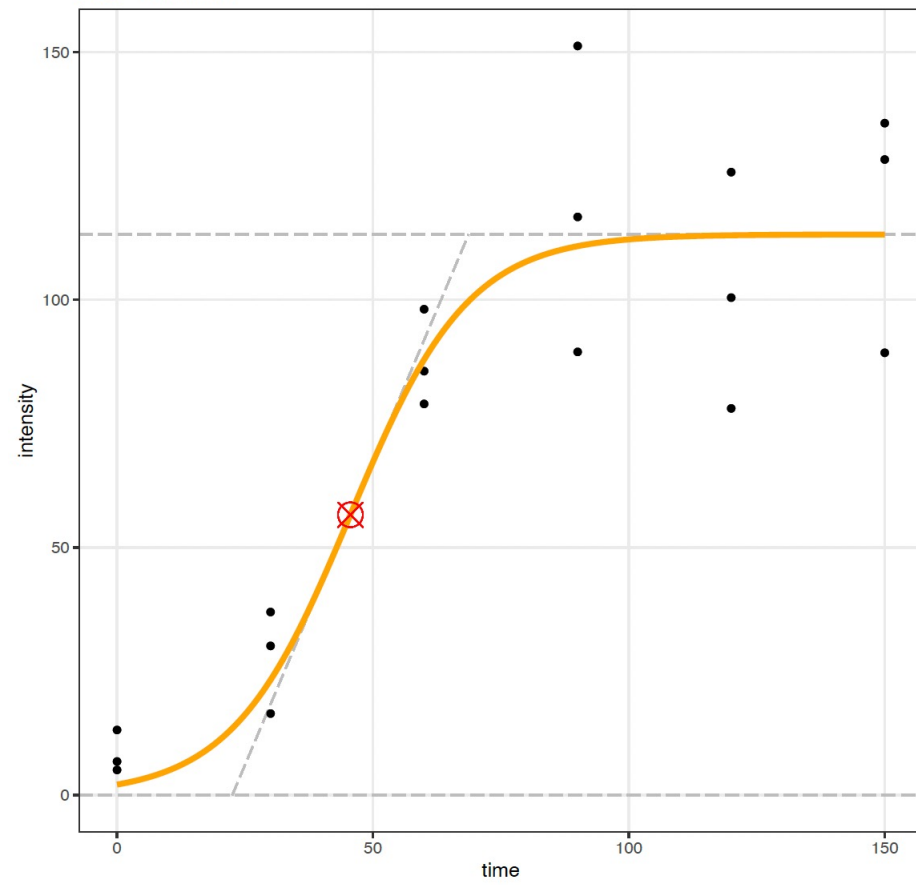


$$f(x) = \frac{1}{h_1} \left(h_0 + \frac{(h_1 - h_0)}{1 + e^{-a_1(x - t_1)}} \right) * \left(h_2 + \frac{(h_1 - h_2)}{1 + e^{a_2(x - t_2)}} \right)$$

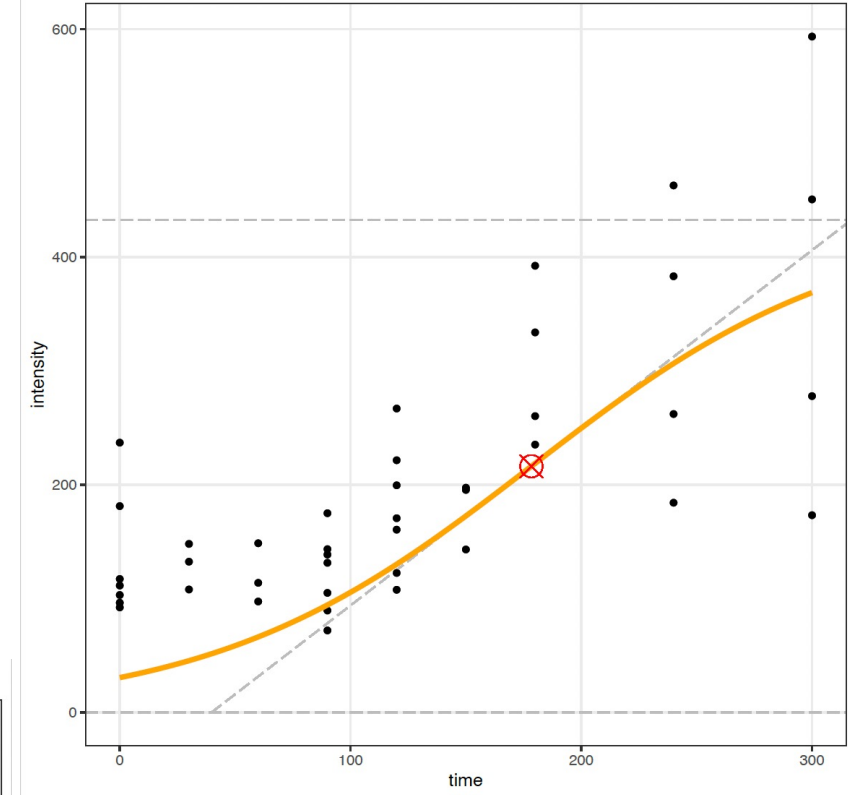
Sigmoid and Double Sigmoid Models

Image credit: doi: [10.7717/peerj.4251](https://doi.org/10.7717/peerj.4251)

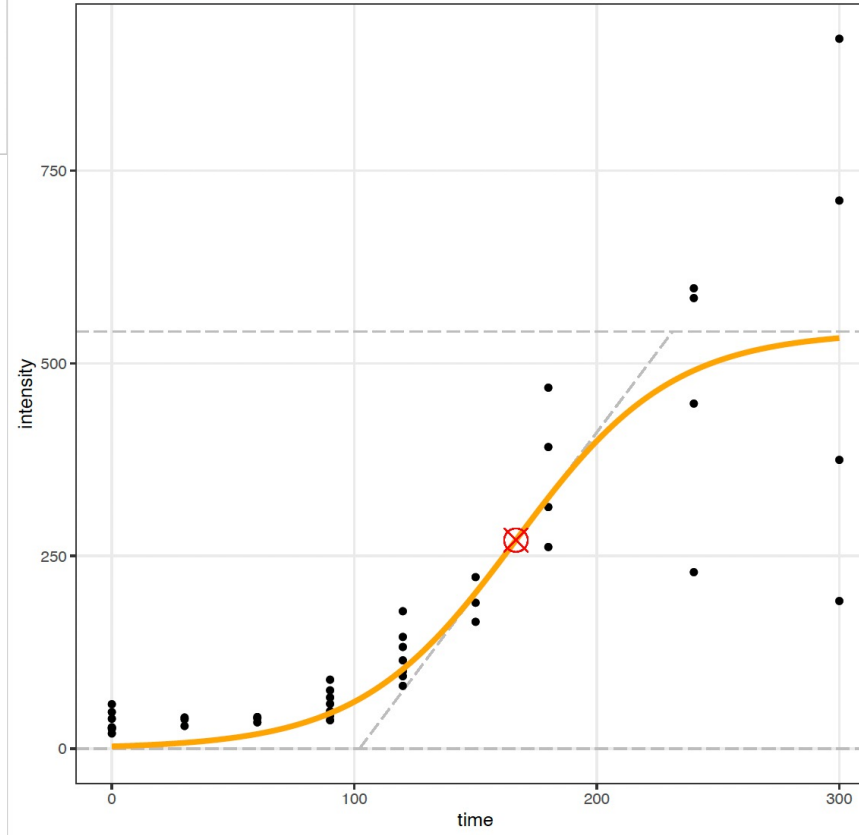
yegP Sigmoid:



ivy Sigmoid:

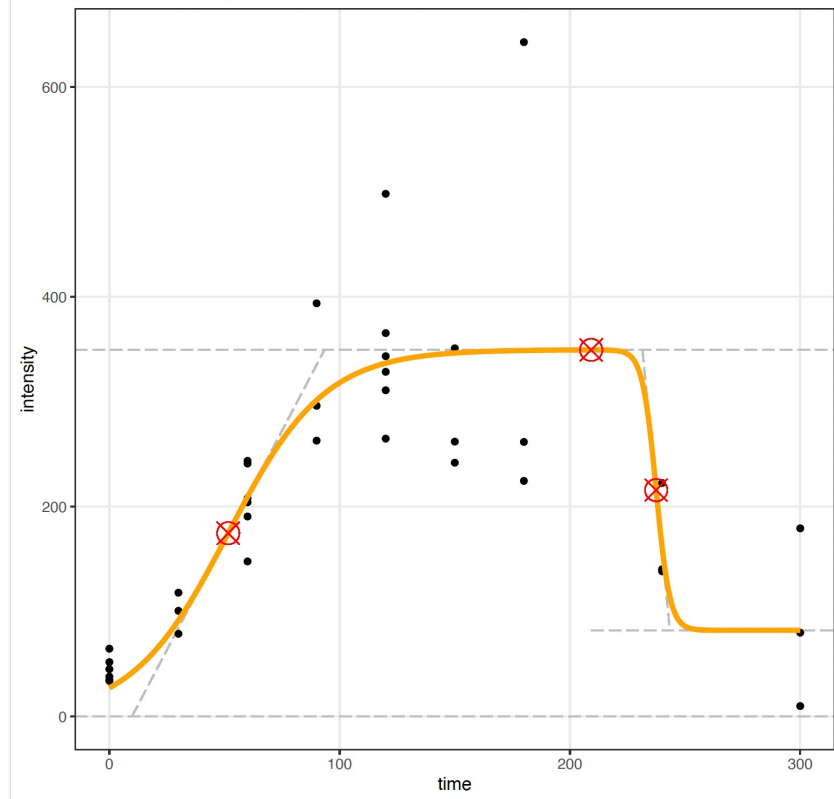


osmC Sigmoid:

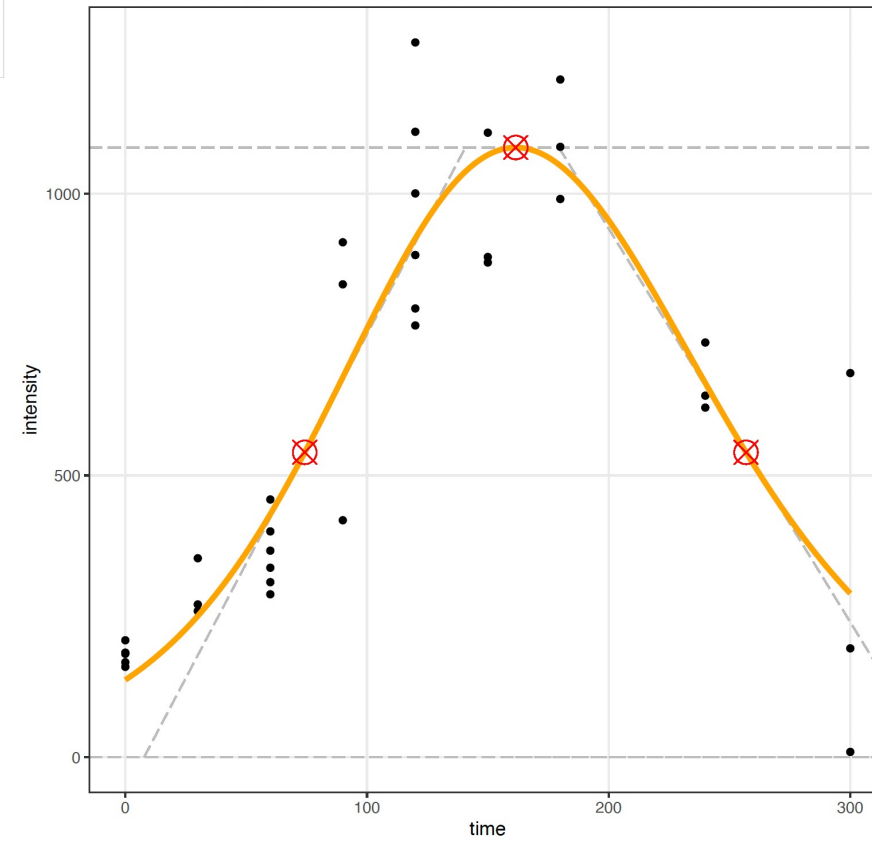


Sigmoidal genes

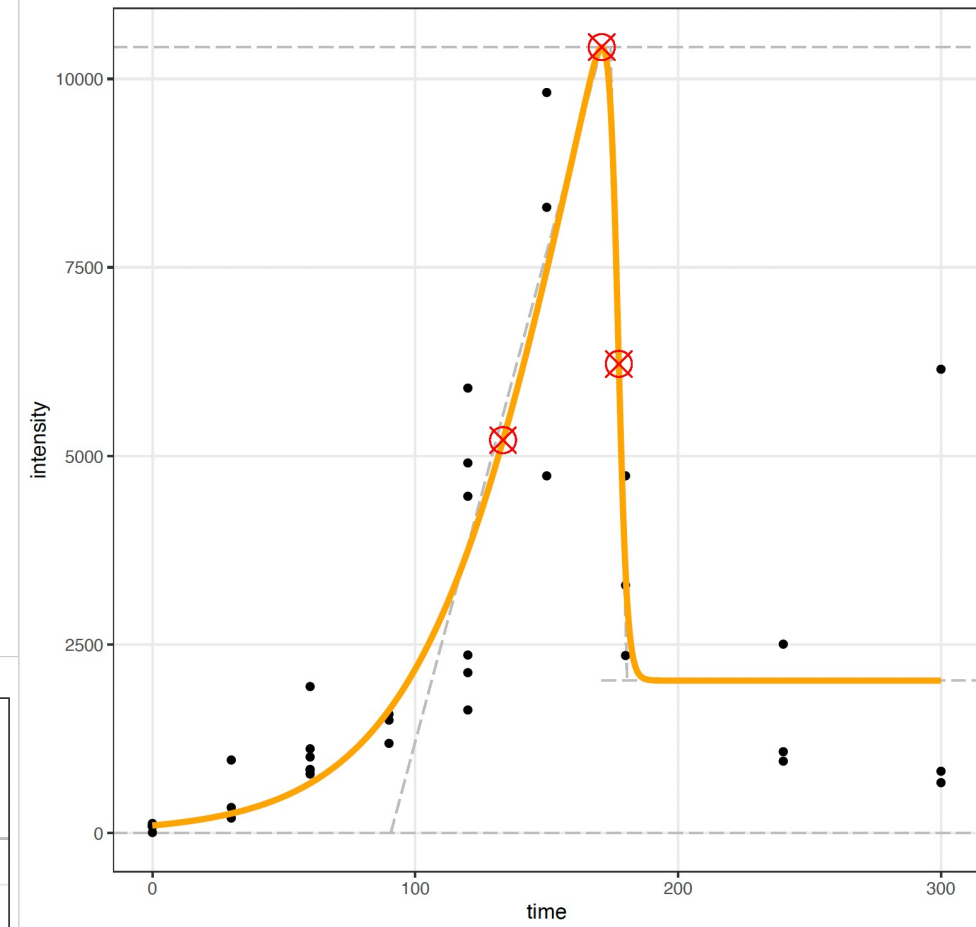
phoA Double Sigmoid:



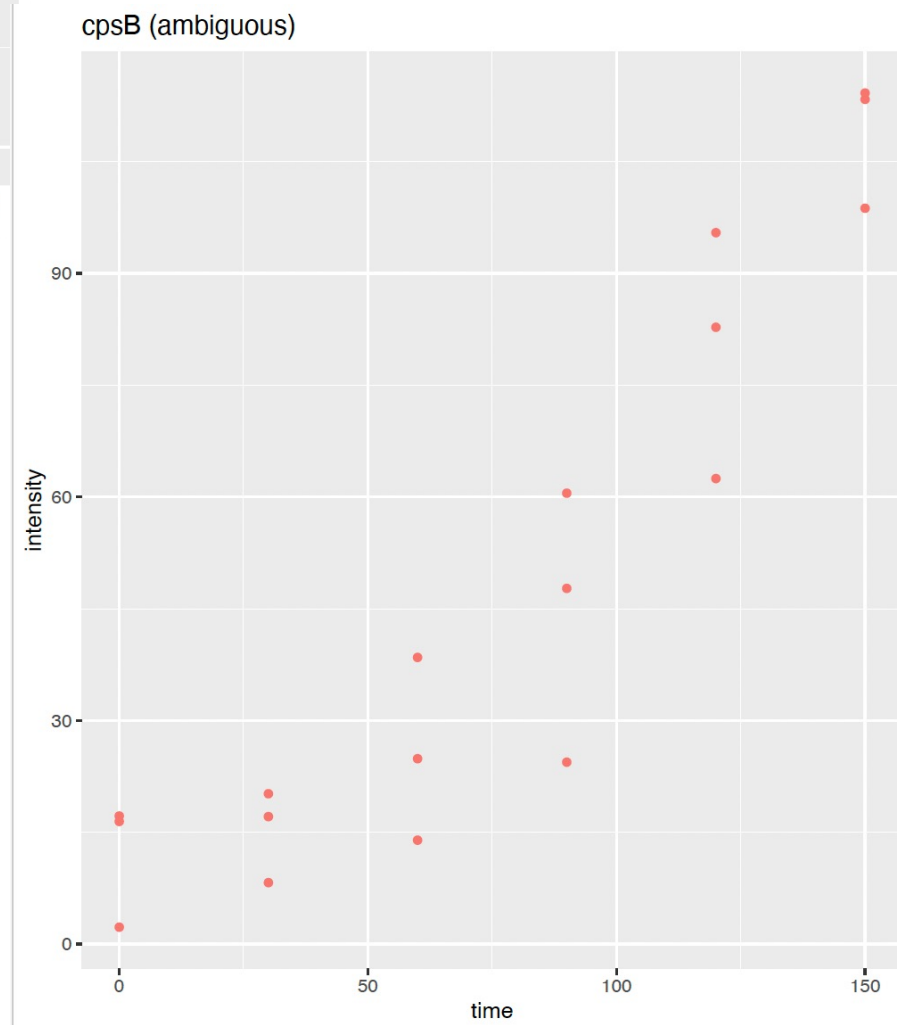
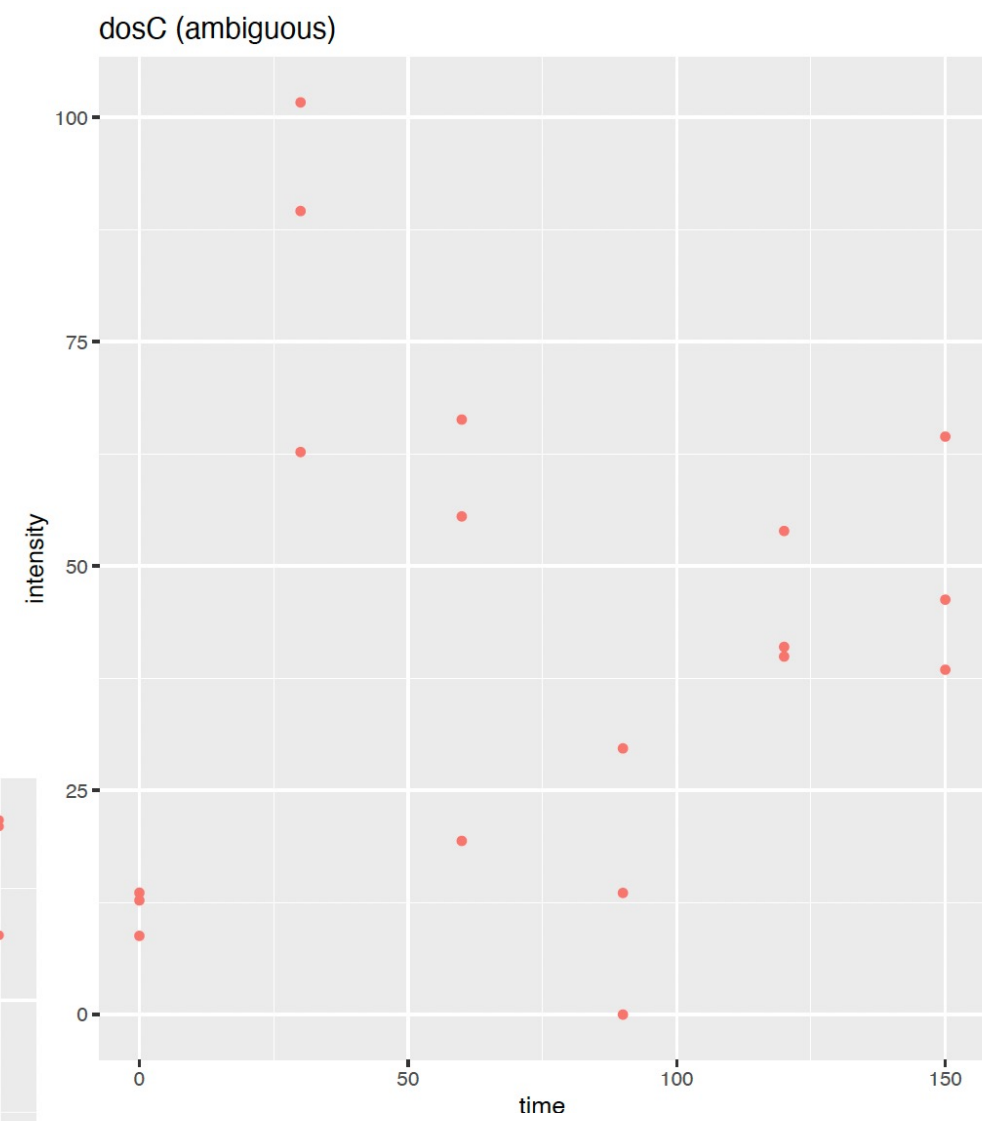
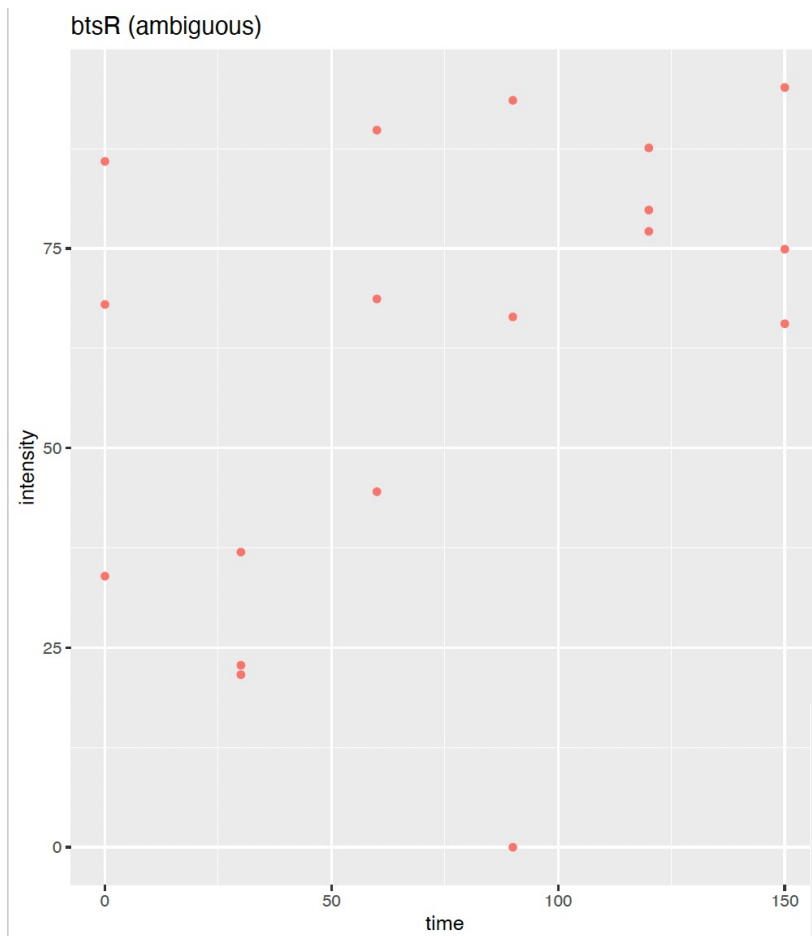
dacC Double Sigmoid:



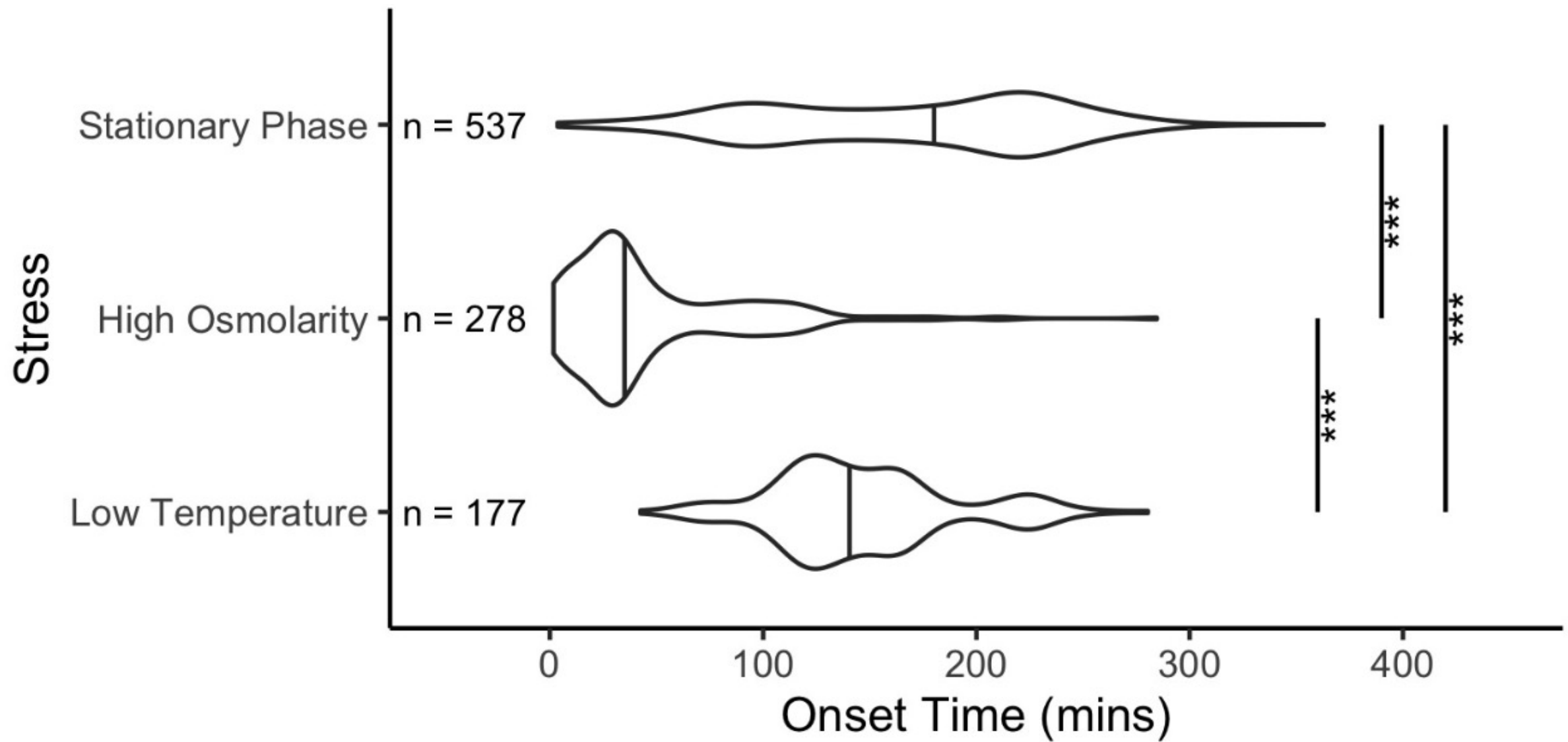
dctA Double Sigmoid:



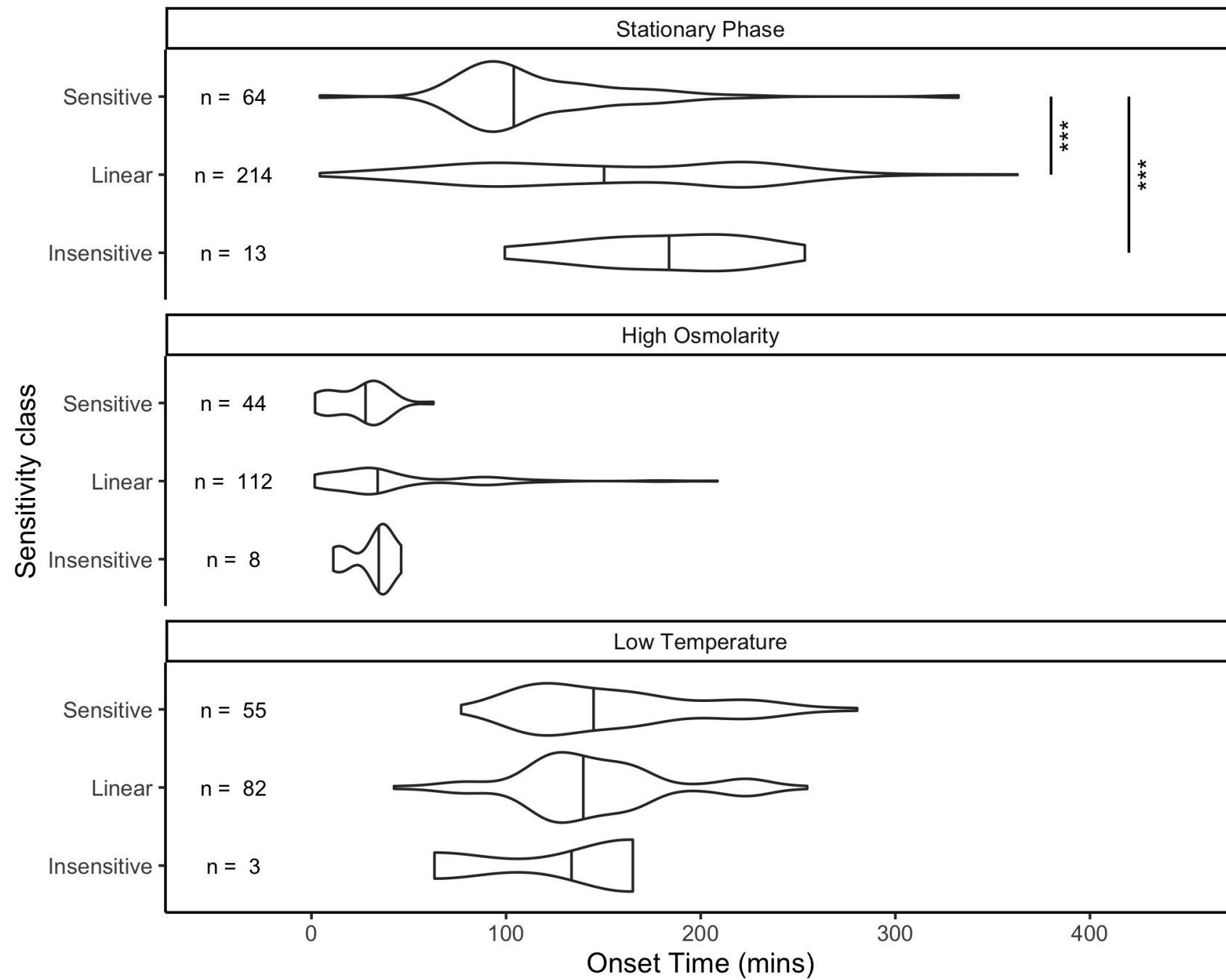
Double sigmoidal genes



Ambiguous genes



Onset time differs across stress

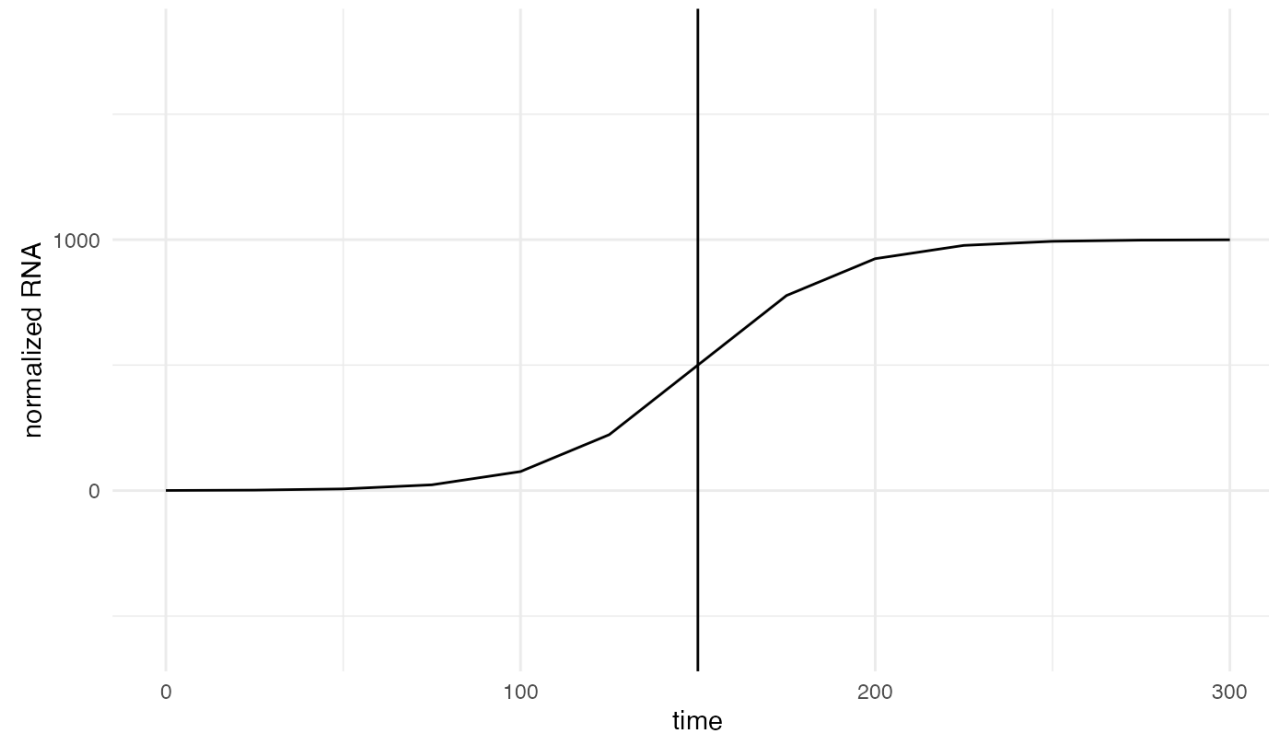


Stationary Phase sensitivity classes do not hold for other stresses

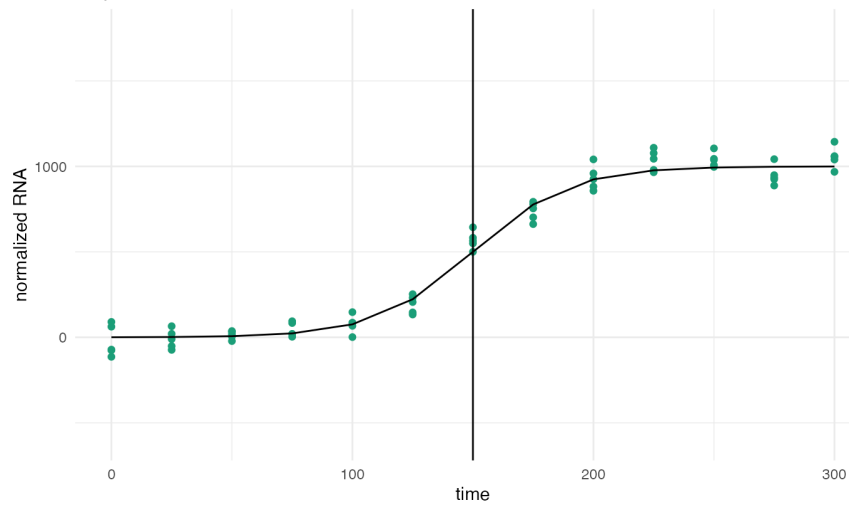
Where does the signal turn on?

1. We want to know when the signal “turns on” (we use the point of inflection).
2. Need to fit a model!
Sigmoid $\rightarrow t_1$
Double Sigmoid $\rightarrow \text{not } t_1$
3. Simulations use functions, what is the true point of inflection?
4. For Sigmoid, does the model fitting algorithm give us back the true point of inflection?
5. For Double Sigmoid, does the model fitting algorithm give us back the true point of inflection? Does it give t_1 ?

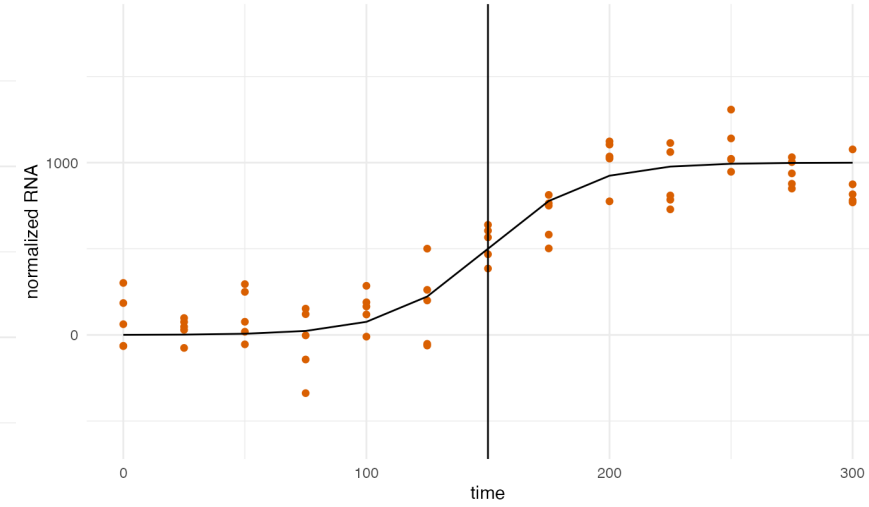
original model



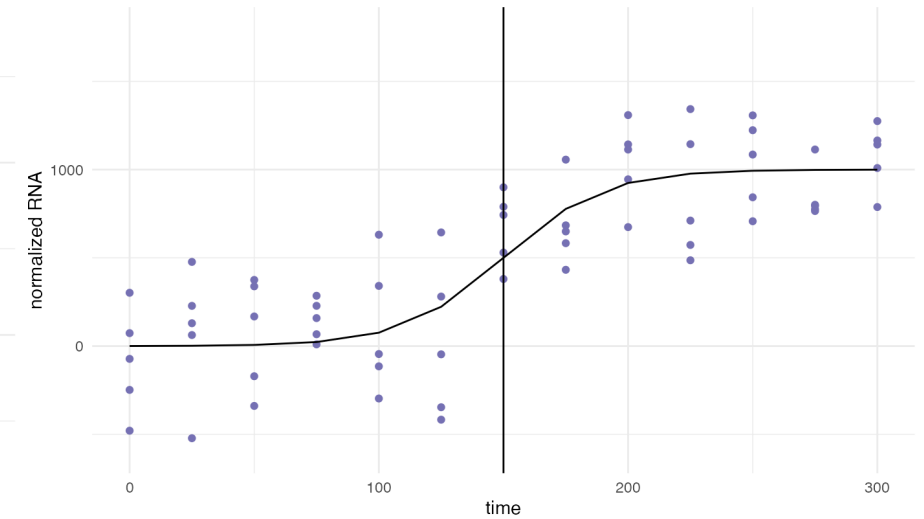
dispersion = 60



dispersion = 180



dispersion = 300

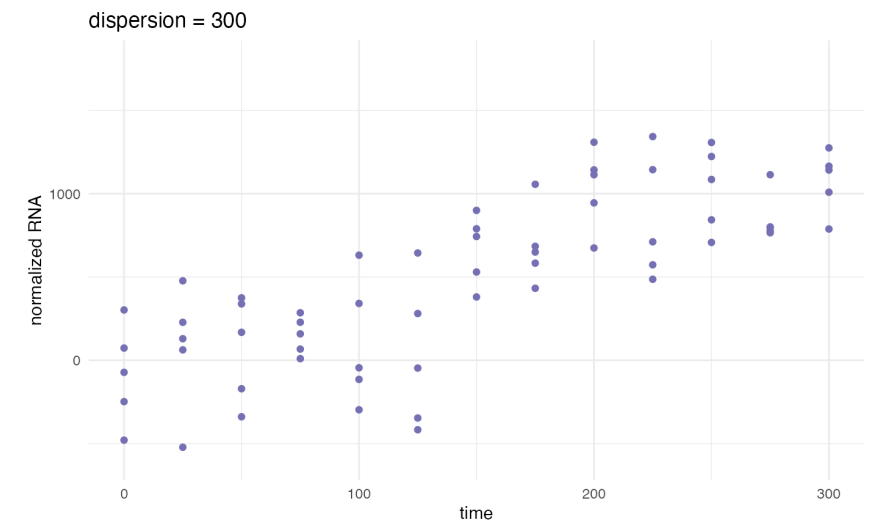
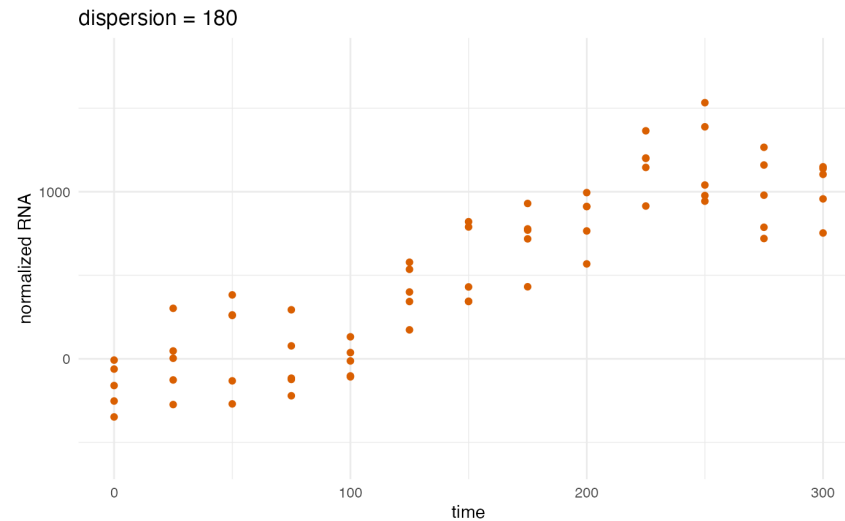
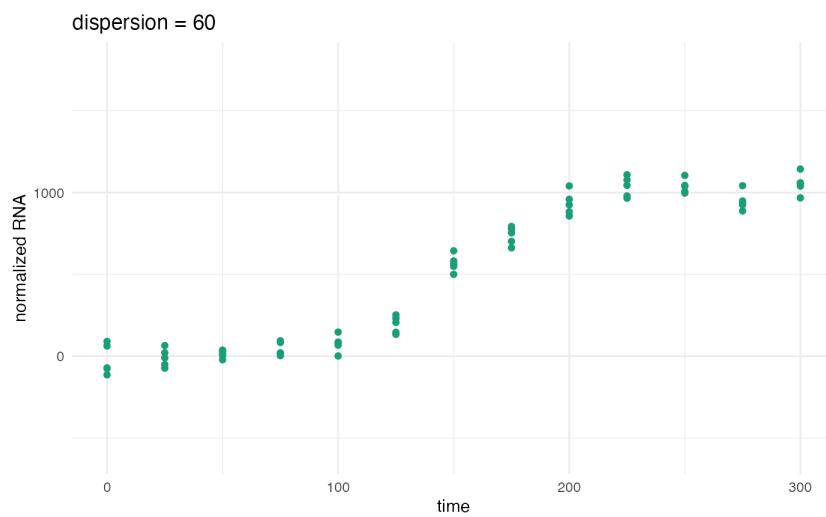
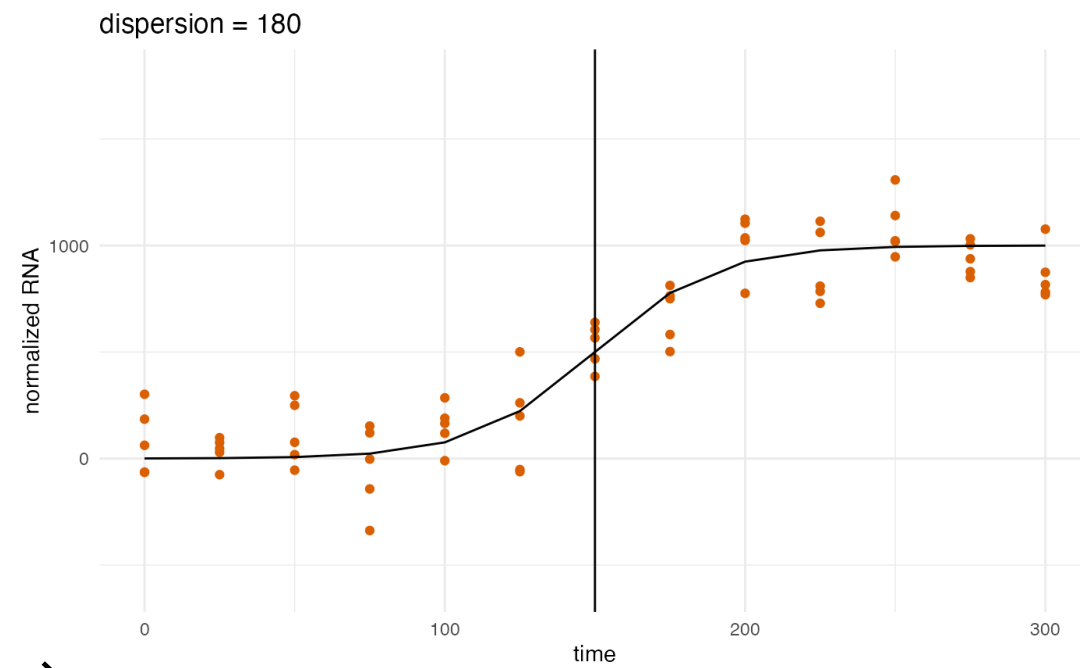


sigmoid

$$f(x) = h_0 + \frac{(h_1 - h_0)}{1 + e^{-a_1(x - t_1)}}$$

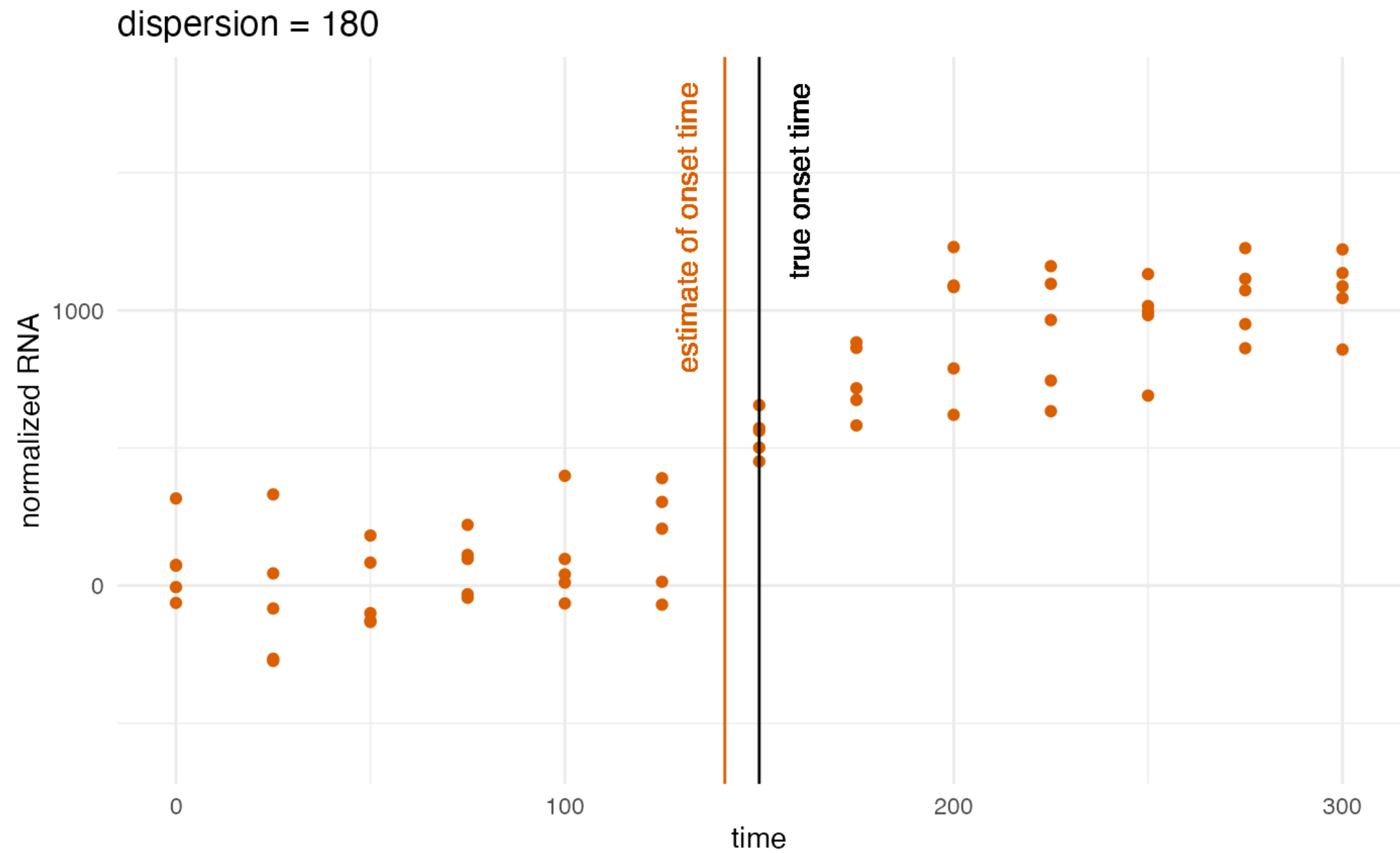
data generation →

data that the computer uses



Changing dispersion

$$f(x) = h_0 + \frac{(h_1 - h_0)}{1 + e^{-a_1(x - t_1)}}$$



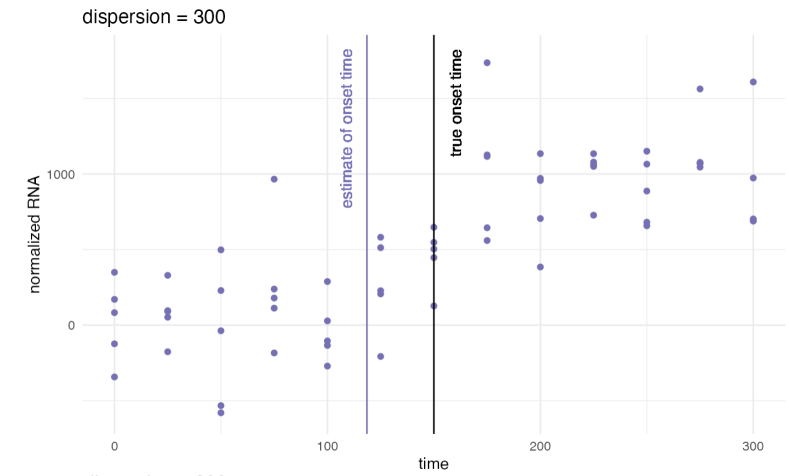
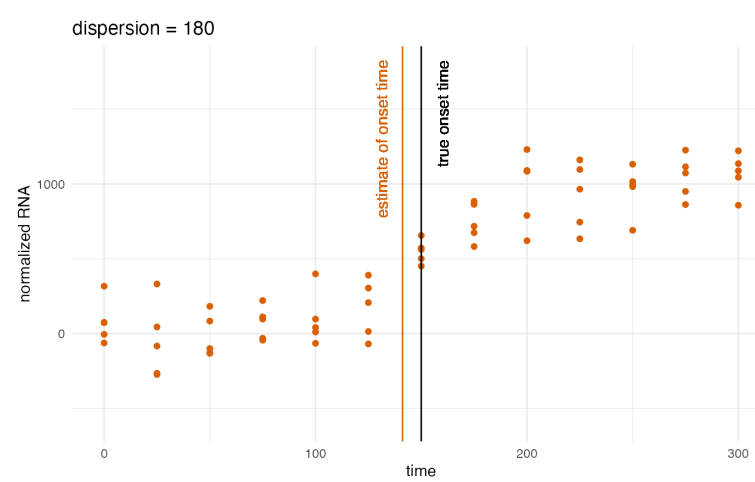
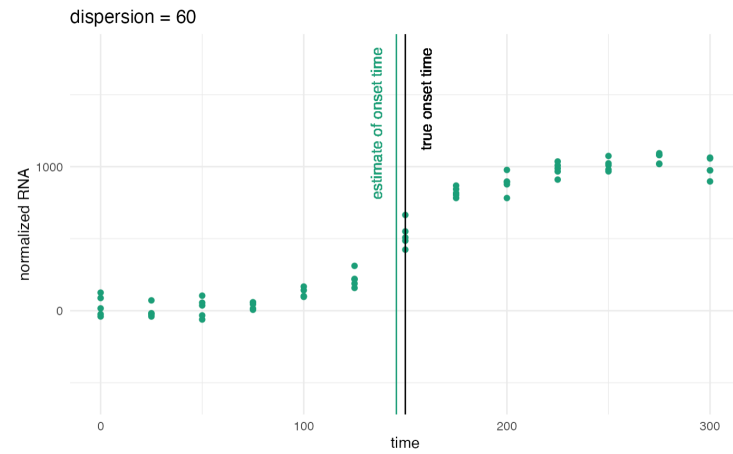
Estimate of onset time

disp = 60

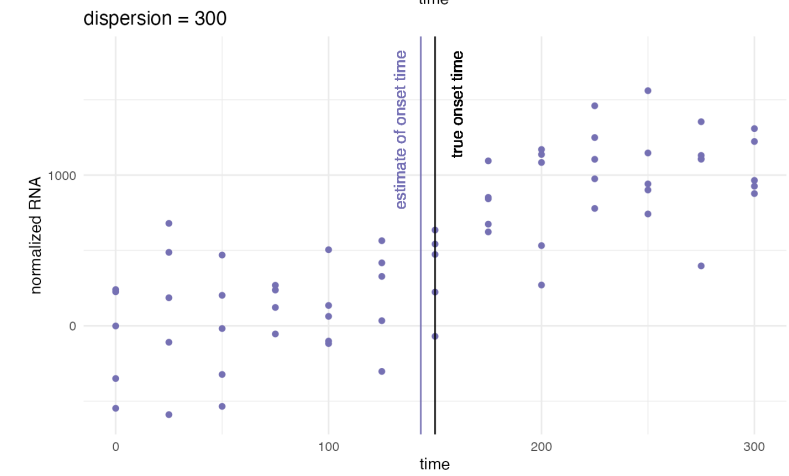
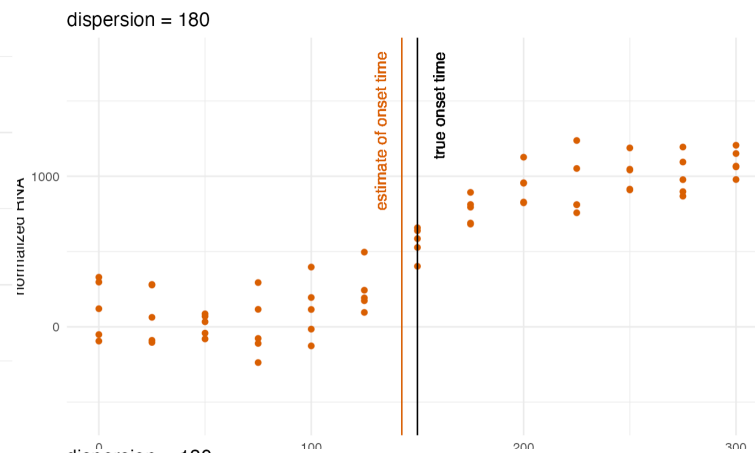
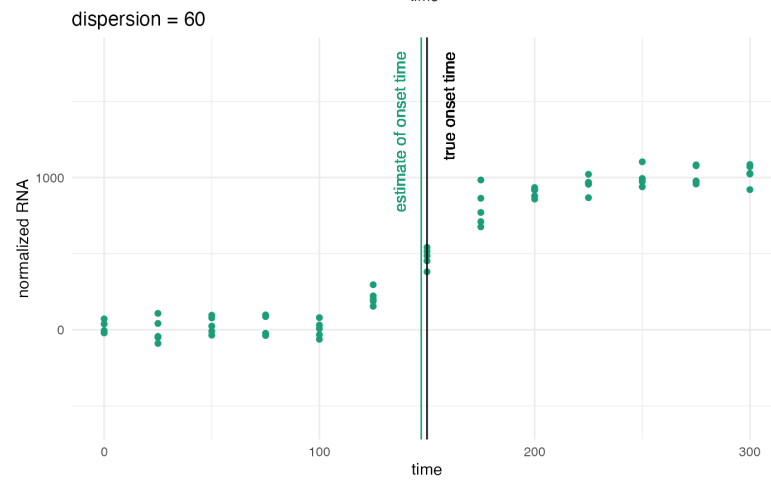
disp = 180

disp = 300

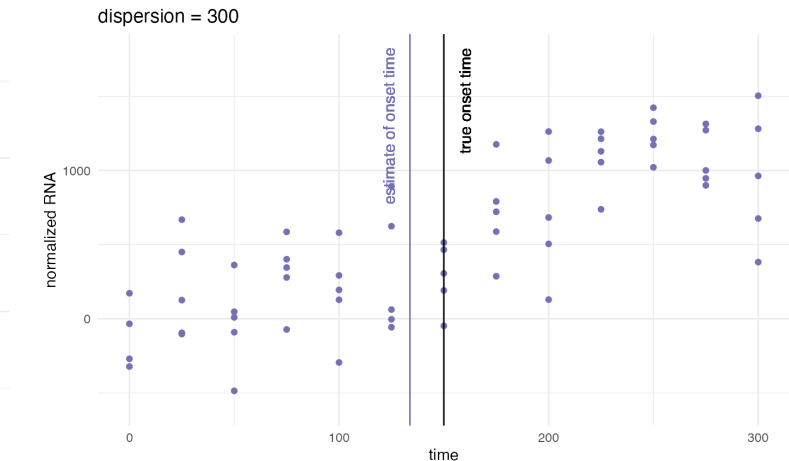
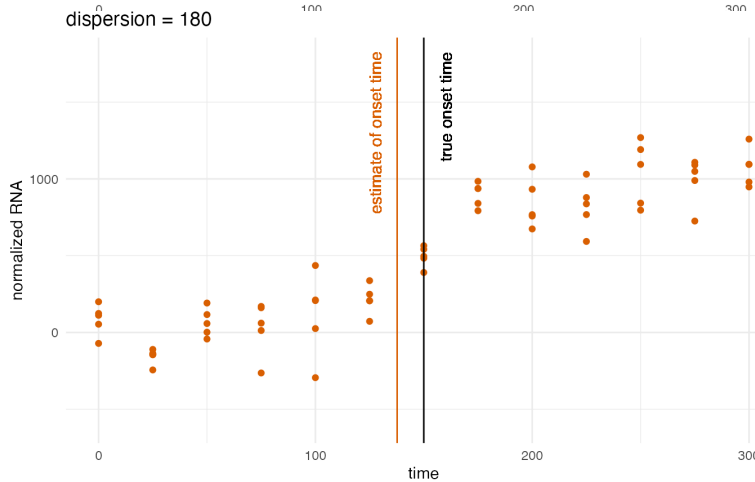
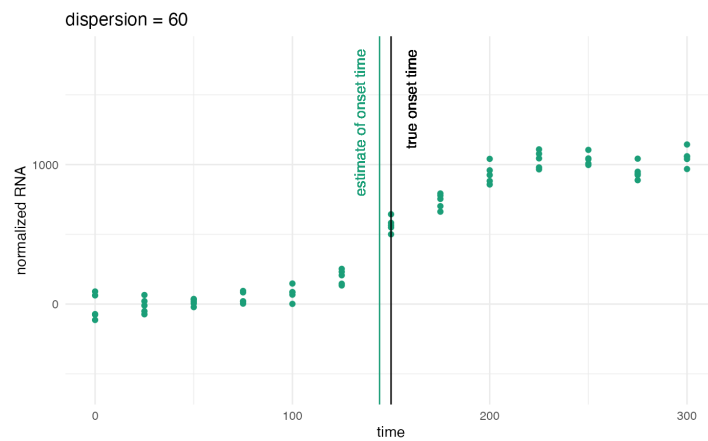
rep 1



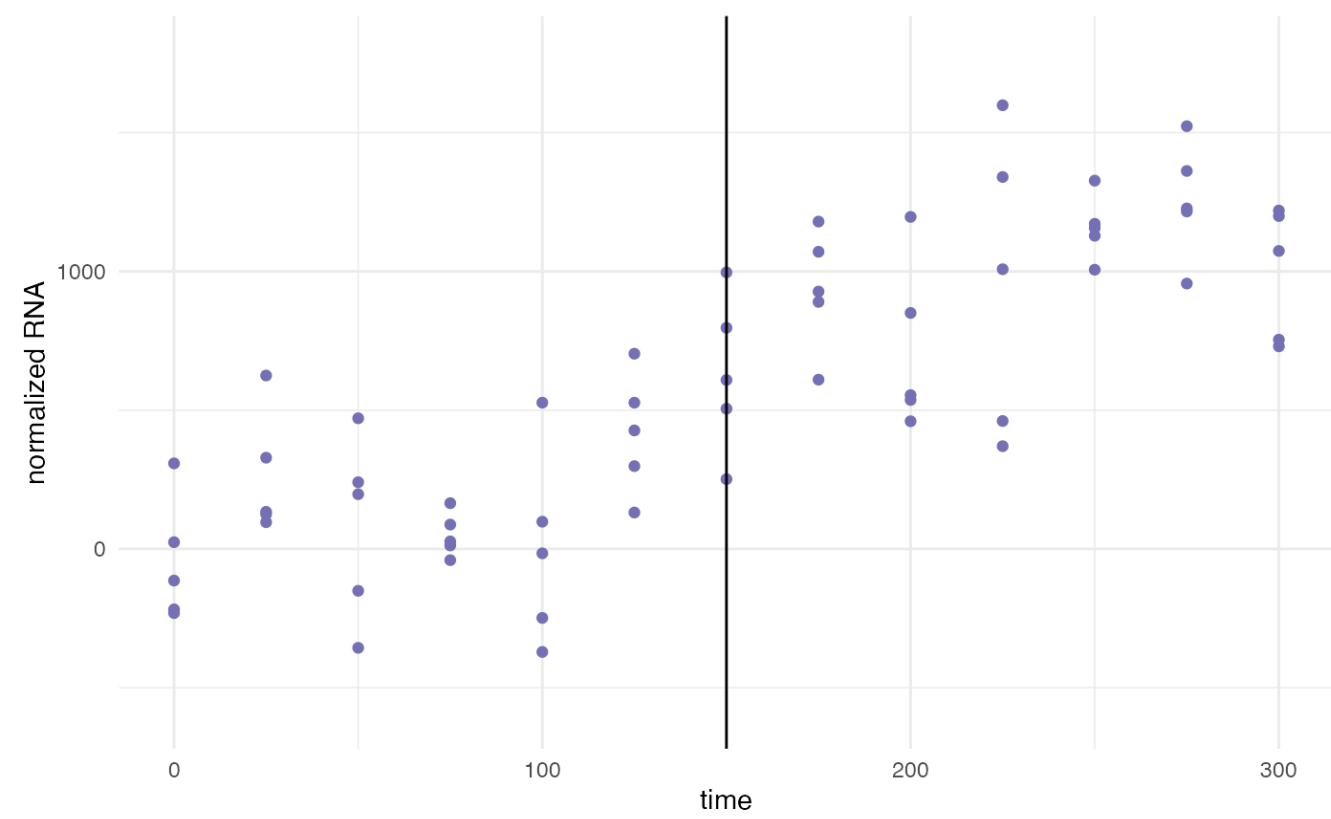
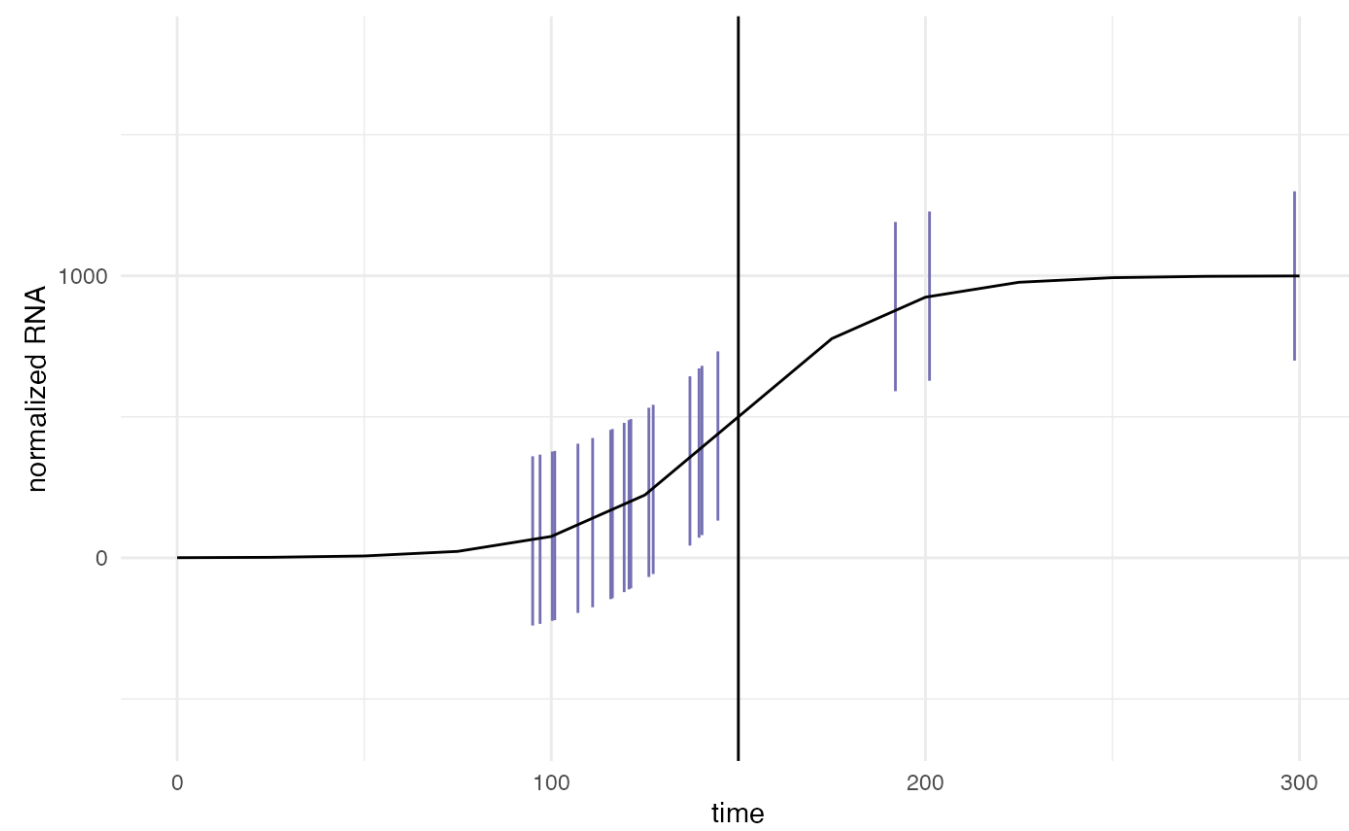
rep 2



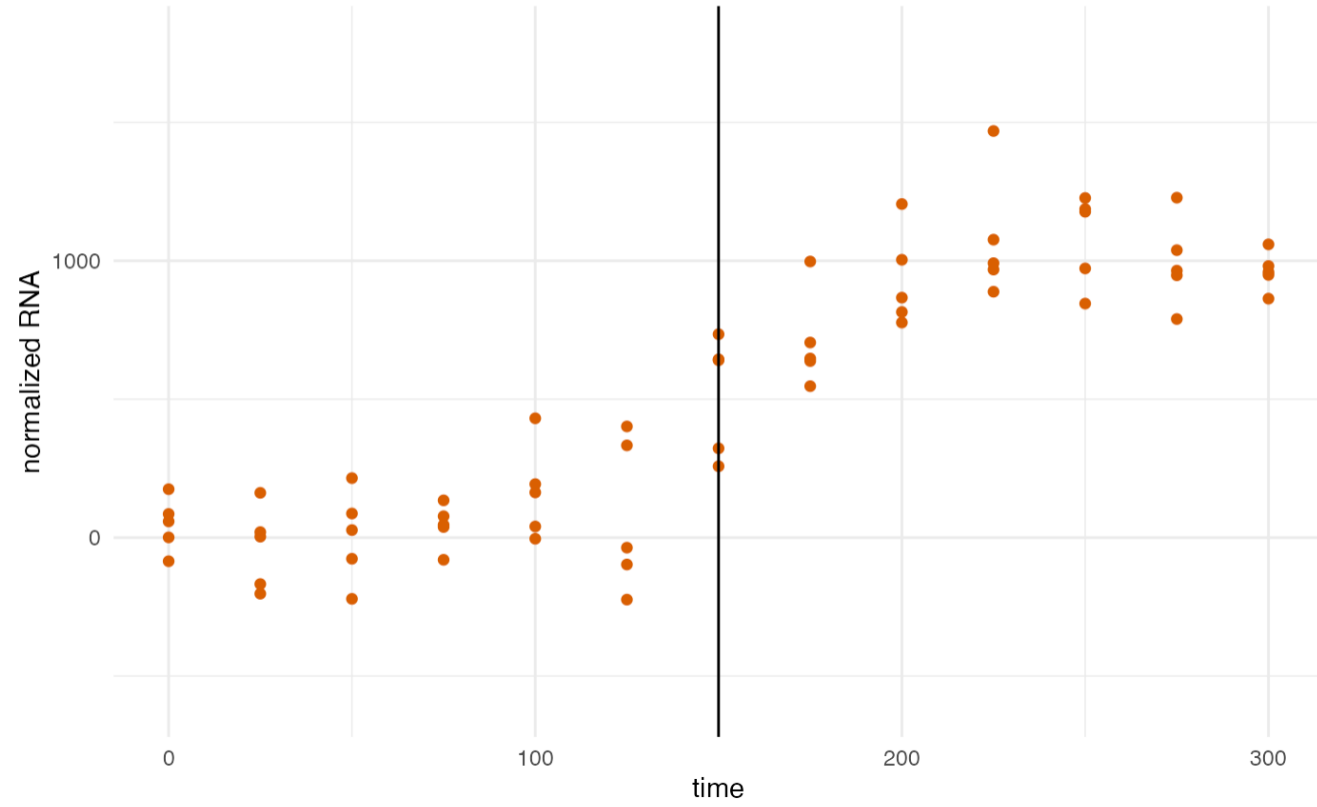
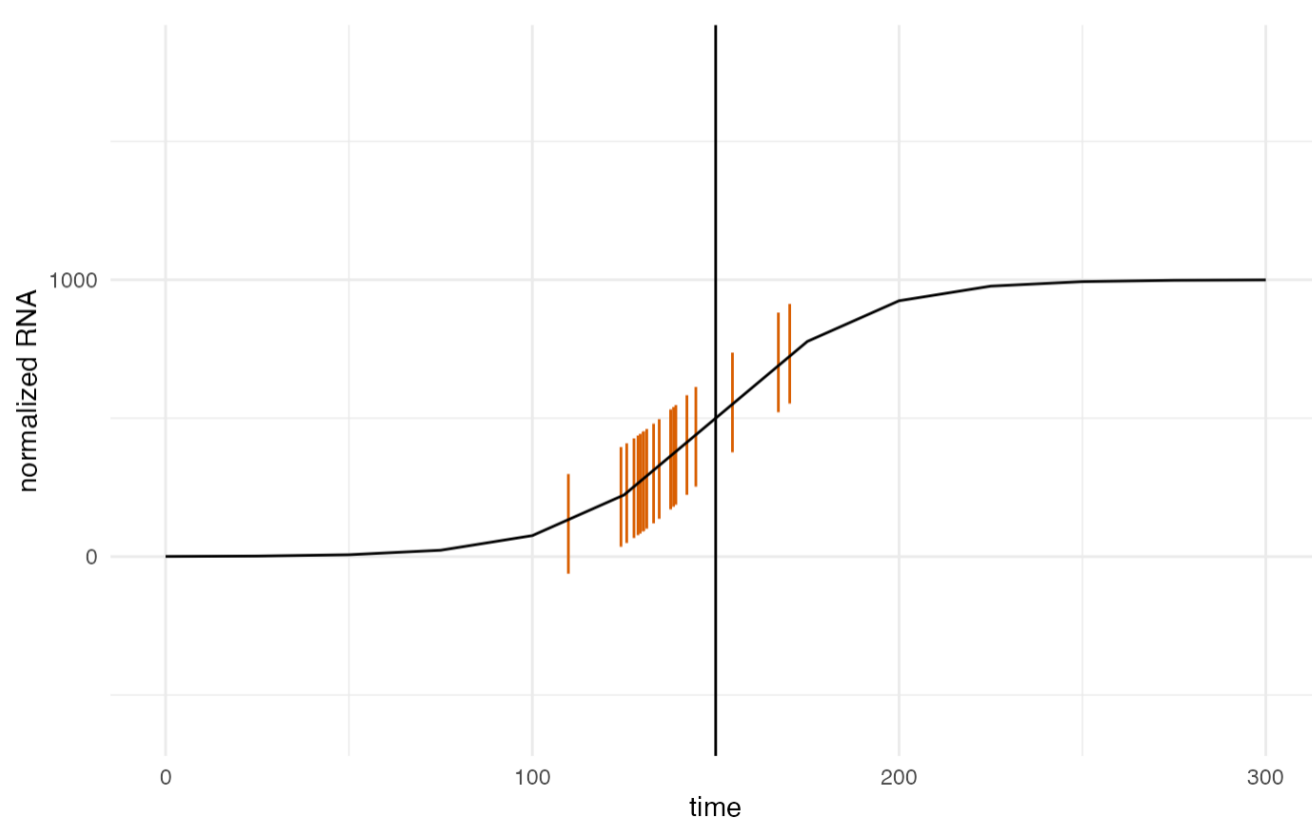
rep 3



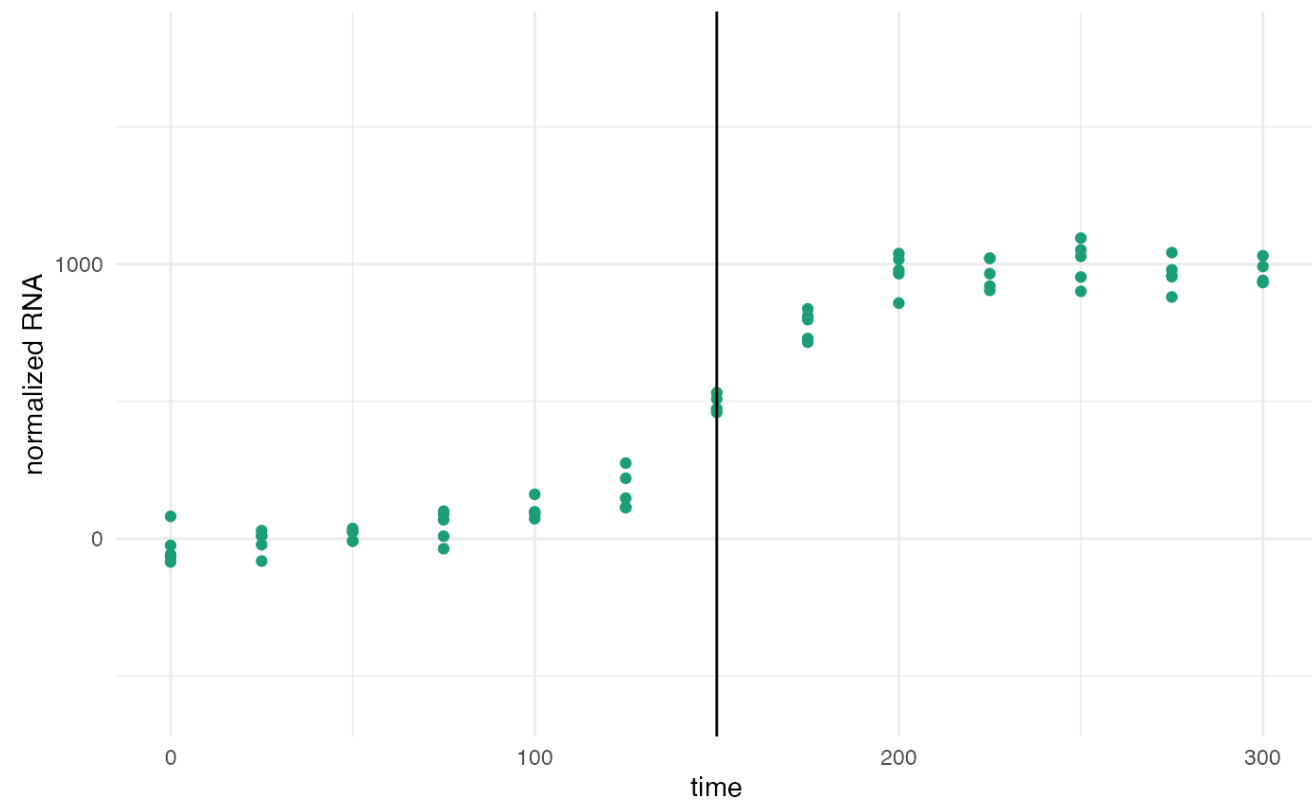
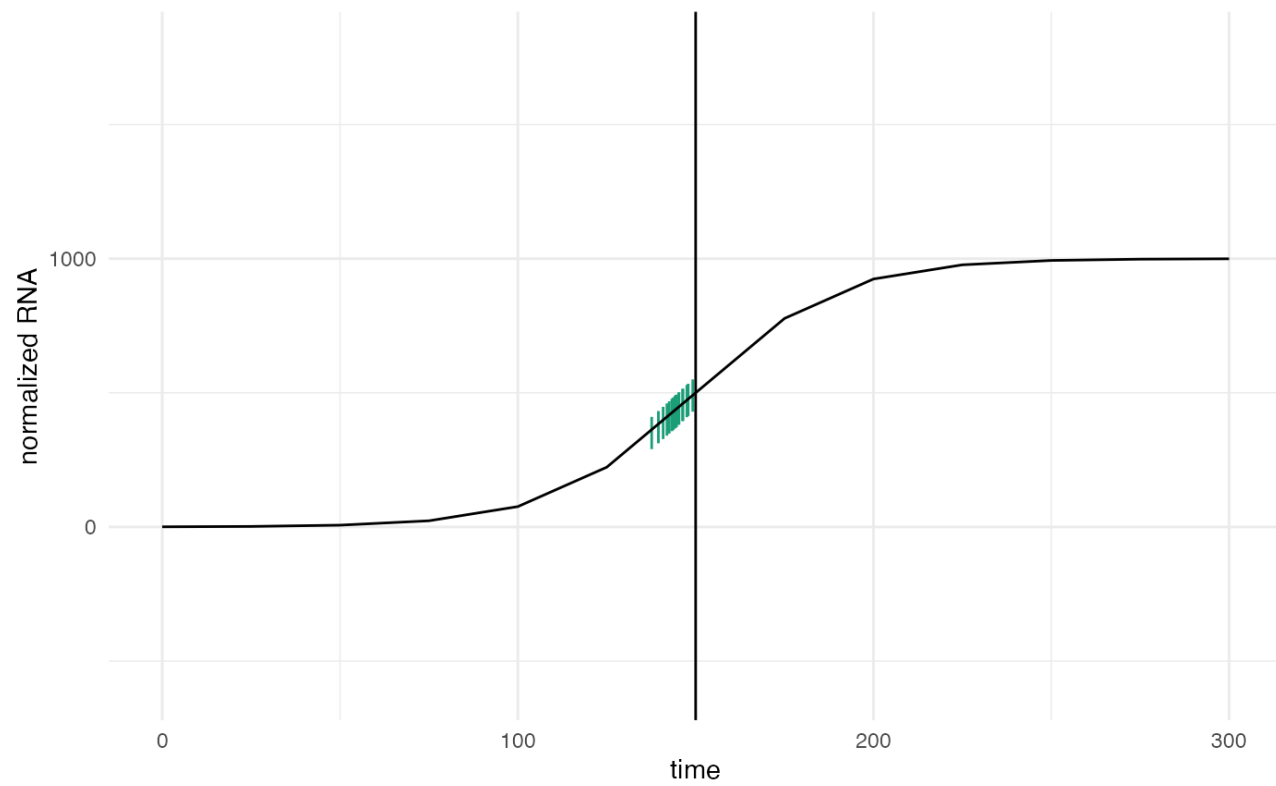
Simulations



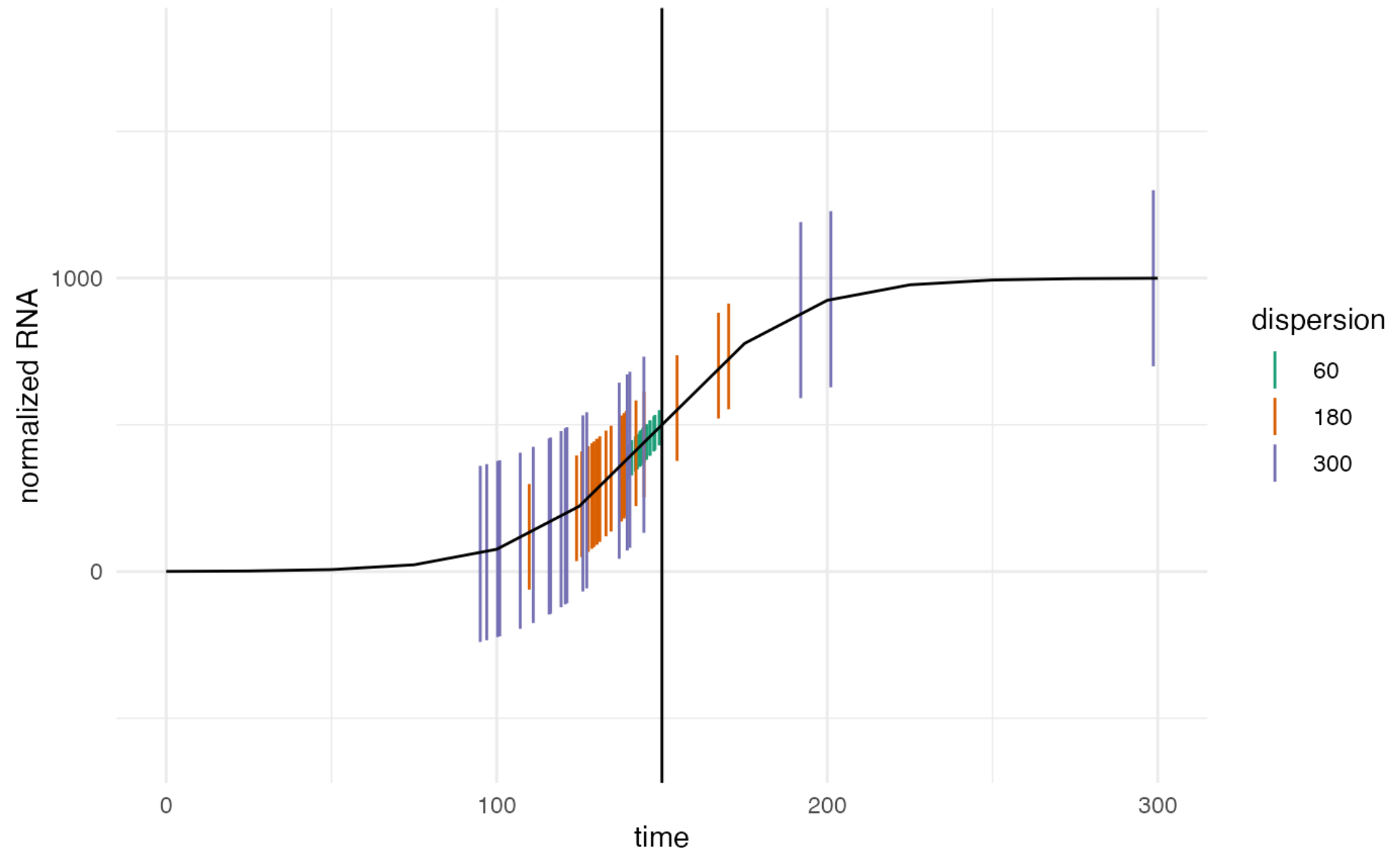
dispersion = 300



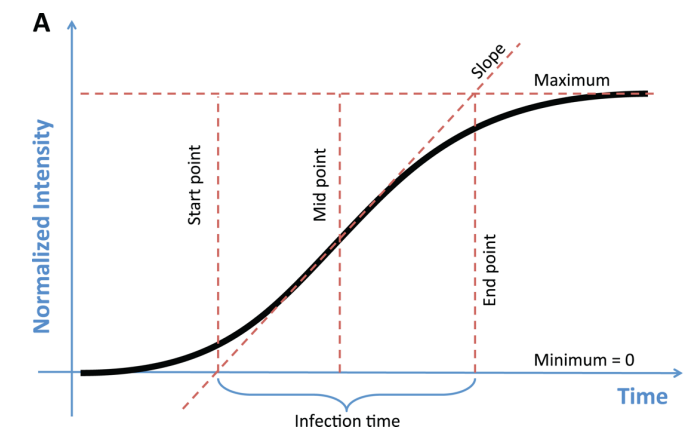
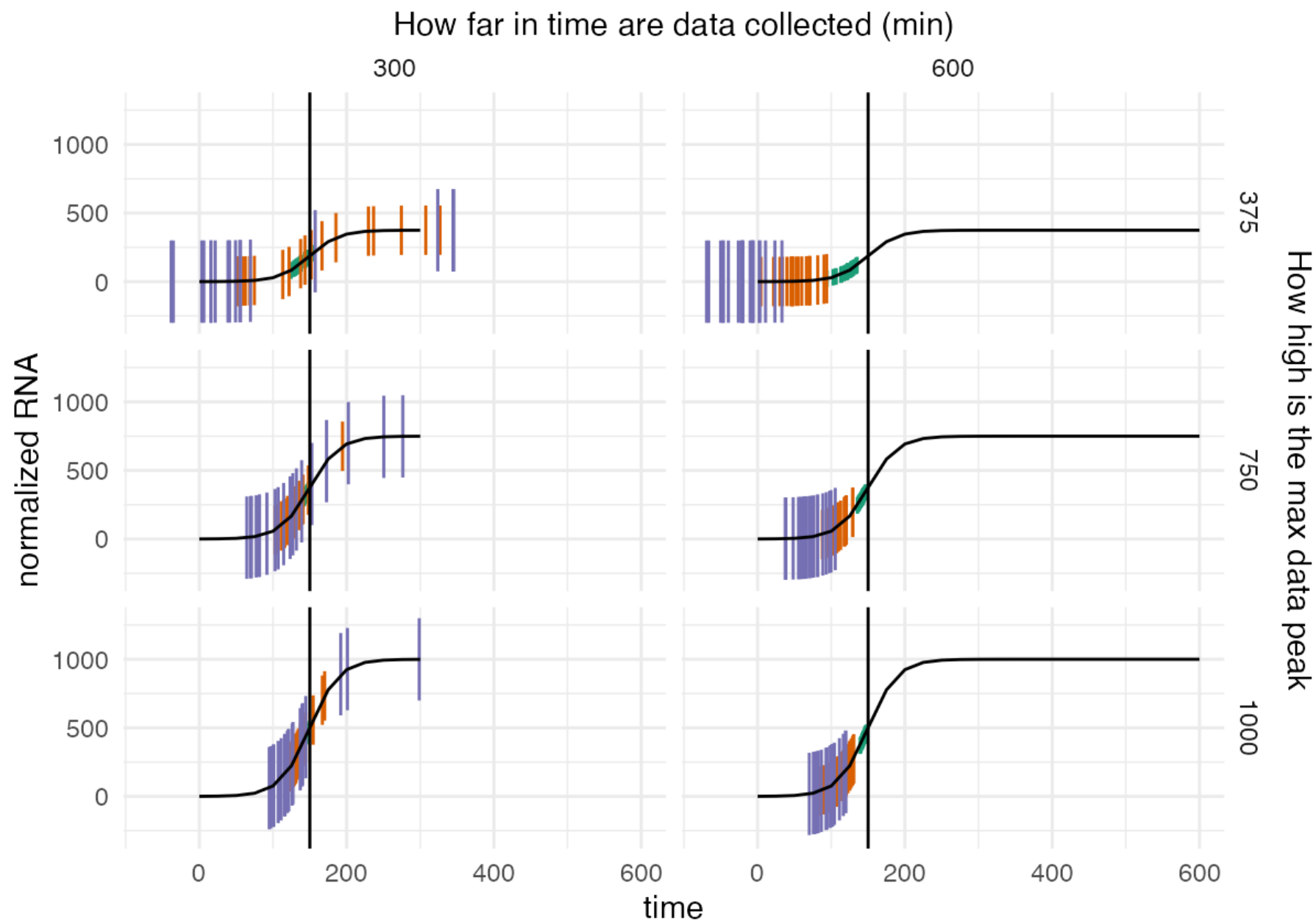
dispersion = 180



dispersion = 60



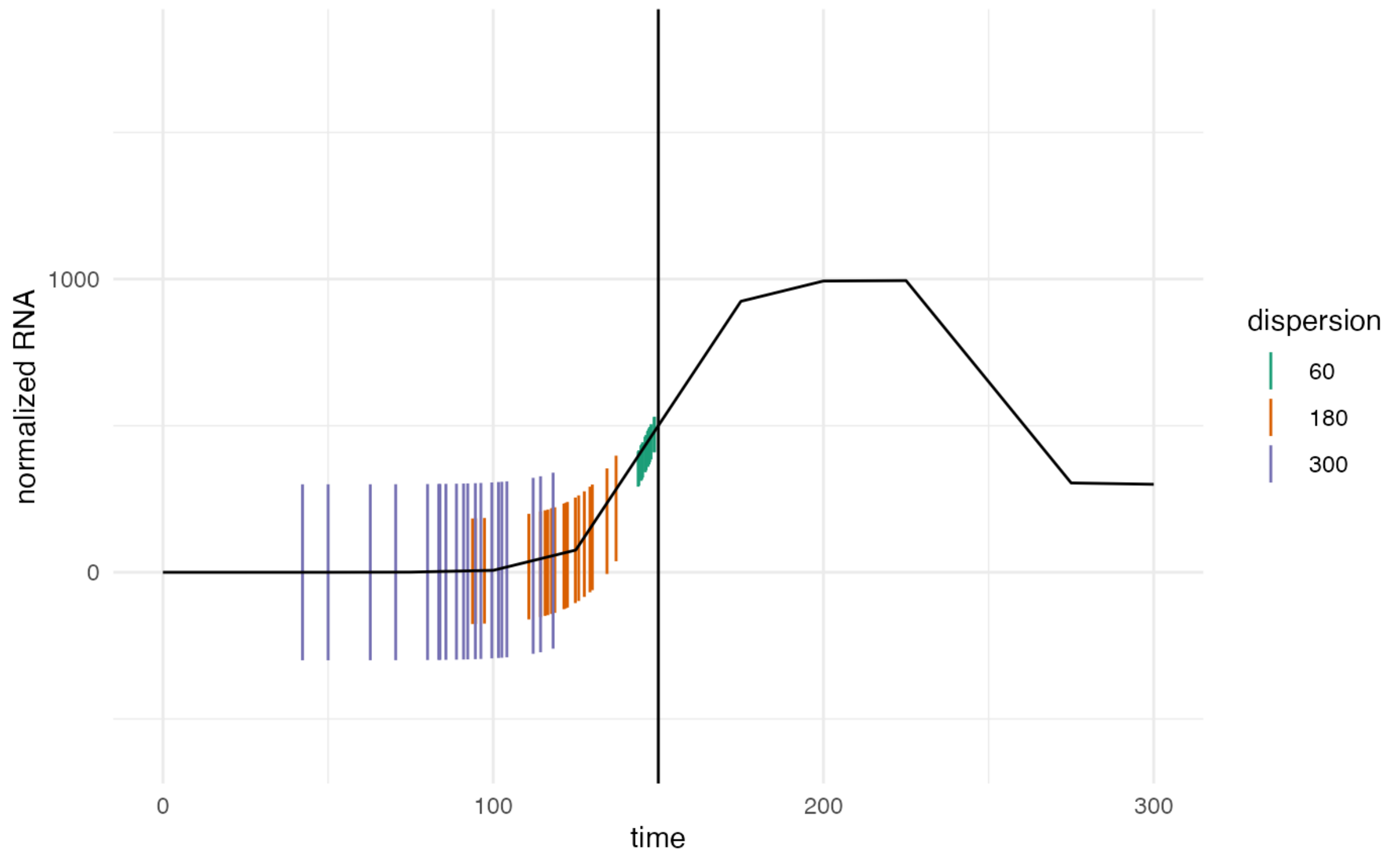
varying dispersion
(sigmoid)



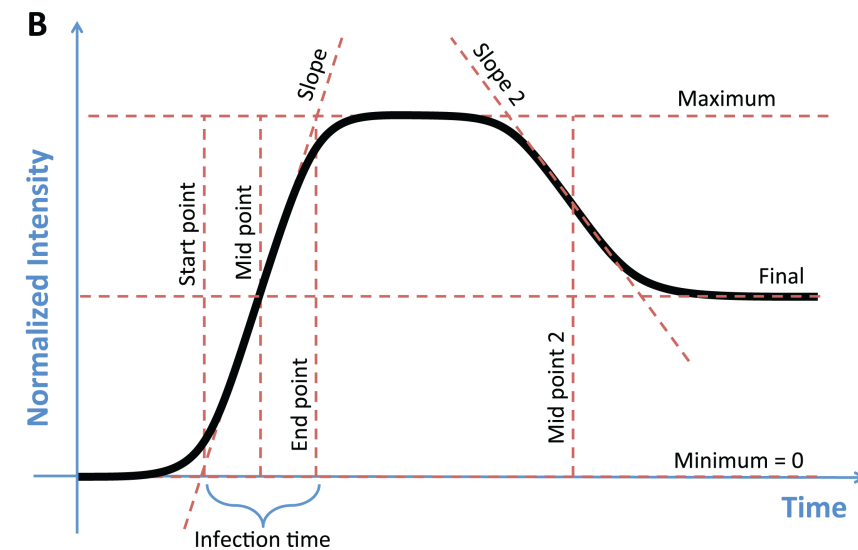
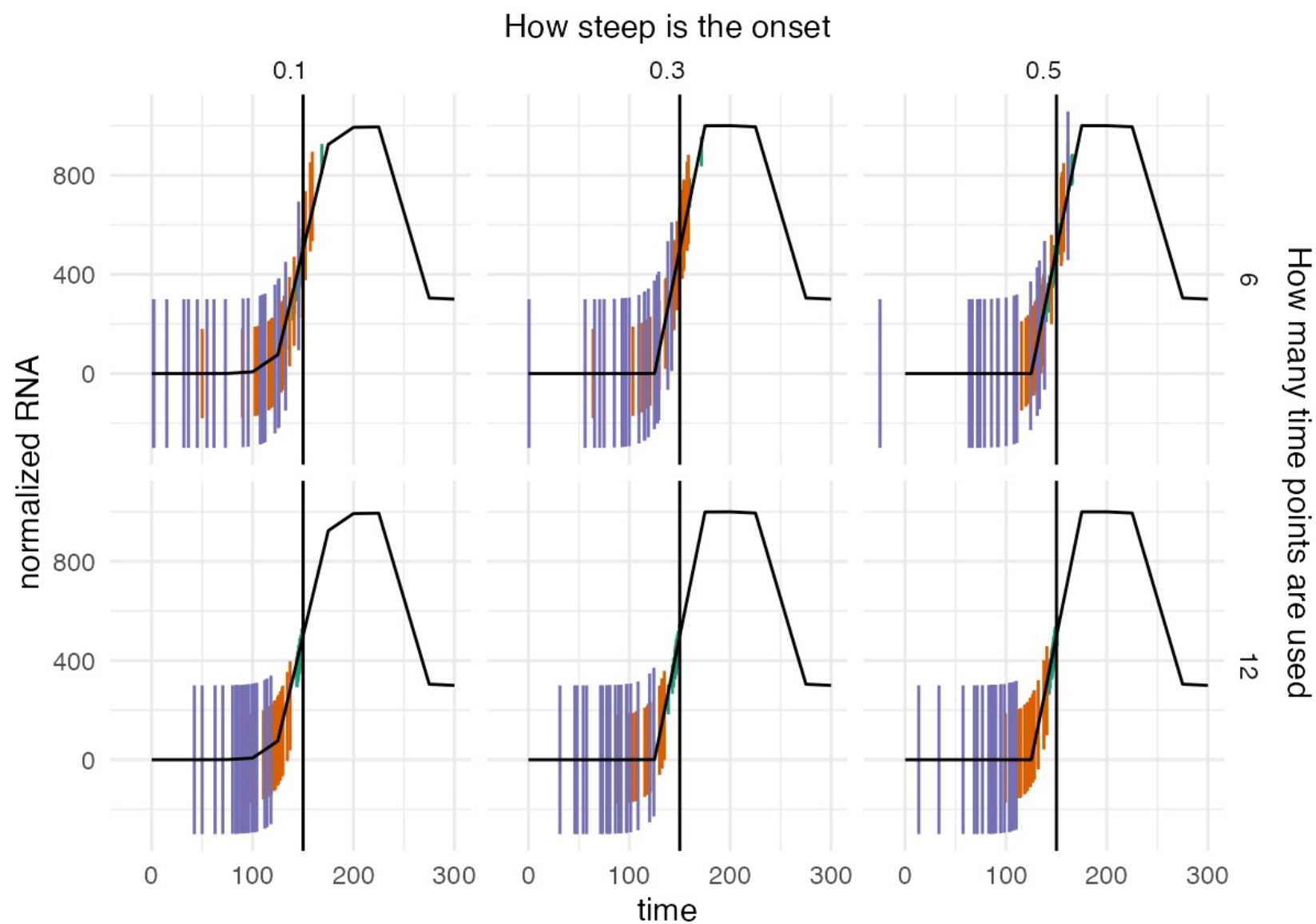
we have no control

we have control

varying time of collection and height of normalized data



varying dispersion
(double sigmoid)



dispersion

60

180

300

we have control

we have no control

varying number of time points
and steepness of onset

What impacts the bias and variability of the onset time?

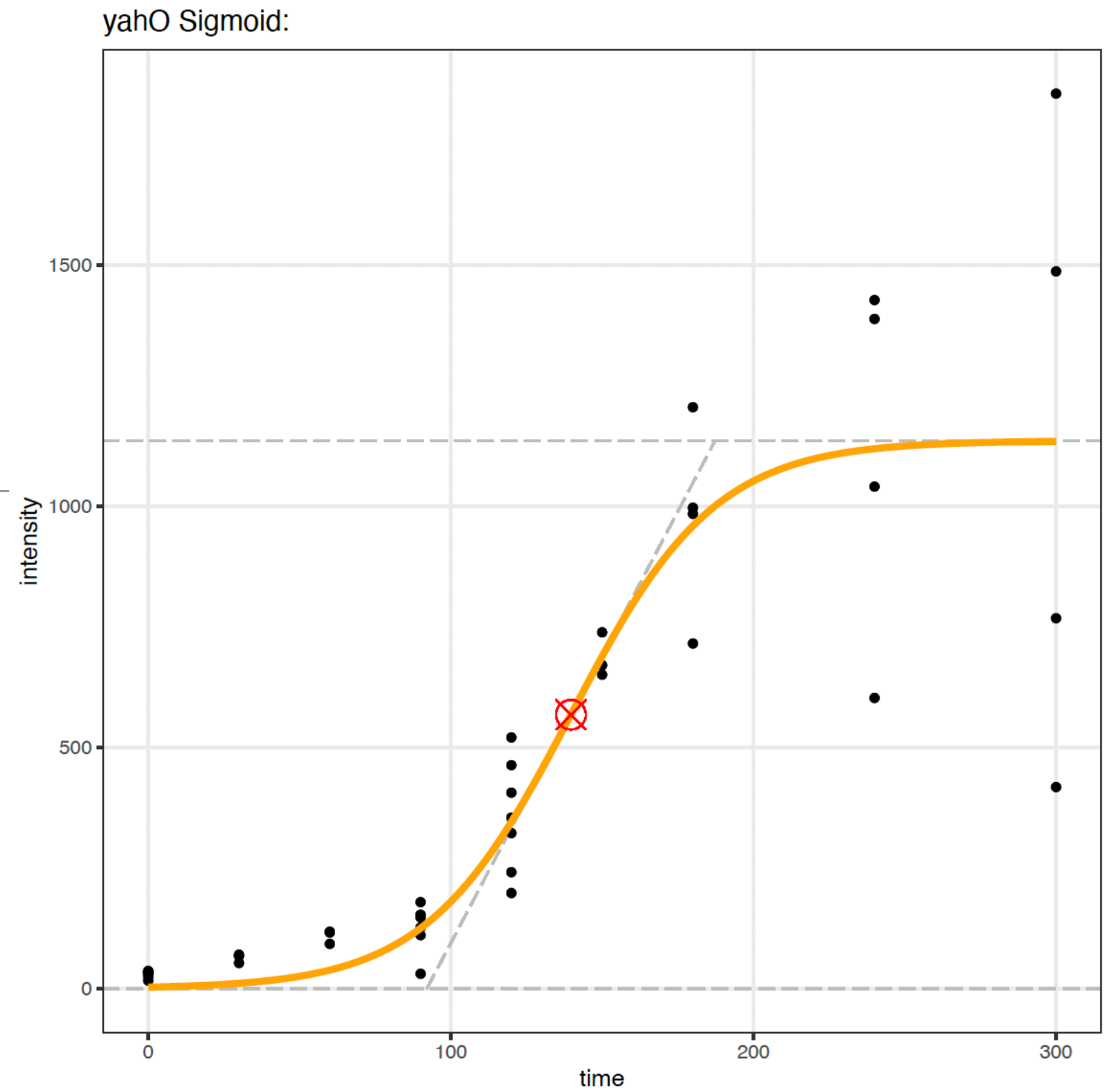
Parameters we cannot experimentally control

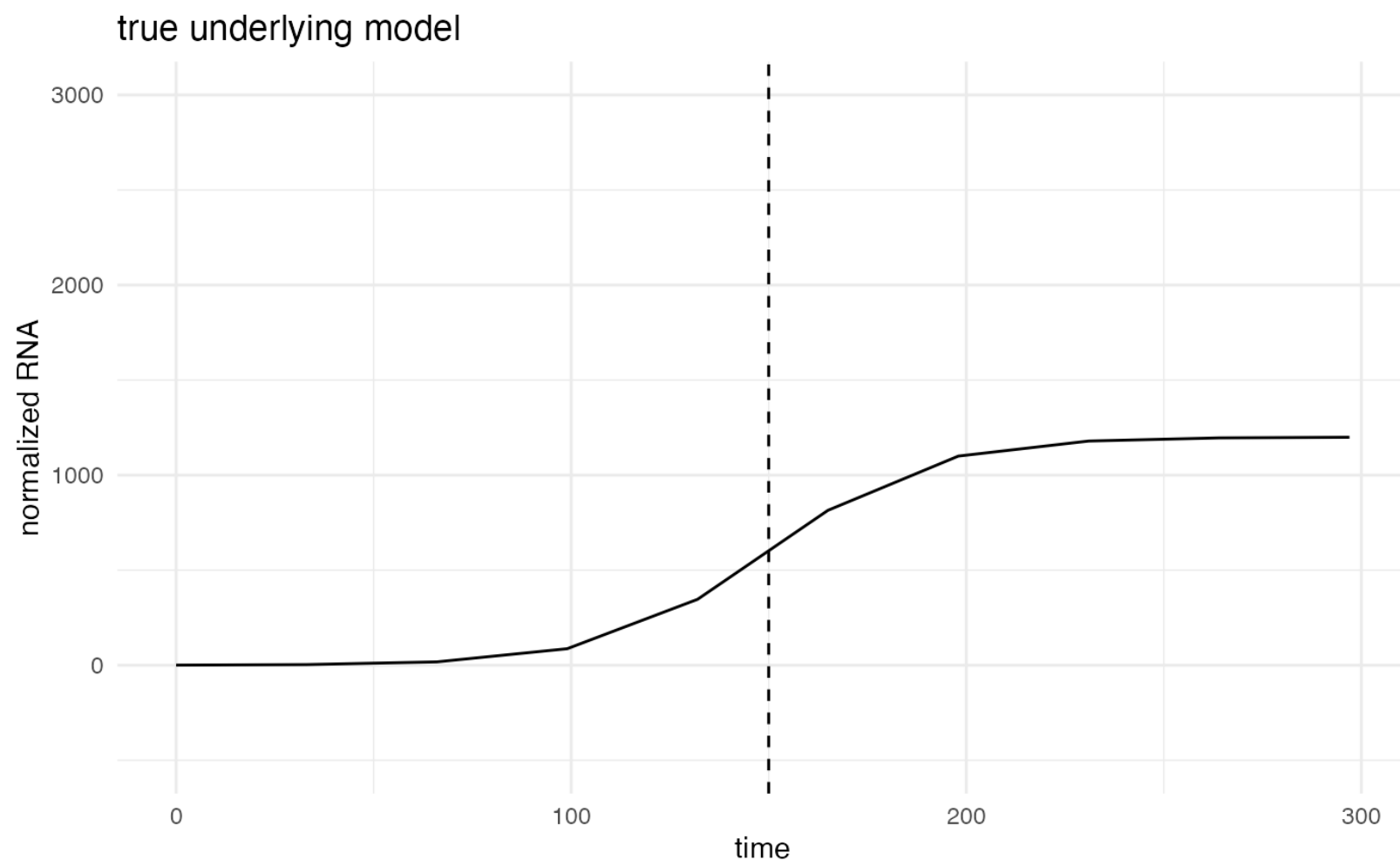
- Steepness of onset
- Difference in expression at time = 0 vs max expression
- Time when expression levels off
- Expression value when expression levels off
- Amount of variability (dispersion) around the model

Parameters we can experimentally control

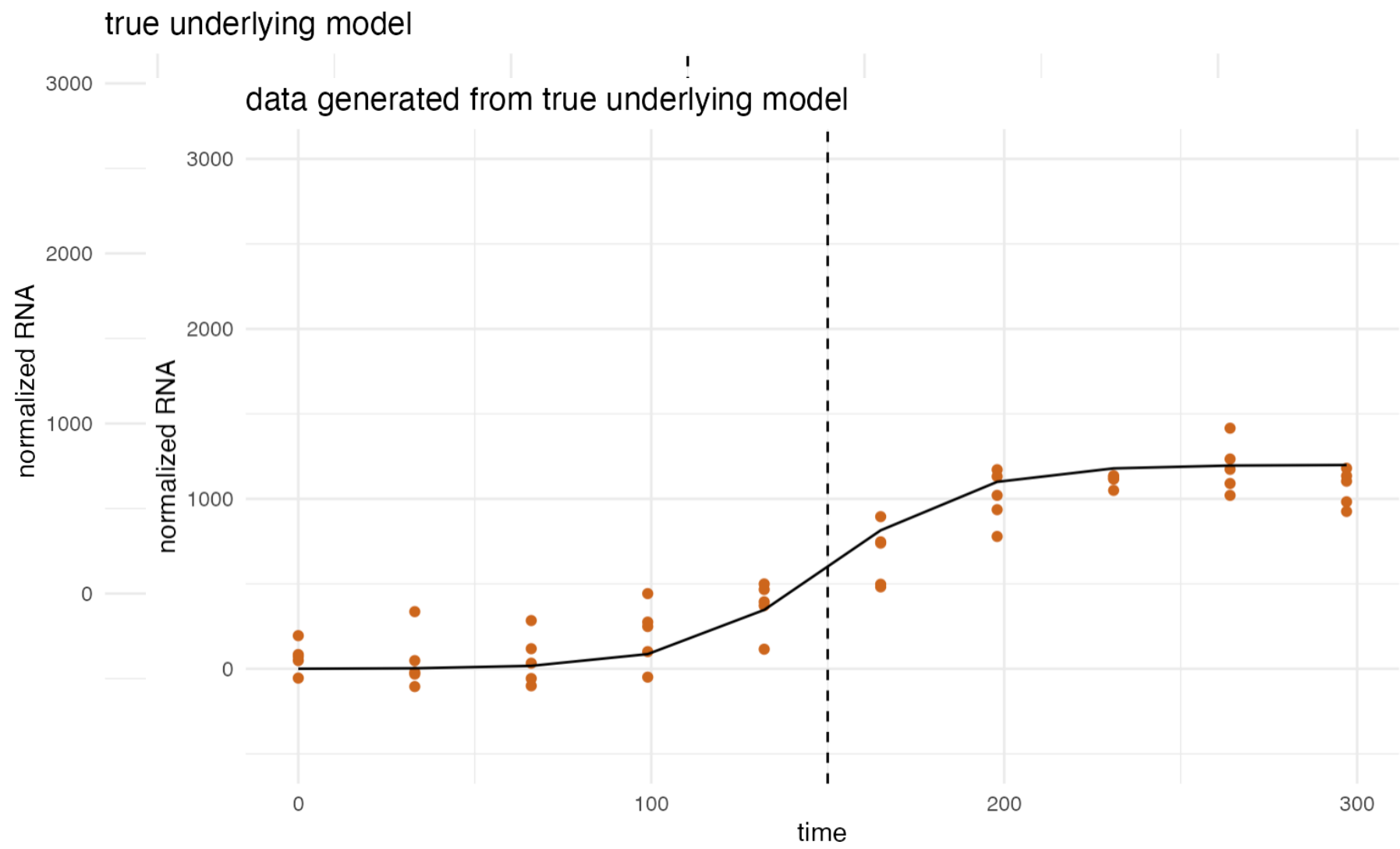
- Number of time points
- Range of time collection
- Number of observations per time point

Bias corrected onset

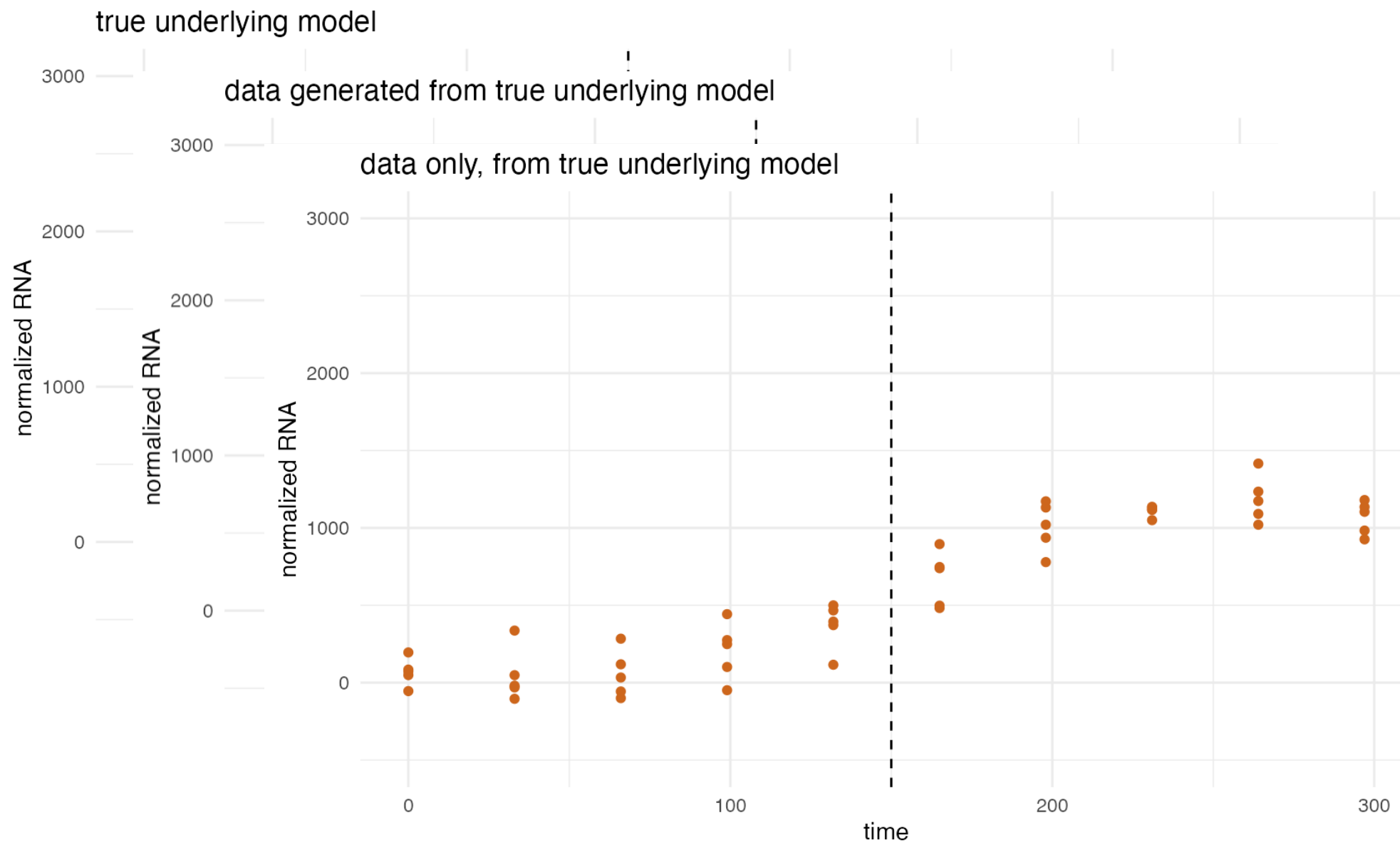




Bootstrapping



Bootstrapping



Bootstrapping

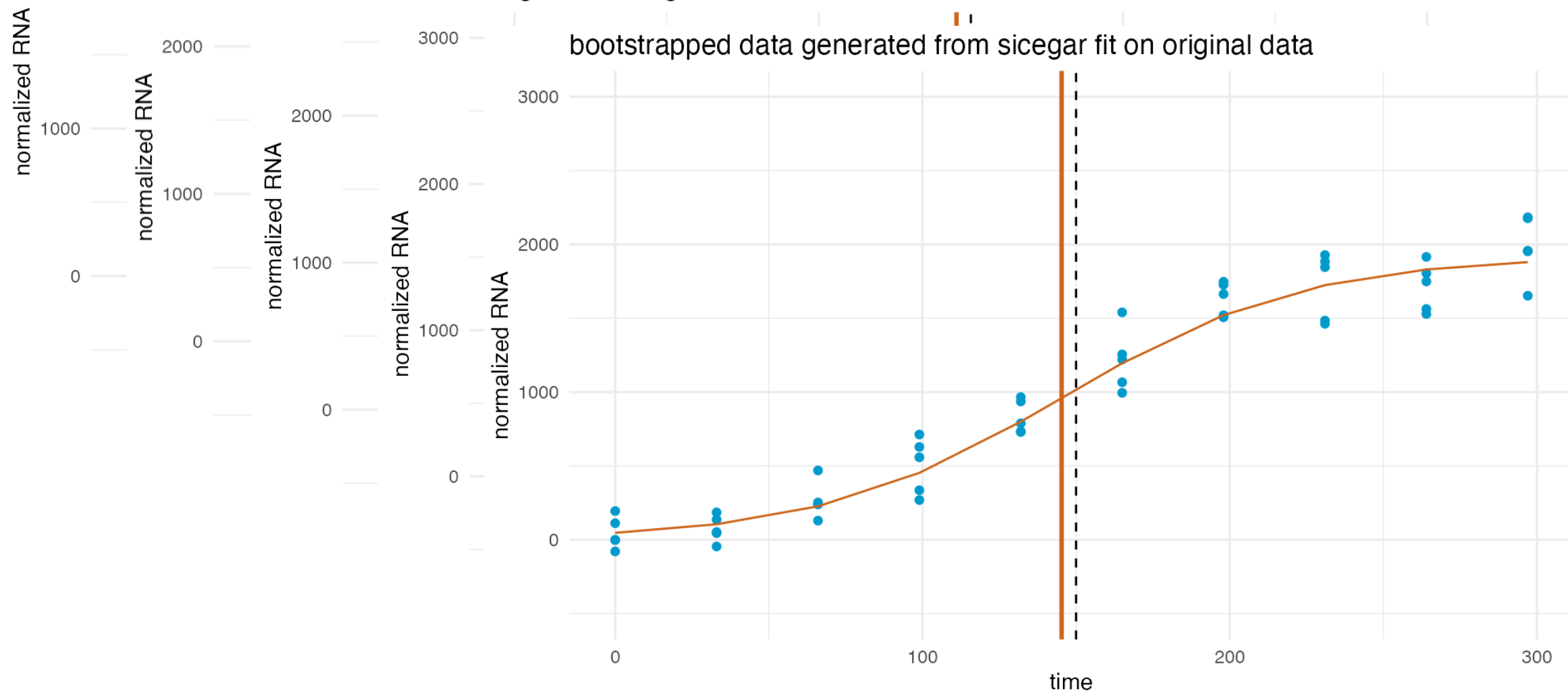
true underlying model

data generated from true underlying model

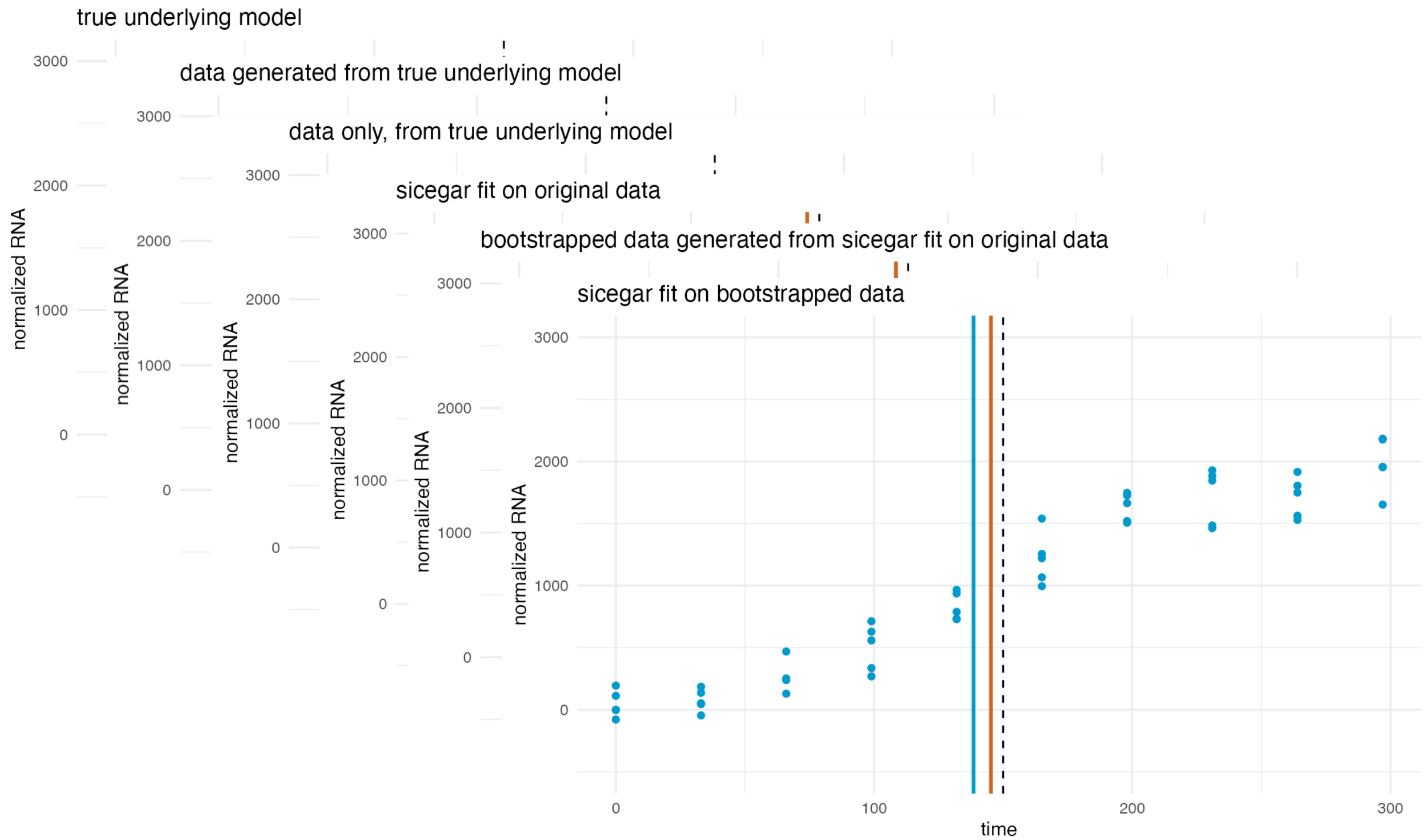
data only, from true underlying model

sicegar fit on original data

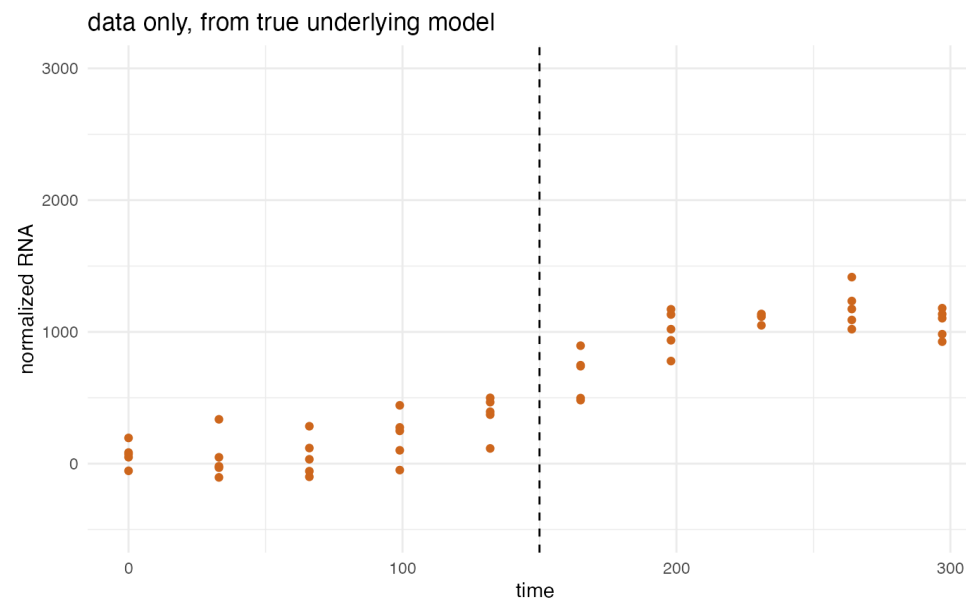
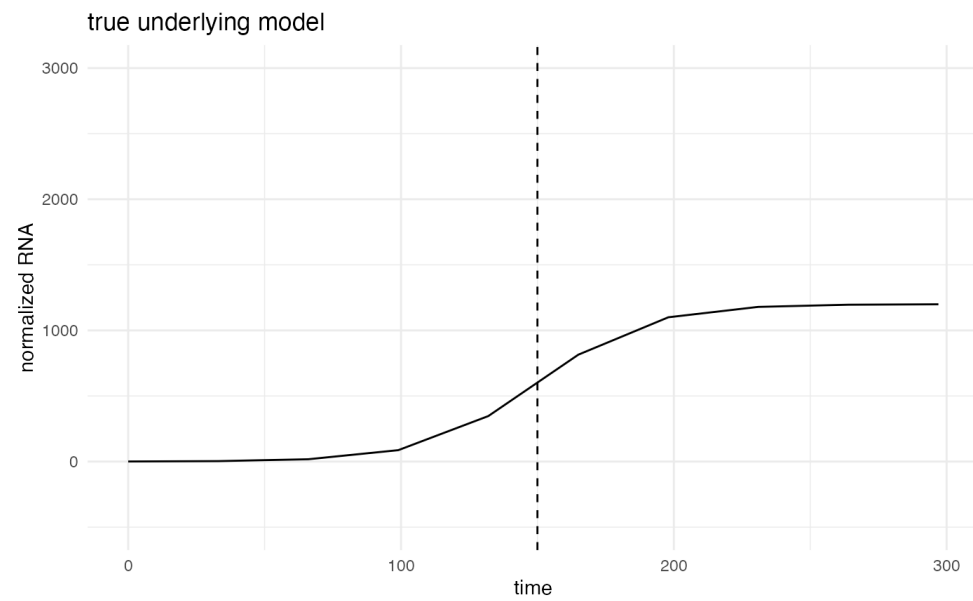
bootstrapped data generated from sicegar fit on original data



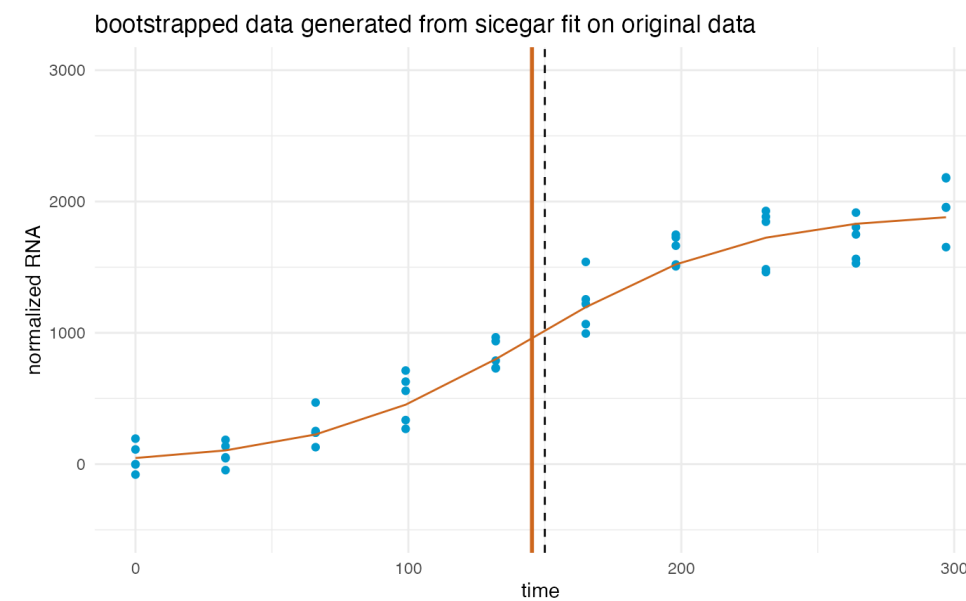
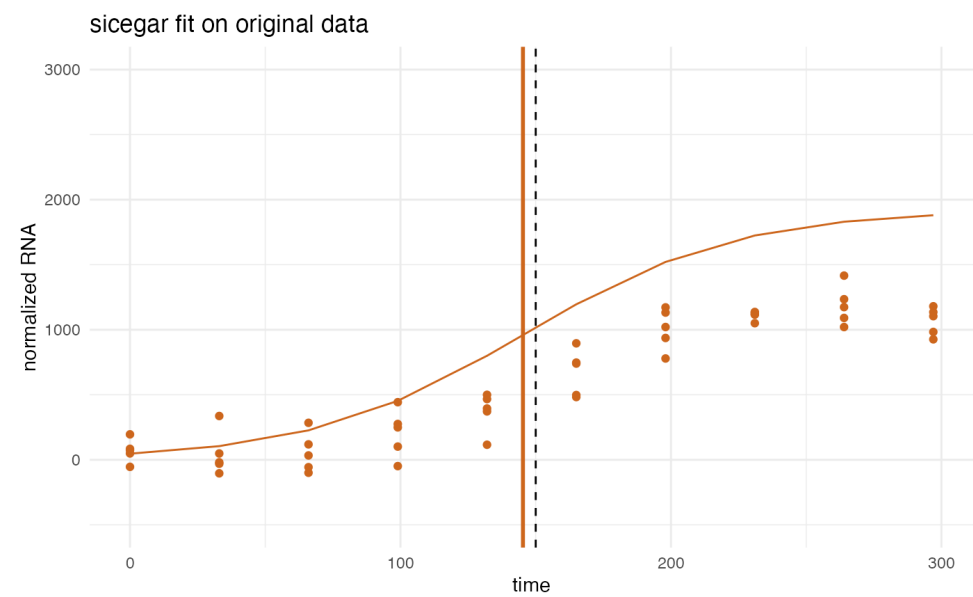
Bootstrapping



Bootstrapping

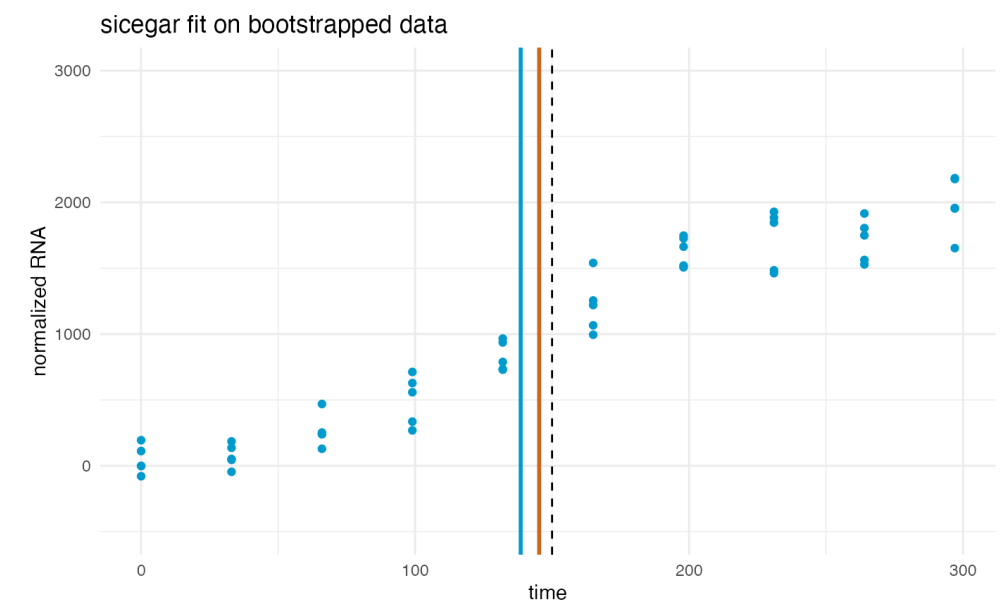


true underlying
model



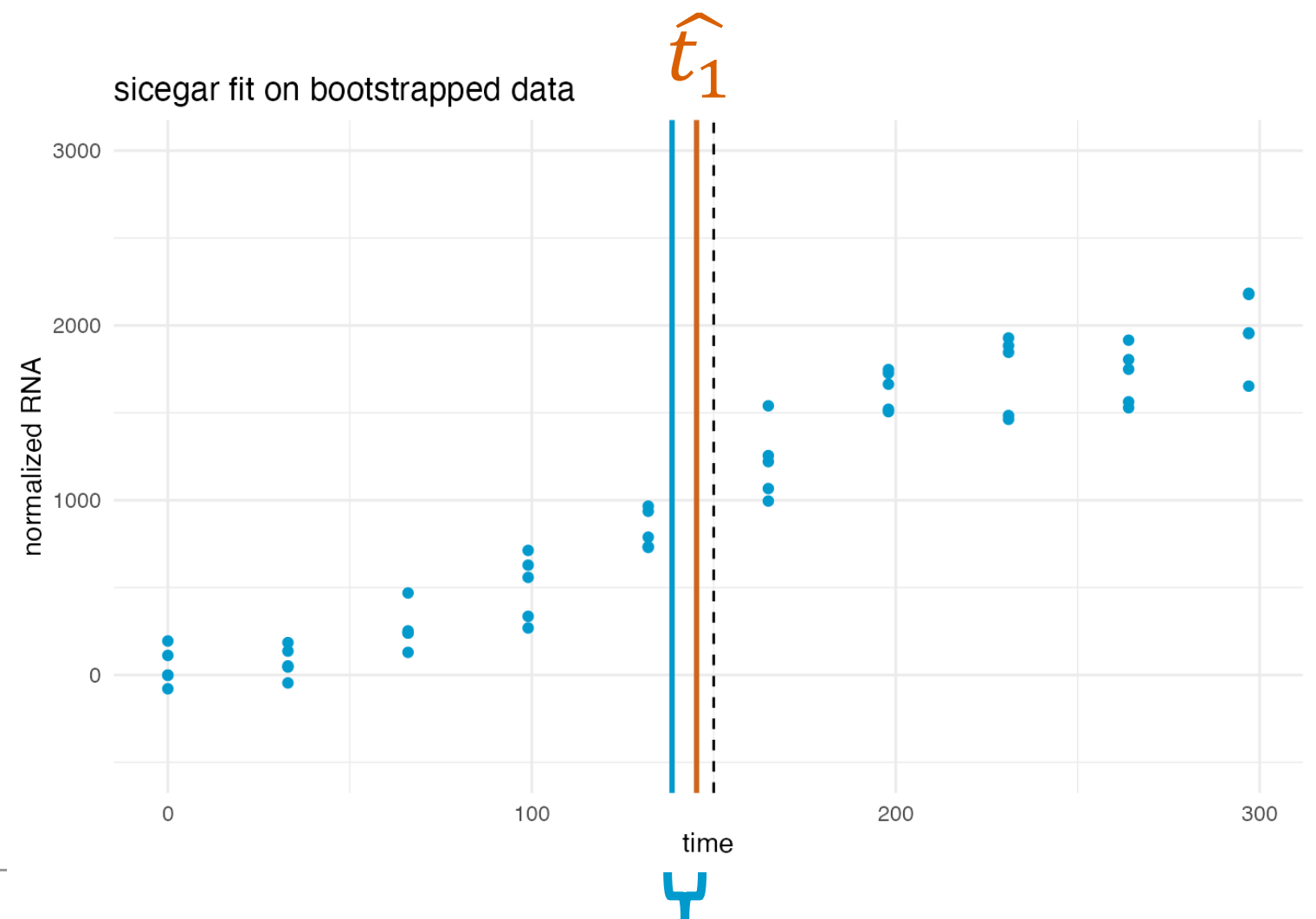
sicegar fit
on original
data

sicegar fit on
bootstrapped
data



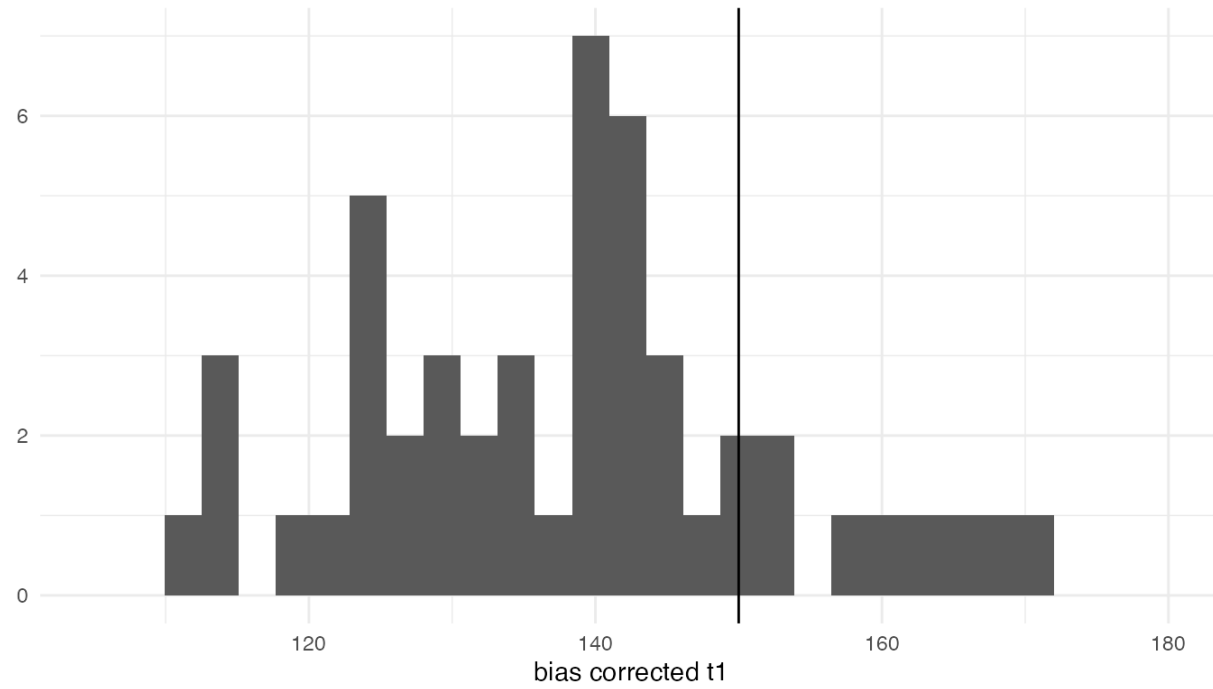
Bootstrapping

Bootstrap Simulations

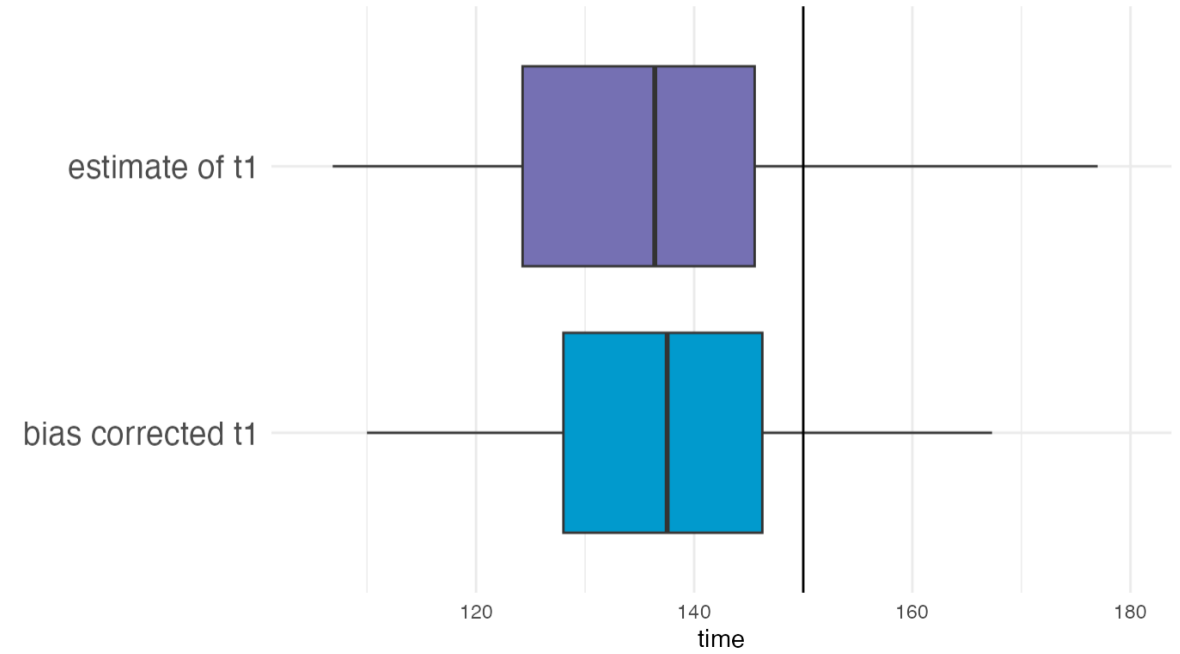


$$\hat{t}_{1boot} = \hat{t}_1 + bias_{boot}$$

Bootstrap bias corrected estimates of onset time
for dispersion 360

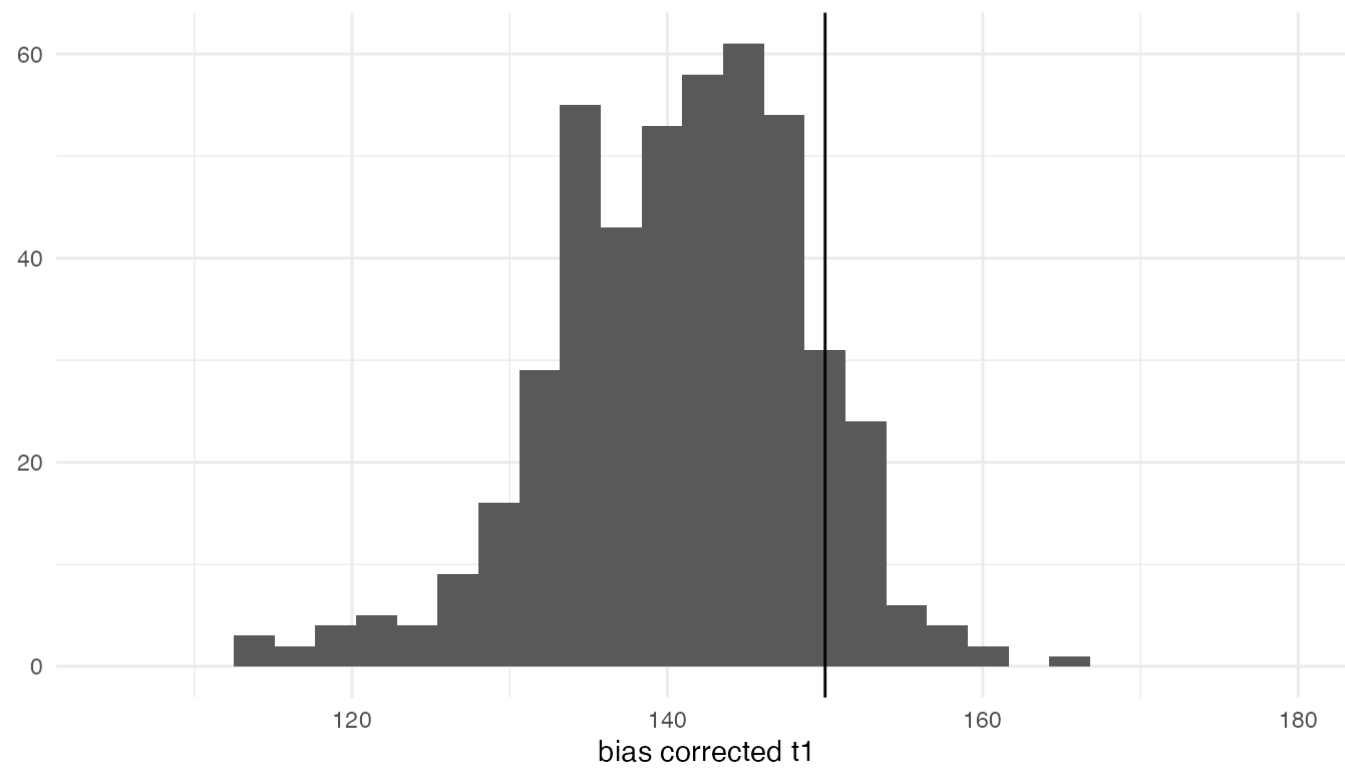


Bootstrap bias corrected estimates of onset time
for dispersion 300

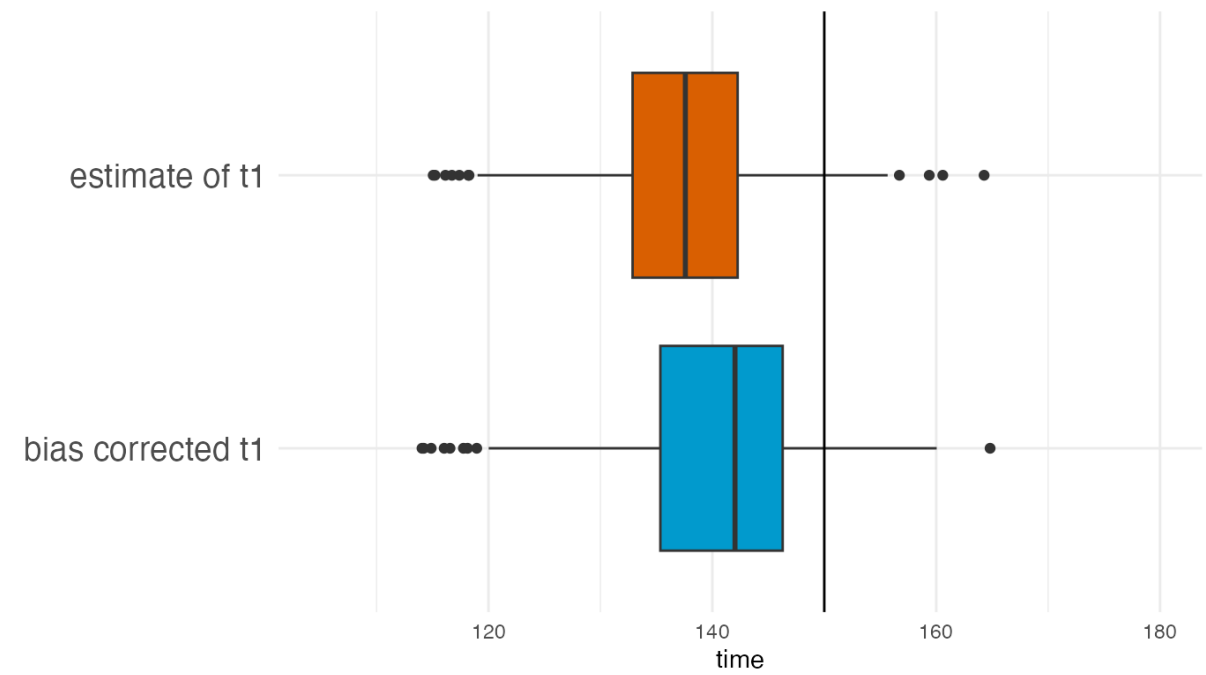


dispersion = 300

Bootstrap bias corrected estimates of onset time
for dispersion 180

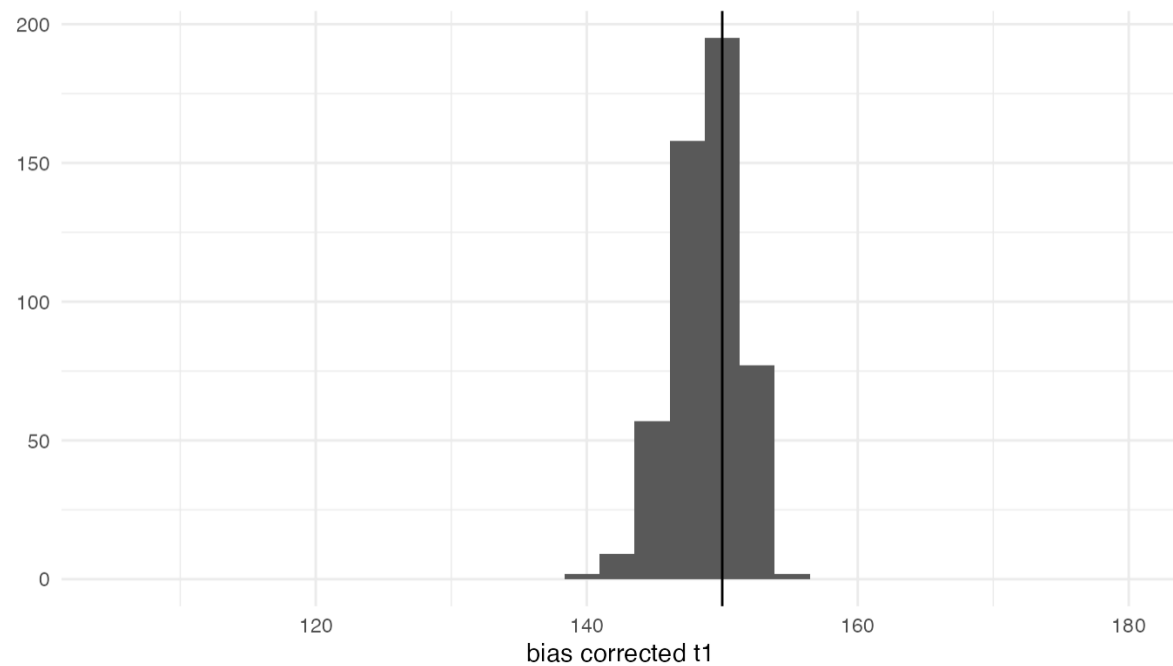


Bootstrap bias corrected estimates of onset time
for dispersion 180

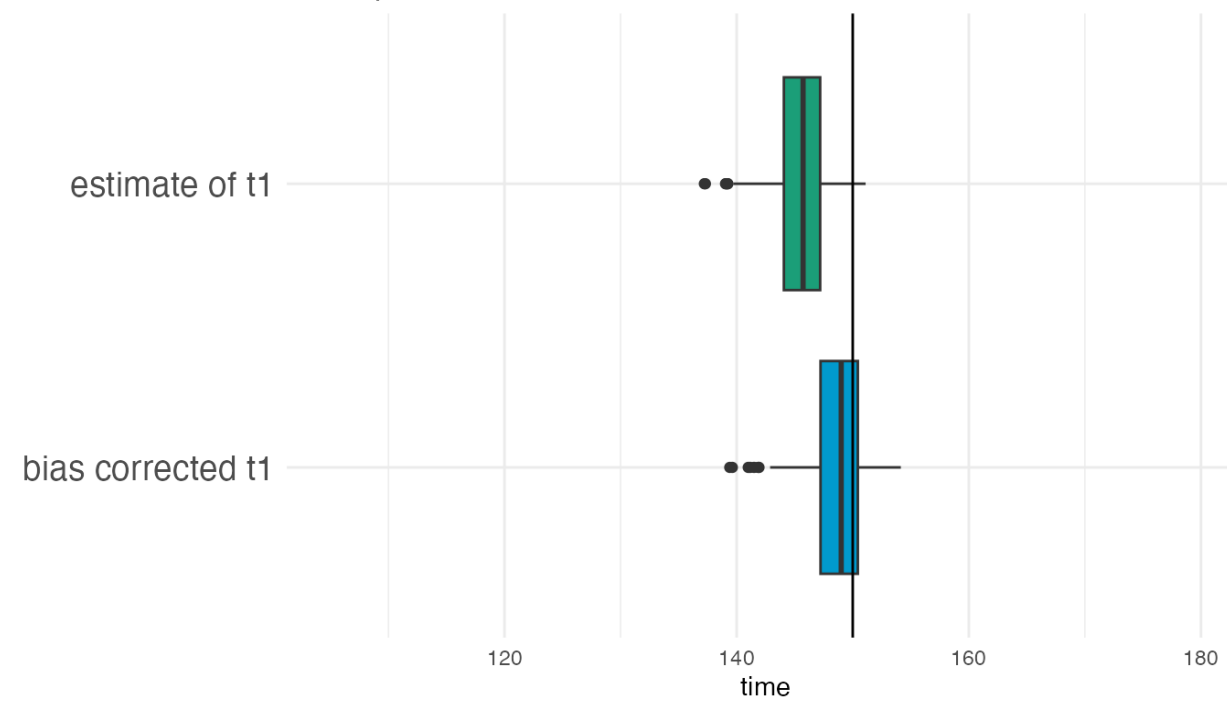


dispersion = 180

Bootstrap bias corrected estimates of onset time
for dispersion 60



Bootstrap bias corrected estimates of onset time
for dispersion 60



dispersion = 60

Next steps

- Bootstrapping the double sigmoid
- Literature review to identify which parameter settings are most common
- Synthesizing results for practitioners
 - focus here has been on t1, but we have also investigated settings where “no model” is reported

Genome-Wide Transcriptional Response to Varying RpoS Levels in *Escherichia coli* K-12



Garrett T. Wong^{a,*}, Richard P. Bonocora^b, Alicia N. Schep^{a,*}, Suzannah M. Beeler^{a,*}, Anna J. Lee Fong^{a,*}, Lauren M. Shull^{a,*}, Lakshmi E. Batachari^a, Moira Dillon^a, Ciaran Evans^{c,*}, Carla J. Becker^a, Eliot C. Bush^a, Johanna Hardin^c, Joseph T. Wade^{b,d}, Daniel M. Stoebel ^a

The timing of transcription of RpoS-dependent genes varies across multiple stresses in *Escherichia coli* K-12

Josephine Adams¹, Johnson Hoang¹, Emily Petroni¹, Ethan Ashby², Johanna Hardin², Daniel M. Stoebel ¹

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Thank you!

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